# Self-Adjoint Linear Operators

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The theory of *self-adjoint linear operators* in Hilbert space proves to be one of the first well-elaborated subjects within the mathematical area of Functional Analysis. Its corner stones have mainly been developed by the mathematicians D. Hilbert, F. Riesz, E. Schmidt, J. von Neumann, M. H. Stone and others in the first half of the 20th century.

A great portion of motivation to develop an abstract theory of self-adjoint linear transformations came from questions and problems arising in mathematical physics. For example, the mathematical formalism for quantum mechanical systems leads in a very natural way to self-adjoint operators acting in $L^2(\mathbb{R}^d)$, the Hilbert space of all measurable and square-integrable functions $f : \mathbb{R}^d \to \mathbb{C}$. Other problems, like the Sturm-Liouville eigenvalue problem and the Dirichlet and the Neumann boundary value problem for the negative Laplacian $-\Delta$, also lead to the theory of self-adjoint operators in the respective Hilbert spaces $L^2(I)$ and $L^2(G)$, where $I \subset \mathbb{R}$ is a finite interval and $G$ is a bounded domain in some Euclidean space $\mathbb{R}^d$.

Self-adjoint operators are very suitable to be studied by *spectral theory*, hence the spectral properties of such an operator are well-understood. Moreover, the spectral theorem for unbounded self-adjoint operators is one of the most celebrated assertions in Functional Analysis and can be seen as the highlight of our subject. But besides this, a further fact will be of interest: if a self-adjoint operator is even *half-bounded* in a sense, then it appears as the infinitesimal generator of a strongly continuous and analytic semigroup, respectively, which is of great importance in the solution theory of evolution equations of hyperbolic and parabolic type governed by a half-bounded self-adjoint operator.

Troughout we will work with the sets $\mathbb{R}$ and $\mathbb{C}$ as the underlying fields of the normed linear spaces under consideration. Furthermore, the Hilbert spaces mentioned in the sequel are always supposed to be *complex* if nothing else is said. This is profitable, since spectral theoretical methods can only be applied to operators acting in complex Banach and Hilbert spaces.

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The content of this treatise is divided into three parts. The first one describes some fundamental tools and concepts in Functional Analysis and will be presented without proofs, but often with references to them, which can be found in any standard textbooks on Functional Analysis (see bibliography). The second part contains the abstract theory of symmetric and self-adjoint linear operators in a separable Hilbert space with the *spectral theorem* as its main assertion. Further topics, like perturbation theory, extention theory and the functional calculus for self-adjoint operators are included. Finally, in part III we apply the results to *realizations* of formally self-adjoint partial differential expressions given in divergence form

$$\mathcal{A}(x,u) := \sum_{|\alpha|\leq m, |\beta|\leq m} (-1)^{|\alpha|}\, \partial^{\alpha}(\, a_{\alpha\beta}(x)\, \partial^{\beta} u(x)) \quad (x \in G),$$

where formally self-adjointness of $\mathcal{A}$ means $a_{\alpha\beta}(x) = a_{\beta\alpha}(\mathrm{x})$ for all $x \in G$ and all functions $a_{\alpha\beta}$. Hence the realizations of such expressions are nothing but self-adjoint linear operators in one of the spaces $L^2(G)$ for $G = \mathbb{R}^d$ or $G \subsetneq \mathbb{R}^d$ and some $d \in \mathbb{N}$.

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# Part I
# Functional Analysis

Let us start our review of Functional Analysis with a short exposition of normed linear spaces. Since we do not need very much knowledge on this general class of spaces, we want to restrict ourselves to the definition of a normed linear space, the standard topology on such a space and the notion of *completeness*.

Throughout this text let $\mathbb{K}$ be one of the fields $\mathbb{R}$ and $\mathbb{C}$. By a *normed space* $X$ we mean a $\mathbb{K}$-linear space that differs from other abstract linear spaces by the presence of a particular mapping $\|\cdot\| : X \to \mathbb{R}$, denoted the *norm* on $X$, which fulfills for all $x, y \in X$ and $\lambda \in \mathbb{K}$

(N-1) $\|x\| \geq 0$,
(N-2) $\|\lambda x\| = |\lambda| \cdot \|x\|$,
(N-3) $\|x + y\| \leq \|x\| + \|y\|$ *(triangle inequality)*,
(N-4) $\|x\| = 0$ if and only if $x = 0$.

Linear spaces can of course be furnished with different norms. Arbitrary norms $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ defined on a linear space $X$ are called *equivalent* to each another, if there are constants $c_1, c_2 > 0$ such that $c_1 \|x\|_\alpha \leq \|x\|_\beta \leq c_2 \|x\|_\alpha$ for all $x \in X$.

Each normed linear space $X$ may be equipped with a very natural topology $\mathcal{T}$, called the *norm topology* on $X$. The definition of $\mathcal{T}$ is done either by the assertion that $\mathcal{T} := \mathcal{T}_{\|\cdot\|}$ is the smallest topology on $X$ such that the norm mapping remains continuous, or by the metric topology $\mathcal{T}_d$, when $X$ is considered as a metric space with metric function $d(x, y) := \|x - y\|$ for all $x, y \in X$. In fact, both definitions lead to the same collection of so-called "open" subsets in $X$, so we have $\mathcal{T} := \mathcal{T}_{\|\cdot\|} = \mathcal{T}_d$, which justifies to speak of *the* topology of a normed linear space. Hence a multiplicity of topological concepts, like the *closure* of a set, the *neighbourhood* of a point, the *convergence* of a series and many others are meaningful in the setting of normed spaces. Notice that equivalent norms on $X$ are creating one and the same topology. For example, if $X$ is a finite-dimensional linear space, then all norms are equivalent and consequently there is exactly one norm topology on $X$.

Recall that a sequence $(x_n)$ in $X$ is called *Cauchy*, if to each $\epsilon > 0$ there is $N_\epsilon \in \mathbb{N}$ such that $m, n > N_\epsilon$ implies $\|x_m - x_n\| < \epsilon$. Any

normed linear space $X$ is called *sequentially complete* (or shortly *complete*), if for any Cauchy sequence $(x_n)$ in $X$ there is $x \in X$ such that $\lim_{n\to\infty} \|x_n - x\| = 0$. Equivalently, $X$ is complete iff for every absolutely convergent series $\sum_{n=1}^{\infty} \|x_n\| < \infty$ there is $x \in X$ such that $\|x - x_n\| \to 0$ for $n \to \infty$. A complete normed space is called a *Banach space*.

It should also be noticed that every uncomplete normed space $X$ may continuously be embedded into a Banach space $\overline{X}$, denoted the *completion* of $X$. The procedure to construct such an embedding is an abstract version of the embedding of the rational numbers into the real ones.

## §1. Continuous Linear Operators

In the following let $X$ and $Y$ be $\mathbb{K}$-linear spaces. By a *linear transformation* or *linear operator* (shortly denoted an *operator*) is meant any mapping $A : X \to Y$ fulfilling

$$A(\lambda x + \mu y) = \lambda Ax + \mu Ay \qquad (x, y \in X,\ \lambda, \mu \in \mathbb{K}),$$

called the *linearity property* of $A$. The spaces $X$ and $Y$ are said to be the *domain space* and the *range space* of $A$, respectively. In the special case, where $Y = X$, it is convenient to say that $A$ is an operator *in* $X$.

We next review the concepts of continuity and boundedness for a linear operator $A : X \to Y$. First, the operator $A$ is called *continuous*, if the preimage $A^{-1}(\mathcal{O})$ of each open set $\mathcal{O} \subset Y$ is also open in $X$. Let $\mathscr{L}(X, Y)$ be the linear space of all continuous linear operators $A : X \to Y$, which forms a subspace of $L(X, Y)$, the space of *all* linear but not necessarily continuous mappings. Second, the operator $A$ is said to be *bounded* if and only if it is *norm-bounded*, that is, there is $c \geq 0$ such that $\|Ax\| \leq c \cdot \|x\|$ for each $x \in X$. In this case we define the *operator norm* of $A$ according to

$$\|A\| := \sup_{\|x\|=1} \|Ax\|. \tag{1.1}$$

The following result on continuous linear operators is well-known:

THEOREM 1.1. Let $A : X \to Y$ be a linear operator between the normed linear spaces $X$ and $Y$. Then the subsequent statements are equivalent to each other:

(a) $A$ is continuous;

(b) $A$ is sequentially continuous in every $x \in X$: if $(x_n)$ is a sequence in $X$ with $\lim_{n\to\infty} x_n = x \in X$, then $\lim_{n\to\infty} Ax_n = Ax$;
(c) $A$ is sequentially continuous at the origin $0 \in X$;
(d) $A$ is uniformly continuous in $X$;
(e) $A$ is bounded.

The space $\mathscr{L}(X, Y)$ of bounded linear operators from a normed linear space $X$ to a Banach space $Y$ and endowed with the norm given by (1.1) is complete, since $Y$ is supposed to be complete, hence $\mathscr{L}(X, Y)$ forms a Banach space in its own right.

In addition, in case of $Y = X$ the space $\mathscr{L}(X) := \mathscr{L}(X, X)$ constitutes a *normed algebra* with composition $(A_1, A_2) \mapsto A_2 \circ A_1$ as its multiplicative operation and owns the property $\|A_1 \circ A_2\| \leq \|A_1\| \cdot \|A_2\|$. It is called the *operator algebra* associated with the normed linear space $X$.

Finally, we turn to the notion of *embedding*, which is a special kind of a continuous linear operator and leads to the concept of *norm isomorphism.*

A linear mapping $\iota : X \to Y$ is called a (continuous) *embedding* from a Banach space $X$ into a Banach space $Y$, if it is injective (and continuous). If $\iota$ is continuous and even surjective, it is said to be a *linear homeomorphism* between $X$ and $Y$ (in the sense of topological linear spaces). Then BANACH*'s isomorphism theorem* claims that the inverse map $\iota^{-1}$ is also continuous, so it forms a norm isomorphism, too. This statement is a direct consequence of the *open mapping theorem* (see 3.1). Furthermore, if the mapping $\iota$ is *norm-preserving*, that is,

$$\|\iota x\|_Y = \|x\|_X \qquad (x \in X),$$

we denote this mapping a *Banach space isomorphism* between $X$ and $Y$, which in this case are called *norm-isomorphic* to each other (notation: $X \cong Y$).

## §2. Hilbert Spaces

A *Hilbert space* is a special kind of a Banach space, where the norm is derived from an *inner product.* It is a matter of fact that certain investigations done in the setting of Hilbert spaces are more transparent than the comparable ones in arbitrary Banach spaces.

Given a real or complex linear space $H$, the *inner product* in $H$ is a mapping $\langle \cdot, \cdot \rangle : H \times H \to \mathbb{K}$, which is

(I-1) linear in the first argument,

(I-2) either linear or semilinear in the second argument (conditional upon, whether $\mathbb{K}$ is the real or the complex number field),

together with the additional properties

(I-3) $\langle y, x \rangle = \langle x, y \rangle$ ($\mathbb{K} = \mathbb{R}$) and $\langle y, x \rangle = \overline{\langle x, y \rangle}$ ($\mathbb{K} = \mathbb{C}$), respectively,

(I-4) $\langle x, x \rangle \geq 0$ for all $x \neq 0$,

(I-5) $\langle x, x \rangle = 0$ implies $x = 0$.

Property (I-3) is denoted the *Hermitean symmetry* of the inner product, while (I-4) and (I-5) together mean positive definiteness. The presence of the inner product gives rise to define a norm in $H$ according to

$$\|x\| := \langle x, x \rangle^{1/2} \qquad (x \in H), \tag{2.1}$$

so each linear space endowed with an inner product is normed in a natural way. Inner product and norm are related to each other by the *Cauchy-Schwarz inequality*

$$|\langle x, y \rangle| \leq \|x\| \cdot \|y\| \qquad (x, y \in H), \tag{2.2}$$

thus the inner product is continuous with respect to the product topology on $H \times H$. An inner product space $H$ is called a *Hilbert space*, if it is complete with respect to the norm defined by (2.1).

The pleasant geometric properties of inner product spaces may be attributed to the *parallelogram equation*

$$2\|x\|^2 + 2\|y\|^2 = \|x + y\|^2 + \|x - y\|^2 \qquad \text{for all } x, y \in H. \tag{2.3}$$

In general, the norm $\|\cdot\|$ on a linear space $X$ can be induced by an inner product if and only if (2.3) holds. In this case, and if $X$ is a complex linear space, the inner product $\langle \cdot, \cdot \rangle_{\|\cdot\|}$ can be recovered from the norm by the *polarization formula*

$$\langle x, y \rangle_{\|\cdot\|} = \frac{1}{4} \left( \|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2 \right).$$

If $X$ is a real linear space, however, the polarization formula simplifies to

$$\langle x, y \rangle_{\|\cdot\|} = \frac{1}{4} \left( \|x + y\|^2 - \|x - y\|^2 \right).$$

In the following we list a couple of pleasant properties of Hilbert spaces which are not shared with more general Banach spaces. Let $H$ be a real or complex Hilbert space.

1. *Orthogonality.* Any elements $x, y$ of a Hilbert space $H$ are called *orthogonal* to each other, if $\langle x, y\rangle = 0$ (notation: $x \perp y$). A subset $\mathfrak{O} \subset H$ is said to be *orthogonal* or an *orthogonal system*, if all elements in $\mathfrak{O}$ are orthogonal to each other. Finally, given an arbitrary set $\mathcal{A} \subset H$, the *orthogonal space* $\mathcal{A}^\perp$ contains the elements $y \in H$ being orthogonal to every $x \in \mathcal{A}$. This set always forms a linear subspace of $H$.

2. *The Projection Theorem.* Let $\mathcal{A} \subset H$ be a *closed* linear subspace. Then for each element $x \in H$ there is exactly one $x_0 \in \mathcal{A}$ such that
$$\|x - x_0\| \;=\; \inf\{\, \|x - y\| : y \in \mathcal{A}\,\}.$$
The element $x_0$ is called the *projection* of $x$ onto $\mathcal{A}$. It minimizes the distance between $\mathcal{A}$ and $x$. The mapping $\pi_{\mathcal{A}} : x \in H \mapsto x_0 \in \mathcal{A}$ is also called the *projection* of $H$ onto $\mathcal{A}$. In general, by a projection is meant a continuous linear operator $\pi : H \to H$ satisfying the conditions $\pi \circ \pi = \pi$ and $\langle \pi x, y\rangle = \langle x, \pi y\rangle$ for all $x, y \in H$.

3. *The Decomposition Theorem.* Using the notation of orthogonal space $\mathcal{A}^\perp$, the projection of $x$ onto $\mathcal{A}$ is characterized as follows: $x_0 = \pi_{\mathcal{A}} x$ iff $x - x_0 \in \mathcal{A}^\perp$. Hence each $x \in H$ can be uniquely represented by the formula $x = x_0 + x^\perp$, where $x_0 \in \mathcal{A}$ and $x^\perp \in \mathcal{A}^\perp$. This means that the space $H$ is isometrically isomorphic to the direct sum $\mathcal{A} \oplus \mathcal{A}^\perp$ of the closed subspaces $\mathcal{A}$ and $\mathcal{A}^\perp$. This fact is known as *decomposition theorem* for Hilbert spaces. In other words, this theorem says that each linear subspace $\mathcal{A}$ of $H$ has a topological complement[1], which is of course the space $\mathcal{A}^\perp$.

4. *Orthonormal Bases.* An orthogonal system $\mathfrak{O} \subset H$ is called an *orthonormal system*, if in addtion $\langle x, x\rangle = 1$ for all $x \in \mathfrak{O}$. If $\mathfrak{O}$ contains only *finitely* many elements, then the span $[\mathfrak{O}]$ forms a *closed* subspace in $H$, and the projection $\pi_{[\mathfrak{O}]}$ of any element $x$ onto $[\mathfrak{O}]$ can be expressed by the formula

[1] A linear subspace $\mathcal{A} \subset X$ of a normed space $X$ is said to be topologically complementable, if there is a further closed subspace $\mathcal{B}$ such that $\mathcal{A} \cap \mathcal{B} = \{0\}$ and $[\mathcal{A} \cup \mathcal{B}] = X$, hence $\mathcal{B}$ is the algebraic complement of $\mathcal{A}$.

$$\pi_{[\mathfrak{O}]}\, x \;=\; \sum_{i=1}^{n} \langle x, e_i\rangle\, e_i.$$

However, if $\mathfrak{O}$ consists of arbitrarily many elements, the inequality $\langle x, e_\alpha\rangle \neq 0$ can only be valid for mostly countable many elements $e_\alpha \in \mathfrak{O}$, which enables us to summarize the coefficients $\langle x, e_\alpha\rangle$ about all elements of $\mathfrak{O}$.

An orthonormal system $\mathfrak{O}$ is called an *orthonormal base* in $H$, if its span $[\mathfrak{O}]$ lies dense in $H$, that is, $\overline{[\mathfrak{O}]} = H$. In this case it is possible to represent each $x \in H$ as a formal sum

$$x \;=\; \sum_{e_\alpha \in \mathfrak{O}} \langle x, e_\alpha\rangle\, e_\alpha. \tag{2.4}$$

The inner products $\langle x, e_\alpha\rangle$ are said to be the *Fourier coefficients* of $x$ with respect to $\mathfrak{O}$. Notice that for each $x \in H$ the right hand side of (2.4) consists of mostly countable many non-zero summands and the series is invariant by realignments of the elements $e_\alpha \in \mathfrak{O}$.

There is the need to determine, whether a given orthonormal system $\mathfrak{O}$ forms an orthonormal base. It can be shown that the following statements are equivalent to each other:

(i) $\mathfrak{O}$ is an orthonormal base in $H$.
(ii) $\mathfrak{O}$ is *complete*, that is, $\langle x, e\rangle = 0$ for all $e \in \mathfrak{O}$ implies $x = 0$.
(iii) $\sum_{e_\alpha \in \mathfrak{O}} |\langle x, e_\alpha\rangle|^2 = \|x\|^2$ for all $x \in H$ *(Parseval's equation)*.

It is also natural to ask for the existence of an orthonormal base in arbitrary Hilbert spaces. First of all, the collection of all orthonormal systems in a Hilbert space $H$ forms a partial ordered set, and under the assumption that ZORN's lemma [2] is valid, there is a maximal orthonormal system $\mathfrak{O}$ in $H$ which cannot be increased. Indeed, such a set forms an orthonormal base, which ensures that every Hilbert space $H \neq \{0\}$ has an orthonormal base.

Furthermore, if $\mathfrak{O}_1$ and $\mathfrak{O}_2$ are orthonormal bases in $H$, then there is a bijective mapping between them, thus they own identical cardinal numbers. This legitimates us to introduce the notation of *Hilbert space dimension* $|H|$ being the cardinal number of any

[2] The *axion of choice* claims that if $\mathcal{F}$ is a disjoint family $F_\alpha$ $(\alpha \in I)$ of non-empty sets, then there is a set $A$ that shares exactly one element with each other set $F_\alpha \in \mathcal{F}$.

orthonormal base in $H$.

A Hilbert space $H$ is called *separable* if there is a mostly countable subset $D \subset H$ lying dense in $H$. It can be shown that $H$ is separable if and only if $|H| \leq |\mathbb{N}|$. In fact, each orthonormal base in $H$ forms a mostly countable set in this case, and $H$ proves to be norm-isomorphic to the space $\ell^2$ of square summarizable sequences $f : \mathbb{N} \to \mathbb{C}$.

5. RIESZ*'s Representation Theorem.* Recall that the *conjugate space* $X^*$ of a normed $\mathbb{K}$-linear space $X$ consists of all *continuous* (anti)linear functionals $f : X \to \mathbb{K}$, depending whether $\mathbb{K} = \mathbb{C}$ or $\mathbb{K} = \mathbb{R}$. In the complex case we have $f(\lambda x) = \overline{\lambda} f(x)$ for $\lambda \in \mathbb{C}$, whereas in the real case $f(\lambda x) = \lambda f(x)$ holds for $\lambda \in \mathbb{R}$. It is well-known that $X^*$ is a Banach space again with operator norm

$$\|f\| := \sup_{x \in X,\, \|x\|=1} |f(x)| \qquad (f \in X^*). \tag{2.5}$$

Due to the properties of an inner product, each element $x \in H$ gives rise to a continuous (anti)linear functional $f_x \in H^*$ according to

$$f_x(y) := \langle x, y \rangle \quad (y \in H).$$

The mapping $\Phi : H \to H^*$ defined by $x \mapsto f_x$ is called the RIESZ *map* of $H$. It is linear on $H$, and since the inner product is positive definite, $\Phi$ turns out to be injective. RIESZ*'s representation theorem* claims that $\Phi$ is actually surjective, so there is a canonical correspondence between the elements of a Hilbert space $H$ and those of its conjugate space $H^*$:

THEOREM 2.1. If $H$ is a Hilbert space, then $\Phi : H \to H^*$ is bijective and norm-preserving, that is, $\|\Phi(x)\| = \|x\|$ for all $x \in H$.

This theorem implies that to each $f \in H^*$ there is exactly one element $x_f \in H$, which fulfills the equation $\langle x_f, y \rangle = f(y)$ for every $y \in H$.

The RIESZ map also gives rise to introduce an inner product on the conjugate space $H^*$ by means of

$$\langle f, g \rangle := \langle \Phi^{-1}(f), \Phi^{-1}(g) \rangle \qquad (f, g \in H^*),$$

hence it is not only norm-preserving, but also *isometric.*

## §3. Closed and Closable Linear Operators

It is the purpose of this section to extend the notation of continuous linear transformation to a more general concept, which then will be called a *closed* linear operator. This extended definition is motivated by the fact that realizations of linear partial differential expressions defined on certain function spaces may indeed not be continuous, but turn out to be closed or at least *closable.*

Let $X$ and $Y$ be real rsp. complex Banach spaces. By a *linear operator* from the domain space $X$ to the range space $Y$ is meant a mapping $A : \mathfrak{D}(A) \subset X \to Y$, where the *domain of definition* $\mathfrak{D}(A)$ of the operator $A$ forms a linear subspace of $X$ and $A$ is *linear* on $\mathfrak{D}(A)$, i. e.

$$A(\lambda x + \mu y) = \lambda Ax + \mu Ay \quad \text{for all } x, y \in \mathfrak{D}(A),\ \lambda, \mu \in \mathbb{K}.$$

In case of $Y = X$, we denote $A$ to be an operator *acting in* $X$ or shortly to be an operator *in* $X$.

Of course, this definition includes that of section 1, where the domain of definition $\mathfrak{D}(A)$ is assumed to be the entire space $X$. It our extended definition it is often required that the domain of definition $\mathfrak{D}(A)$ lies dense in $X$, that is, each $x \in X$ can be approximated by a sequence $(x_n)$ of elements in $\mathfrak{D}(A)$. However, if the domain of definition $\mathfrak{D}(A)$ is *not* dense in $X$, it is nearby to consider the closure $X_0 := \overline{\mathfrak{D}(A)} \subset X$ and to pay attention to the slightly modified but densely defined operator $A_0 : \mathfrak{D}(A) \subset X_0 \to Y$.

Rather the domains of definition for operators $A_1, A_2$ with domain space $X$ do not necessarily form the whole of $X$, the *sum* $A_1 + A_2$ and the *composition* $A_1 \circ A_2$ are well-defined with respective domains of definition $\mathfrak{D}(A_1 + A_2)$ and $\mathfrak{D}(A_1 \circ A_2)$ given by

$$\begin{cases} \mathfrak{D}(A_1 + A_2) := \mathfrak{D}(A_1) \cap \mathfrak{D}(A_2), \\ \mathfrak{D}(A_1 \circ A_2) := \{ x \in \mathfrak{D}(A_2) : A_2 x \in \mathfrak{D}(A_1) \}. \end{cases}$$

Especially, if $Y = X$ and $A_2$ is a *dilatation* in $X$, i. e. $A_2 x = \lambda x$ for $x \in X$ and some $\lambda \in \mathbb{K}$, the operator $A_1 + A_2 = A_1 + \lambda 1$ has domain of definition $\mathfrak{D}(A_1 + \lambda) = \mathfrak{D}(A_1)$.

Since the domain of definition $\mathfrak{D}(A)$ of a linear operator $A$ may be a proper linear subspace of $X$, the concept of *extension* is meaningful. Let $A_1$ and $A_2$ be linear operators in $X$ with domains $\mathfrak{D}(A_1)$ and $\mathfrak{D}(A_2)$, respectively. We introduce the partial ordered relation

$$A_1 \prec A_2 \;:\Longleftrightarrow\; \mathfrak{D}(A_1) \subset \mathfrak{D}(A_2) \text{ and } A_1 x = A_2 x \text{ for all } x \in \mathfrak{D}(A_1)$$

and call $A_2$ an *extension* of $A_1$ iff $A_1 \prec A_2$. In addition, the operators $A_1$ and $A_2$ are said to be *equal* (denoted $A_1 = A_2$) iff $A_1 \prec A_2$ and $A_2 \prec A_1$, or equivalently

$$\mathfrak{D}(A_1) = \mathfrak{D}(A_2) \text{ and } A_1 x = A_2 x \quad \text{for all } x \in \mathfrak{D}(A_1).$$

Reversely, let $Y_0$ be a linear subspace of the target space $Y$. We then may restrict the operator $A$ to a linear operator $A_0$ defined by $\mathfrak{D}(A_0) = \{x \in \mathfrak{D}(A) : Ax \in Y_0\}$ and $A_0 x := Ax$ for $x \in \mathfrak{D}(A_0)$. Clearly, $A_0$ has target space $Y_0$, and we say that $A_0$ is the *part of* $A$ *in* $Y_0$.

Finally, with every linear operator $A : \mathfrak{D}(A) \subset X \to Y$ acting between the linear spaces $X$ and $Y$ there are associated three fundamental linear subspaces:

1. the *kernel* $\mathfrak{N}(A) = \{x \in \mathfrak{D}(A) : Ax = 0\} \subseteq X$;
2. the *range* $\mathfrak{R}(A) = \{\, Ax : x \in \mathfrak{D}(A) \,\} \subseteq Y$;
3. the *cokernel* $\mathfrak{C}(A)$, being the algebraic complement of the range $\mathfrak{R}(A)$ within $Y$, that is, $Y = \mathfrak{R}(A) \oplus \mathfrak{C}(A)$.

Recall that a linear operator $A$ is called *continuous* if $x_n \to 0$ implies $Ax_n \to 0$ for any sequence $(x_n)$ settled in the domain of definition $\mathfrak{D}(A)$. If this set is dense in $X$ and $A$ is continuous, then $A$ can be extended to its closure $\overline{\mathfrak{D}(A)}$, which is continuous as well. This is the claim of the so-called *B.L.T. theorem*, which is an abbreviation for "Bounded Linear Transformation theorem" (see [10], page 6).

Unfortunately, many linear operators in applications and especially the differential operators studied in the theory of partial differential equations are often uncontinuous. But the concept of *closed operator* stands for a certain compensation for the missing property of continuity. If $X$ and $Y$ are Banach spaces, then a linear operator $A : \mathfrak{D}(A) \subset X \to Y$ is called *closed*, if the linear subspace

$$\Gamma_A := \{\, (x, Ax) : x \in \mathfrak{D}(A) \,\} \subset X \times Y,$$

denoted the *graph* of $A$, is closed with respect to the product topology $\mathcal{T}_{X \times Y}$ in $X \times Y$, which especially implies that the kernel of $A$ is a closed set in $X$.

The closeness of an operator can be characterized as follows: a linear operator $A : \mathfrak{D}(A) \subset X \to Y$ is closed if and only if

1. the convergence of a sequence $(x_n)$ to an element $x \in X$ and the convergence of $(Ax_n)$ to an element $y \in Y$ implies $x \in \mathfrak{D}(A)$ and $Ax = y$, briefly written

$$\left.\begin{array}{r} x_n \to x \\ Ax_n \to y \end{array}\right\} \Longrightarrow\ x \in \mathfrak{D}(A) \ \text{ and } \ Ax = y.$$

2. the linear subspace $\mathfrak{D}(A)$ endowed with the *graph norm*

$$\|x\|_A := \|x\|_X + \|Ax\|_Y, \quad x \in \mathfrak{D}(A) \tag{3.1}$$

is complete, that is, $\mathfrak{D}(A)$ furnished with norm $\|\cdot\|_A$ is Banach.

It may happen that a linear operator $A$ fails to be closed. However, $A$ is said to be *closable*, if the closure of $\Gamma_A$ is the graph of an extension $\overline{A}$ of $A$, called the *closure* of $A$. Any dense subspace $D \subset X$ is called a *core* of a closed operator $A$, if $D \subset \mathfrak{D}(A)$ and the closure of $A_{|D}$ resembles $A$.

A similar criterion as for closeness is available for closability: the linear operator $A : \mathfrak{D}(A) \subset X \to Y$ is closable if and only if each sequence $(x_n)$ in $\mathfrak{D}(A)$ with $x_n \to 0$ and $Ax_n \to y \in X$ implies $y = 0$, shortly written

$$\left.\begin{array}{r} x_n \to 0 \\ Ax_n \to y \end{array}\right\} \Longrightarrow\ y = 0.$$

There are of course relationships between the presented concepts. In fact, if $\mathfrak{D}(A)$ is closed and $A$ is continuous, then it is closed, and if $A$ is closed, then it is closable, thus we obtain the following chain of implications:

$$A \text{ bounded} \iff A \text{ continuous} \implies A \text{ closed} \implies A \text{ closable}.$$

We are going on with some further questions concerning the correspondence between continuity and closeness of linear operators, which are meaningful in what follows.

Let us start with the *open mapping theorem*, that describes a remarkable property of continuous linear operators acting between Banach spaces. In fact, Baire*'s category theorem* coming from point-set topology leads to this fundamental theorem in Functional Analysis:

Theorem 3.1. If $X$ and $Y$ are Banach spaces and the linear operator $A : X \to Y$ is continuous and surjective, then $A$ is open.

The *closed graph theorem*, which is equivalent to the open mapping theorem, establishes the relationship between continuous and closed operators under our specific assumption on the domain of definition:

THEOREM 3.2. If $X, Y$ are Banach spaces and the domain of definition $\mathfrak{D}(A)$ of a closed linear operator $A : \mathfrak{D}(A) \subset X \to Y$ is closed, then $A$ is continuous.

This theorem especially implies that if $A$ is everywhere defined (so $\mathfrak{D}(A) = X$), then closeness and continuity of $A$ are equivalent to each other.

A collection $\mathfrak{F}$ of linear operators between normed spaces $X$, $Y$ is called *pointwise bounded*, if for each $x \in X$ there is $c_x > 0$ such that $\|Ax\| \leq c_x \|x\|$ for all $A \in \mathfrak{F}$. The following statement, called the *uniform boundedness principle*, can be proved either by theorem 3.1 or by theorem 3.2.

THEOREM 3.3. Let $X$ be a Banach space and $Y$ be a normed space. Then a family $\mathfrak{F}$ of pointwise bounded linear operators $A_\alpha : X \to Y$ is actually *uniformly bounded*, that is, there is a constant $c > 0$ such that $\|A\|_\alpha \leq c$ for each $A_\alpha \in \mathfrak{F}$.

We now turn to the problem of invertibility of operators. A linear operator $A$ is said to be *invertible*, if it is bijective and the inverse $A^{-1}$ is continuous.

If $A$ is not only continuous and surjective, but also injective, then theorem 3.1 is known as the *bounded inverse theorem* or as BANACH'S *isomorphism theorem*. It quotes that under the just mentioned conditions the inverse (linear) operator $A^{-1} : Y \to X$ is even bounded, so $A$ and $A^{-1}$ both are homeomorphisms between $X$ and $Y$.

Similarly, if $A$ is closed and bijective, then the operator $A^{-1} : Y \to \mathfrak{D}(A) \subset X$ is closed, too. This assertion can be proved much easier than the invertibility of a continuous linear operator. In addition, the closed graph theorem implies that the inverse $A^{-1}$ of a closed linear operator $A$ is even continuous, which is of outstanding importance with respect to well-posedness of the linear operator equation $Ax = y$. In fact, $A^{-1}$ being continuous is nothing but the statement that the solution $x$ of this equation depends continuously on the right hand side $y$.

## §4. Adjoint Operators

Given a linear operator $A$ in a Hilbert space $H$ with densely defined domain of definition $\mathfrak{D}(A)$, we want to introduce the concept of the *adjoint operator* $A^*$. This notation can also be defined for densely defined linear operators between ordinary and possibly different Banach spaces, but it is sufficient for our purposes to restrict the considerations to the Hilbert space setting.

By the domain of definition $\mathfrak{D}(A^*)$ of the operator $A^*$ is meant the collection of elements $y \in H$, for which there is $y^* \in H$ such that

$$\langle Ax, y\rangle = \langle x, y^*\rangle \quad \text{for all } x \in \mathfrak{D}(A).$$

The set $\mathfrak{D}(A^*)$ constitutes a linear subspace of $H$. Since $A$ is assumed to be densely defined, the element $y^*$ is uniquely determined by the element $y$ (this is a consequence of Riesz's representation theorem), thus one may define the operator $A^*$ according to $A^*y := y^*$ for $y \in \mathfrak{D}(A^*)$. Then, given the pair of operators $(A, A^*)$,

$$\langle Ax, y\rangle = \langle x, A^*y\rangle$$

holds for all $x \in \mathfrak{D}(A)$ and $y \in \mathfrak{D}(A^*)$.

It is easy to verify that the operator $A^*$ is also linear. Moreover, if $B$ is an extension of $A$, then $A^*$ is an extension of $B^*$, that is, $A \prec B$ implies $B^* \prec A^*$. Indeed, let $y \in \mathfrak{D}(B^*)$, which means that there is $y^* \in H$ such that

$$\langle Bx, y\rangle = \langle Ax, y\rangle = \langle x, y^*\rangle \quad \text{for all } x \in \mathfrak{D}(B)\,,$$

then it is easy to see that $y \in \mathfrak{D}(A^*)$ and $A^*y = B^*y$.

From another point of view, the notion of adjoint operator is of interest when studying the continuity of the sesquilinear form $(x, y) \mapsto \langle Ax, y\rangle$. In fact, the domain of definition $\mathfrak{D}(A^*)$of the adjoint operator consists exactly of those $y \in H$, for which the functional $f_y(x) := \langle Ax, y\rangle$ is continuous. Since $\mathfrak{D}(A)$ is supposed to be dense in $H$, the mapping $f_y$ can be extended to the whole of $H$, and by Riesz's representation theorem 2.1 there is $y^* \in H$ such that $f_y(x) = \langle x, y^*\rangle$ for all $x \in H$.

We especially have $(A - \lambda 1)^* = A^* - \overline{\lambda}1$ for $\lambda \in \mathbb{C}$, where $1_H$ is the identity operator in $H$. From now on we use the abbreviation $A - \lambda$ instead of $A - \lambda 1_H$. Then the domain of definition of the operator $(A - \lambda)^*$ is given by

$$\mathfrak{D}((A - \lambda)^*) = \mathfrak{D}(A^* - \overline{\lambda}) = \mathfrak{D}(A^*).$$

Moreover, for $\lambda \in \mathbb{C} \setminus \{0\}$ the equality $(\lambda A)^* = \overline{\lambda} A^*$ is valid, but this surely is not true in case of $\lambda = 0$.

There are fundamental relationships between the kernels and ranges of the linear operators $A$ and $A^*$, which read as

$$\overline{\mathfrak{R}(A^*)} = \mathfrak{N}(A)^\perp \quad \text{and} \quad \overline{\mathfrak{R}(A)} = \mathfrak{N}(A^*)^\perp. \tag{4.1}$$

They are very useful to investigate the possible property of surjectivity of the operator $A$. Indeed, if one can prove that $A^*$ is injective, then the range of $A$ is dense in $H$, and if in addition $A$ has closed range, for example if $A$ is bounded below, then we can conclude that $A$ is even surjective.

Each adjoint operator is closed, but it is not necessarily densely defined. But if $A$ itself is closed, then $A^*$ is densely defined and $A^{**} := (A^*)^* = A$ holds, so in this case taking the adjoint is a self-dual operation. Furthermore, if $A$ is closable, but not closed, then we have $A^{**} = \overline{A}$, that is, the closure of $A$ can be obtained by taking the adjoint twice.

## §5. Semi-Fredholm Operators

Since the kernel $\mathfrak{N}(A)$ and the range $\mathfrak{R}(A)$ of a linear operator $A : \mathfrak{D}(A) \subset X \to Y$ are linear subspaces of $X$ and $Y$, respectively, it is possible to define the value $\alpha(A)$ and $\beta(A)$ to be the Hamel dimension of $\mathfrak{N}(A)$ and $\mathfrak{C}(A)$, respectively. Both together measure the deviation of $A$ from being bijective. Then the so-called *index*

$$\chi(A) := \begin{cases} +\infty, & \alpha(A) = \infty \\ \alpha(A) - \beta(A), & \alpha(A) < \infty,\ \beta(A) < \infty \\ -\infty, & \beta(A) = \infty \end{cases}$$

of $A$ is well-defined and takes values in the set $\{-\infty\} \cup \mathbb{N} \cup \{+\infty\}$. In the special case, where $\alpha(A) < \infty$, $\beta(A) < \infty$ and consequently $\chi(A) \in \mathbb{Z}$, the operator $A$ is said to have *finite index*.

We denote a densely defined linear operator $A$ with range space $Y$ to be *normally solvable* if its range $\mathfrak{R}(A)$ is closed in $Y$. A closed and normally solvable linear operator $A$ is said to be *semi-Fredholm*, if the kernel $\mathfrak{N}(A)$ or the cokernel $\mathfrak{C}(A)$ is *finite-dimensional*, or equivalently, if $\alpha(A) < \infty$ or $\beta(A) < \infty$.

We refine the concept of semi-Fredholm operator as follows: a normally solvable linear operator $A : \mathfrak{D}(A) \subset X \to Y$ is called

1. *upper semi-Fredholm*, if $\alpha(A) < \infty$ and $\chi(A) < \infty$, respectively,
2. *lower semi-Fredholm*, if $\beta(A) < \infty$ and $\chi(A) > -\infty$, respectively.
3. *Fredholm*, if it is lower and upper semi-Fredholm as well, thus $\chi(A) \in \mathbb{Z}$.

We continue with a further class of linear operators, which is of importance with respect to solvability of equations of the form $Ax = y$, where $x, y$ are elements of the Banach spaces $X$ and $Y$.

A linear operator $A : \mathfrak{D}(A) \subset X \to Y$ is said *to have closed range*, if $\mathfrak{R}(A)$ is closed in $Y$. Moreover, $A$ said to be *bounded below* if there is $\theta > 0$ such that

$$\|Ax\| \geq \theta \|x\| \quad \text{for all } x \in \mathfrak{D}(A). \tag{5.1}$$

The following assertion is one of the fundamental theorems in operator theory:

Theorem 5.1. A closed linear operator $A : \mathfrak{D}(A) \subset X \to Y$ is bounded below if and only if $A$ is injective and has closed range.

For closed operators $A$ being bounded below, the inverse operator $A^{-1}$ is then defined on the Banach space $\mathfrak{R}(A)$ and proves to be continuous by the closed graph theorem, so the linear equation $Ax = y$ is well-posed for $y \in \mathfrak{R}(A)$.

The *closed range theorem* (in its Hilbert space version) due to S. Banach claims that the property of a closed linear operator $A$ having closed range is strongly related to that property of the adjoint operator $A^*$. Moreover, it also quotes that, concerning the orthogonality relations (4.1), the closures of the ranges of $A$ and $A^*$ may be substituted by the ranges themselves, supposed one of the operators $A$ and $A^*$ has closed range.

Theorem 5.2. Let $A$ be a closed and densely defined linear operator in Hilbert space $H$. Then the following assertions are equivalent:

1. $A$ has closed range.
2. $A^*$ has closed range.
3. $\mathfrak{R}(A^*)$ is the annihilator of $\mathfrak{N}(A)$, i. e. $\mathfrak{R}(A^*) = \mathfrak{N}(A)^\perp$.
4. $\mathfrak{R}(A)$ is the pre-annihilator of $\mathfrak{N}(A^*)$, i. e. $\mathfrak{R}(A) = {}^\perp\mathfrak{N}(A^*)$.

We next quote a feature of semi-Fredholm operators with respect to the functions $\xi \mapsto \alpha(A - \xi)$ and $\xi \mapsto \beta(A - \xi)$, where $\xi \in \mathbb{C}$ is close to zero. It is known as *Gohberg*'s *punctured neighbourhood*

*theorem.*

THEOREM 5.3. Let the operator $A$ be upper semi-Fredholm. Then:
1. the mappings $\xi \mapsto \alpha(A-\xi) =: c_l$ and $\xi \mapsto \beta(A-\xi) =: c_u$ are constant for $\xi \neq 0$ and $\xi$ belonging to a sufficiently small punctured environment of $0 \in \mathbb{C}$,
2. $c_l \leq \alpha(A)$ and $c_u \leq \beta(A)$ holds,
3. the operator $A-\xi$ is upper semi-Fredholm with index that of $A$.

In the Hilbert space setting the concept of *numerical range* is useful to study the semi-Fredholm property of operators. We define the numerical range of a linear operator $A$ acting in a Hilbert space $H$ and with domain of definition $\mathfrak{D}(A)$ by

$$\Theta(A) := \{ \langle Ax, x\rangle : x \in \mathfrak{D}(A), \|x\| = 1 \}.$$

It is a known fact that $\Theta(A)$ is a convex set within the complex plane. Moreover, let $\Delta(A) := \mathbb{C} \setminus \overline{\Theta(A)}$ be the complement of the closure of $\Theta(A)$. If $\Delta(A) \neq \emptyset$, then $\Delta(A)$ consists of one or two connected components. To be precise, if this set has two components, then the numerical range $\Theta(A)$ must be a vertical or horizontal strip.

For $\xi \in \Delta(A)$ we of course have $\langle Ax, x\rangle \neq \xi$ for $x \in \mathfrak{D}(A)$ and $\|x\| = 1$. Then

$$0 < \delta(\xi, A) \leq |\langle Ax, x\rangle - \xi| = |\langle (A-\xi)x, x\rangle| \leq \|(A-\xi)x\|,$$

where $\delta(\xi, A)$ is the positive distance between the point $\xi$ and the closed set $\overline{\Theta(A)}$. Hence the operator $A-\xi$ is bounded below, which implies that $A-\xi$ is semi-Fredholm with $\alpha(A-\xi) = 0$. By this reason the set $\Delta(A)$ is sometimes called the *field of external regularity* of $A$. By theorem 5.3, the index of $A-\xi$ is constant in each connected component of $\Delta(A)$.

## §6. Resolvent Set and Spectrum

Throughout this section let $X$ be a *complex* Banach space and $1_X$ the identity operator in $X$. Notice that each real Banach space $X$ can be made into a complex Banach space $X_{\mathbb{C}}$, and the procedure for this is denoted the *complexification* of $X$. It can be done as follows: let $X_{\mathbb{C}} := X + iX = \{x + iy : x, y \in X\}$ and furnished with the addition and multiplication derived from those in $X$. The norm in $X_{\mathbb{C}}$ is defined according to $\|x + iy\| := \sup\{\|x\| \cos\theta + \|y\| \sin\theta : \theta \in [0, 2\pi)\}$, in which this space is complete.

*Spectral theory* is a branch in operator theory and deals with the obstacles for a linear operator $A - \lambda 1 : \mathfrak{D}(A) \subset X \to X$ to be bijective. Its purpose is to obtain assertions for the solvability of the equation

$$(A - \lambda 1)\, x \;=\; y \qquad (x \in \mathfrak{D}(A),\; y \in X)$$

conditional upon the complex parameter $\lambda \in \mathbb{C}$. Thus one may consider spectral theory as a far-reaching generalization of the eigenvalue problem for complex $(d,d)$-matrices $(d \in \mathbb{N})$ to linear operators acting between infinite-dimensional Banach spaces. At the same time it is a useful tool in the functional analytic study of linear evolution equations.

In the sequel we shall use consistently the abbreviation $A - \xi$ instead of $A - \xi 1$. Given a densely defined and closed linear operator $A : \mathfrak{D}(A) \subset X \to X$, each complex number $\xi \in \mathbb{C}$ is called a

1. *regular value* of $A$, if $A - \xi : \mathfrak{D}(A) \to X$ is bijective. The set $\rho(A)$ of regular values is denoted the *resolvent set* or *domain of regularity* of $A$;
2. *spectral value* of $A$, if $\xi \notin \rho(A)$. The set $\sigma(A)$ of spectral values of $A$ is called the *spectrum* of $A$, thus $\sigma(A) = \mathbb{C} \setminus \rho(A)$.

If $\xi \in \rho(A)$, then the inverse operator $R(A,\xi) := (A - \xi)^{-1} : X \to \mathfrak{D}(A)$ is everywhere defined in $X$. Due to the closeness of $A$, the operators $A - \xi$ and $R(A,\xi)$ are closed, too, and the closed graph theorem 3.2 implies that $R(A,\xi)$ is continuous, that is, $R(A,\xi) \in \mathscr{L}(X)$. The operator $R(A,\xi)$ is said to be the *resolvent* of $A$ with respect to $\xi \in \rho(A)$, and the operator-valued mapping

$$\begin{cases} R(A,\cdot) : \rho(A) \to \mathscr{L}(X)\,, \\ \xi \mapsto (A - \xi)^{-1} \end{cases} \tag{6.1}$$

is denoted the *resolvent function* of $A$.

The question, whether a given complex number $\xi$ belongs to the resolvent set or the spectrum of $A$, is answered by the following criterion: let $(\xi_n)$ be a sequence in $\rho(A)$ with $\lim_{n\to\infty} \xi_n = \xi$. Then $\xi \in \rho(A)$ if and only if

$$\sup_{n \in \mathbb{N}} \|R(A,\xi_n)\| < \infty.$$

On the other hand, if $\xi$ lies in the boundary of $\rho(A)$, then $\|R(A,\xi_n)\| \to \infty$ for each sequence $\xi_n \to \xi$.

Resolvents related to a given operator are not independent from each other. In particular, for all $\xi_1, \xi_2 \in \rho(A)$ the *first resolvent equation*

$$R(A,\xi_1) - R(A,\xi_2) = (\xi_2 - \xi_1)\, R(A,\xi_1) \circ R(A,\xi_2) \tag{6.2}$$

holds, which implies that the resolvents $R(A,\xi_1)$ and $R(A,\xi_2)$ commute with each other. Furthermore, if $\xi_0 \in \rho(A)$, then we have $\xi \in \rho(A)$ for all $\xi \in \mathbb{C}$ satisfying

$$|\xi - \xi_0| < \|R(A,\xi_0)\|^{-1},$$

which implies that $\rho(A)$ is open in $\mathbb{C}$. For values $\xi$ in this disk one may represent the resolvent as a power series with coefficients in the Banach algebra $\mathscr{L}(X)$:

$$R(A,\xi) = \sum_{n=0}^{\infty} R(A,\xi_0)^{n+1}\, (\xi_0 - \xi)^n.$$

In fact, the mapping $R(A,\cdot)$ as defined in (6.1) is locally holomorphic in $\rho(A)$.

We now turn to the properties of the spectrum of $A$. First of all, the set $\sigma(A)$ is closed, since $\rho(A)$ is open. If $A$ is closable but not closed, it may happen that the spectrum consists of the entire set $\mathbb{C}$. For a closed linear operator $A$, however, the resolvent set is non-empty, so $\sigma(A)$ cannot be the whole of $\mathbb{C}$. It is also possible that $\sigma(A)$ degenerates to the empty set. But in the case, where $A$ is everywhere defined and *bounded*, the spectrum of $A$ is non-empty.

We now consider the spectral values of a closed linear operator $A$ and study possible reasons, which for $\lambda \in \sigma(A)$ prevent the operator $A - \lambda$ from being bijective.

1. Recall from linear algebra that a complex number $\lambda$ is called an *eigenvalue* of a linear operator $A$, if there is an element $x \in X$ such that $x \neq 0$ and $Ax = \lambda x$. In this case the element $x$ is called an *eigenvector* of $A$ with respect to $\lambda$, and all these eigenvectors form a linear subspace of $X$, called the *eigenspace* of $A$ for $\lambda$. The *point spectrum* of $A$ is defined to be the set $\sigma_p(A)$ of all eigenvalues of $A$. Clearly, for $\lambda \in \sigma_p(A)$ the operator $A - \lambda$ is not injective, since its kernel is non-trivial.
   If $A - \lambda$ is injective, but not surjective, then the inverse operator may be defined on the range of $A - \lambda$. We distinguish between

two cases:

2a. if $\mathfrak{R}(A-\lambda)$ is dense in $X$, then $\lambda$ is said to be an element of the set $\sigma_c(A)$, called the *continuous spectrum* of $A$, otherwise

2b. the number $\lambda$ belongs to the *residual spectrum* $\sigma_r(A)$. There is an interesting identity between the spectrum of a closed and densely defined linear operator $A$ and that of its transposed $A^t$, namely $\sigma_r(A) = \sigma_p(A^t)$ (see [5], p. 161).

In summary, for each densely defined operator $A : \mathfrak{D}(A) \subset X \to X$ the set of complex numbers forms a disjoint decomposition of the resolvent set and the different kinds of spectra, namely

$$\mathbb{C} = \rho(A) \sqcup \sigma_p(A) \sqcup \sigma_c(A) \sqcup \sigma_r(A).$$

The *approximate point spectrum* $\sigma_{ap}(A)$ is defined to be the set of $\lambda \in \mathbb{C}$, for which $A-\lambda$ is not injective or $\mathfrak{R}(A-\lambda)$ is not closed in $X$. The elements of $\sigma_{ap}(A)$ are called *approximate eigenvalues* of $A$. In fact, any number $\lambda$ belongs to the set $\sigma_{ap}(A)$ if and only if there is a sequence $(x_n) \subset \mathfrak{D}(A)$ such that $\|x_n\| = 1$ for all $n$ and $\lim_{n\to\infty} \|Ax_n - \lambda x_n\| = 0$. For a closed linear operator $A$, the boundary $\partial\sigma(A)$ of the spectrum forms a subset of the approximate point spectrum of $A$.

The approximate point spectrum generalizes the point spectrum $\sigma_p(A)$, but it can be empty only in the case, where $\sigma(A) = \emptyset$ or $\sigma(A) = \mathbb{C}$.

It is also possible to breakup the spectrum into several parts by using the notion of semi-Fredholm operator. The *essential spectrum* $\sigma_{ess}(A)$ of a closed operator $A$ consists of those points $\lambda$ that are either accumulation points of $\sigma(A)$ or isolated eigenvalues withe infinite-dimensional eigenspaces. The essential spectrum $\sigma_{ess}(A)$ forms a closed subset of $\sigma(A)$.

Semi-Fredholm operators can also be used to characterize the essential spectrum $\sigma_{ess}(A)$ of a closed operator $A$: a number $\lambda \in \mathbb{C}$ belongs to $\sigma_{ess}(A)$ if and only if $A-\lambda$ is not semi-Fredholm. The main property of the essential spectrum is given by the fact that it does not change under compact perturbations of $A$.

On the other hand, the relative complement $\sigma_d(A) := \sigma(A) \setminus \sigma_{ess}(A)$ is called the *discrete spectrum* of $A$ and purely contains eigenvalues of finite multiplicity.

For fixed $\xi \in \mathbb{C}$, the spectrum of the resolvent $(A-\xi)^{-1}$ is related to the spectrum of $A$ itself by the identity

$$\sigma(R(A,\xi)) \setminus \{0\} = \{(\lambda - \xi)^{-1} : \lambda \in \sigma(A)\},$$

known as *spectral mapping theorems* for resolvents, see [5], pp. 161-162. Similar identities are also valid for sets $\sigma_p(A)$, $\sigma_{ap}(A)$ and $\sigma_r(A)$:

$$\sigma_p(R(A,\xi)) \setminus \{0\} = \{(\lambda - \xi)^{-1} : \lambda \in \sigma_p(A)\}.$$

$$\sigma_{ap}(R(A,\xi)) \setminus \{0\} = \{(\lambda - \xi)^{-1} : \lambda \in \sigma_{ap}(A)\}.$$

$$\sigma_r(R(A,\xi)) \setminus \{0\} = \{(\lambda - \xi)^{-1} : \lambda \in \sigma_r(A)\}.$$

If $A$ is bounded, we may define its *spectral radius* by $r(A) := \sup\{\lambda \in \sigma(A)\}$. It can be shown that $r(A) \leq \|A\|$, and since each resolvent $R(A,\xi)$ is bounded, we have

$$\|R(A,\xi)\| \geq r(R(A,\xi)) \geq \delta(\xi, \sigma(A))^{-1} \tag{6.3}$$

for any $\xi \in \rho(A)$, where $\delta(\xi, \sigma(A))$ is the minimal distance between $\xi$ and the spectrum of $A$. Note that inequality (6.3) is a consequence of the spectral mapping theorem.

We finally describe the basic sets of spectral theory for adjoint operators in Hilbert space. Let $A : \mathfrak{D}(A) \subset H \to H$ be closed and densely defined. Then the following identities hold:

1. $\rho(A^*) = \{\, \overline{\xi} \in \mathbb{C} : \xi \in \rho(A) \,\}$.
2. $\sigma(A^*) = \{\, \overline{\lambda} \in \mathbb{C} : \lambda \in \sigma(A) \,\}$.
3. $\sigma_c(A^*) = \{\, \overline{\lambda} \in \mathbb{C} : \lambda \in \sigma_c(A) \,\}$.
4. If $\lambda \in \sigma_p(A) \setminus \sigma_c(A)$, then $\overline{\lambda} \in (\sigma_p(A^*) \setminus \sigma_c(A^*)) \cup \sigma_r(A^*)$.
5. If $\lambda \in \sigma_r(A)$, then $\overline{\lambda} \in \sigma_p(A^*) \setminus \sigma_c(A^*)$.

We finally clarify the relations between the spectrum and the numerical range rsp. the external field of regularity of operators acting in a Hilbert space.

THEOREM 6.1. Let $H$ be a Hilbert space and $A : \mathfrak{D}(A) \subset H \to H$ be a linear operator. Then the following assertions are true:

(a) If $A$ is bounded, then $\sigma(A) \subseteq \overline{\Theta(A)}$.

(b) $\sigma_p(A) \subseteq \Theta(A)$.

(c) $\sigma_c(A) \subseteq \overline{\Theta(A)}$.

(d) If $\zeta \in \Delta(A)$, then $\|R(A,\zeta)\| \leq \operatorname{dist}(\zeta, \Theta(A)^{-1}$.

(e) If $U \subseteq \Delta(A)$ is connected and $U \cap \rho(A) \neq \emptyset$, then $U \subseteq \rho(A)$.

The last assertion of this theorem is nothing but to say that if a subset $U$ of $\Delta(A)$ contains at least one element of $\rho(A)$, then all elements of $U$ belong to $\rho(A)$.

## §7. Compact Linear Operators and Fredholm Operators

A linear operator $k : X \to Y$ between Banach spaces $X$ and $Y$ is called *compact*, if the range of each bounded set under $k$ is relative compact with respect to the norm topology in $Y$, or equivalently, if for every bounded sequence $\{x_n\}$ in $X$ the sequence $\{k\,x_n\}$ in $Y$ has a convergent subsequence.

For example, a linear operator $k : X \to Y$ is compact, if at least one of the spaces $X$ and $Y$ is finite-dimensional. In case of $X = Y$, the identity operator $1_X : X \to X$ is compact if and only if $X$ is finite-dimensional.

Compact operators have the following general properties:

1. they are closed under taking finite linear combinations of them, and the limit of a norm convergent sequence of compact operators is compact again;
2. the composition $A_1 \circ A_2$ of bounded linear operators is compact, if at least one of them is so;
3. each compact linear operator is continuous, which can easily be shown by contradiction;
4. $\mathscr{C}(X)$, the set of compact operators $k : X \to X$ constitutes a *closed* ideal within the operator algebra $\mathscr{L}(X)$;
5. $\mathscr{C}(X)$ is complete and thus a Banach algebra in its own right.

A linear operator $F : X \to Y$ is said to have *finite rank*, if the range $\mathfrak{R}(F)$ is finite-dimensional, which of course implies compactness of $F$. Assume that $(F_n)$ is a sequence of finite-rank operators converging to a further operator $A$ with respect to the operator norm, then $A$ also proves to be compact. Thus compact operators are part of the closure of the set $\mathscr{F}(X, Y)$ of finite-rank operators, which itself lies within the set $\mathscr{C}(X, Y)$ of compact operators.

Reversely, in the setting of Hilbert spaces each compact operator $k$ appears in such a way, that is, $k$ is the norm-limit of a convergent sequence of finite-rank operators. But this assertion may not be true for compact operators acting between general Banach spaces.

With respect to the question of compactness of a linear operator and its transposed, SCHAUDER*'s theorem* is valid:

THEOREM 7.1. A continuous linear operator $A: X \to Y$ between Banach spaces $X$ and $Y$ is compact if and only if $A^t$ is compact.

For a proof of this theorem, see for example [14], pp. 282-283.

We now turn to the spectral properties of compact linear operators. Historically, the spectral theory of such an operator was developed by F. RIESZ and J. SCHAUDER. The essential results are known as the RIESZ-SCHAUDER *spectral theorem*, which resembles those of $(d, d)$-matrices acting in the finite-dimensional spaces $\mathbb{C}^d$:

THEOREM 7.2. Let $X$ be a complex Banach space and $k: X \to X$ be compact. Then the following assertions hold:

1. The spectrum $\sigma(k) \subset \mathbb{C}$ is mostly countable.
2. $\xi = 0$ is the only possible limit point of $\sigma(k)$.
3. If $X$ is infinite-dimensional, then $0 \in \sigma(k)$.
4. Each spectral value $\lambda \neq 0$ is an eigenvalue of $k$ with finite multiplicity.
5. Every $0 \neq \lambda \in \sigma(k)$ is a pole of the resolvent function $R(k, \cdot) : \rho(k) \to \mathscr{L}(X)$.

In the second part of this section we shall deal with the class of Fredholm operators. A continuous linear operator $A: X \to Y$ is called *Fredholm*, if it is lower and upper semi-Fredholm at the same time, or equivalently, if $\chi(A) \in \mathbb{Z}$. We collect a couple of non-trivial properties of Fredholm operators:

(a) $A^*$ is Fredholm if and only if $A$ is so.
(b) If one of $A$ or $A^*$ is Fredholm, then the dimensions of the kernels and ranges of $A$ and $A^*$ are related to each other by $\alpha(A^*) = \beta(A)$ and $\beta(A^*) = \alpha(A)$, which implies $\chi(A^*) = -\chi(A)$.
(c) Continuous Fredholm operators acting in $X$ and belonging to one and the same connected component set within the operator algebra $\mathscr{L}(X)$ have identical index. Thus, if $A$ is Fredholm and $B$ can be reached by a continuous path with starting point $A$, then $B$ is Fredholm, too, and $\chi(B) = \chi(A)$.

A Fredholm operator can be represented by means of a *left* and *right parametrix*, respectively.

Preliminary, if $A: X \to Y$ is a continuous linear operator, then a linear mapping $P_l : Y \to X$ is called a *left parametrix* if $A \circ P_l = 1_X + k_l$ for some compact operator $k_l \in \mathscr{L}(X)$. In a similar way, a linear mapping $P_r : Y \to Y$ is said to be a *right parametrix* of $A$, if $P_r \circ A = 1_Y + k_r$ for some compact operator $k_r \in \mathscr{L}(Y)$. Every Fredholm operator owns a left and right parametrix, where the involved compact operators $k_l$, $k_r$ are of *finite-rank*, that is, they have finite-dimensional ranges.

We next cite the Fredholm *alternative theorem* as the outstanding result in Fredholm theory with respect to the solvability of the linear equation $Ax = y$. Preliminary, the *rank-nullity theorem* known from linear algebra claims that if a linear space $V$ is finite-dimensional, then a linear mapping $A: V \to V$ is surjective iff it is injective. Fredholm's alternative theorem is a considerable generalization to general Banach spaces and Fredholm operators with vanishing index:

Theorem 7.3. Let $X$, $Y$ be Banach spaces and the continuous linear operator $A: X \to Y$ be Fredholm with index $\chi(A) = 0$. According to the solvability of the linear equation $Ax = y$, exactly one of the following cases occurs:

(a) $Ax = y$ is uniquely solvable for all $y \in X$, so $A$ is bijective,
(b) $Ax = y$ is not for all $y \in X$ solvable, and in this case the operator $A$ is neither injective nor surjective. If the equation is solvable for an element $y_0 \in X$, then the set of solutions forms a finite-dimensional subspace of $X$. In addition, the equation $Ax = y$ owns a solution $x \in X$ if and only if $y \in {}^{\perp}\mathfrak{N}(A^*)$.

A great portion of Fredholm theory deals with the study of linear operators of type $1 + k$, where $k : X \to Y$ is compact. Such operators are indeed Fredholm, to which Fredholm's alternative theorem can be applied. On the other hand, a continuous semi-Fredholm operator can be represented by means of a *left* and *right parametrix*, respectively. Given a continuous linear operator $A: X \to Y$, a linear mapping $P_l : Y \to Y$ is called a *left parametrix* if $A \circ P_l = 1_X + k_l$ for some compact operator $k_l \in \mathscr{L}(X)$. Similarly, a linear mapping $P_r : Y \to Y$ is said to be a *right parametrix* of $A$, if $P_r \circ A = 1_Y + k_r$ for some compact operator $k_r \in \mathscr{L}(Y)$.

We emphasize the usage of Fredholm theory by the following consideration, which is helpful for applications in partial differential

equations. Consider the linear equation $Ax = y$, where $X$ is Banach and $A : X \to X$ is a continuous linear operator. We assume that there is $\zeta \in \mathbb{C}$ such that $A - \zeta$ is invertible and the mapping $(A - \zeta)^{-1}$ is compact. Then we obtain

$$A x \;=\; (A - \zeta)\, x + \zeta\, x \;=\; y,$$

and applying the mapping $(A - \zeta)^{-1}$ from the left yields

$$x \;=\; \zeta (A + \zeta)^{-1}\, x \;=\; \tilde{y} := (A - \zeta)^{-1}\, y$$

or equivalently $x + \tilde{k}\, x = \tilde{y}$, where $\tilde{k} := -\zeta (A - \zeta)^{-1}$ is compact.

The equations $A\, x = y$ and $(1 + \tilde{k})\, x = \tilde{y}$ are called *equivalent to each other*, since $x$ is a solution of the first one if and only if it is a solution of the second one. By ATKINSON's theorem, the operator $1 + \tilde{k}$ is Fredholm with vanishing index, so theorem 7.3 is applicable and provides assertions also for the solution of the equation $Ax = y$. We summarize this statement to the following theorem:

THEOREM 7.4. Let $A : \mathfrak{D}(A) \subset X \to X$ be an operator, for which there is $\zeta \in \mathbb{C}$ such that $A - \zeta$ is invertible. Then the equation $Ax = y$ is equivalent to a equation of the form $(1 + k)x = y$, and FREDHOLM's alternative theorem for $Ax = y$ is valid.

We have discussed the (semi-)Fredholm property of linear operators under the basic assumption that they are continuous and everywhere defined in their domain spaces. If $A$ is only closed instead of continuous however, we define the possible Fredholm property by regarding $A$ as a continuous linear operator $A : \mathfrak{D}(A) \to Y$, where $\mathfrak{D}(A)$ is endowed with the graph norm $\| \cdot \|_A$ defined by (3.1).

We close this section with a collection of concepts regarding solvability of linear equations of the form $Ax = y$, where $A$ is a closed linear operator acting between Banach spaces $X$ and $Y$. Following [8] and [3], the equation $Ax = y$ is called

(a) *uniquely solvable* ([8]), if $A$ is injective;
(b) *everywhere solvable* ([8]), if $A$ is surjective;
(c) *solvable* ([3]), if it is uniquely and everywhere solvable;
(d) *well-posed*, if it is solvable and $A^{-1}$ is continuous;
(e) *completely solvable* ([3]), if it is solvable and $A^{-1}$ is compact;
(f) *densely solvable* ([8]), if $\overline{\mathfrak{R}(A)} = Y$;
(g) *normally solvable* ([8]), if $\overline{\mathfrak{R}(A)} = \mathfrak{R}(A)$;
(h) *correctly solvable* ([8]), if $A$ is bounded below;

(i) *semi-solvable* ([3]), if $A$ is Fredholm;
(j) *almost solvable* or *regularly solvable* ([3]), if $A$ is semi-solvable and $\chi(A) = 0$.

To the end, remember again the important result based on the closed graph theorem: if $A$ is continuous and closed, then (a) + (b) implies (d).

## §8. Sesquilinear Forms and Associated Operators

Remember that a *sesquilinear form* (*bilinear form*) in a complex Hilbert space $V$ is a mapping $\mathfrak{a} : V \times V \to \mathbb{C}$ which is linear in the first and antilinear in the second argument. Such forms are in one-to-one relationship with mappings $\mathbb{A} : V \to V^*$, where $V^*$ is the conjugate space of continuous and (anti)linear functionals $\phi : V \to \mathbb{K}$. In fact, we define $\mathbb{A}(x) := f_x \in V^*$ with $f_x(y) := \mathfrak{a}(x, y)$ for $y \in V$, and the form $\mathfrak{a}$ can be recovered from the operator $\mathbb{A}$ in a similar way. One often says that $\mathbb{A}$ is the *representation operator* associated with the form $\mathfrak{a}$.

Moreover, any form $\mathfrak{a}$ is called *bounded*, if there is $\Theta \geq 0$ such that

$$|\mathfrak{a}(x, y)| \leq \Theta \|x\| \, y \qquad (x, y \in V).$$

It can be shown by the *uniform boundedness principle*, that $\mathfrak{a}$ is continuous if and only if $\mathfrak{a}$ is bounded. A fundamental representation theorem for continuous forms claims that there is a bounded linear operator $B \in \mathscr{L}(V)$ such that $\mathfrak{a}(x, y) = \langle Bx, y \rangle$ for all $x, y \in V$.

The concept of *coerciveness* is of outstanding meaning in what follows. Let $\mathfrak{a}$ be a continuous sesquilinear form in Hilbert space $V$. Then $\mathfrak{a}$ is called *coercive*, if there is $\theta > 0$ such that

$$\operatorname{Re} \mathfrak{a}(x, x) \geq \theta \|x\|^2 \qquad (x \in V).$$

This property especially implies

$$\|Bx\| \, \|x\| \geq |\langle Bx, x \rangle| = |\mathfrak{a}(x, x)| \geq \operatorname{Re} \mathfrak{a}(x, x) \geq \theta \|x\|^2 \quad (x \in V),$$

hence the operator $B$ is bounded below with lower bound $\theta$.

The Lax-Milgram *lemma* is one of the key theorems in applied functional analysis and ensures the well-posedness of the variational problem

$$\mathfrak{a}(x, y) = \phi(y) \quad \text{for all } y \in V. \tag{8.1}$$

for any right-hand side $\phi \in V^*$.

THEOREM 8.1. If the form $\mathfrak{a} : V \times V \to \mathbb{C}$ is coercive, then problem (8.1) has exactly one solution $x_\phi \in V$ for each $\phi \in V^*$. In addition, $\| x_\phi \| \leq \theta^{-1} \|\phi\|$ holds.

In other words: if $\mathfrak{a}$ is coercive, then the representation operator $\mathbb{A} : V \to V^*$ is bjective, the inverse operator $\mathbb{A}^{-1} : V^* \to V$ is bounded and the solution $x_\phi$ of (8.1) continuously depends on $\phi$.

The presented theory of coercive sesquilinear forms in a Hilbert space $V$ is often not sufficent for applications. Thus it is indicated to study coercive forms in the setting of *pairs* $(V, H)$ of Hilbert spaces, where the space $V$ is continuously and densely embedded into a second space $H$. More precisely, we assume that

(T-1) there is a linear mapping $\iota_1 : V \to H$ such that $\overline{\mathfrak{R}(\iota_1)} = H$;

(T-2) there is $c > 0$ such that $\|\iota_1 v\|_H \leq c \|v\|_V$ for all $v \in V$.

By means of the embedding $\iota_1$, one can also think of $V$ as a dense linear subspace within the larger space $H$.

Moreover, each Hilbert pair $(V, H)$ gives rise to introduce the *transposed pair* $(H^*, V^*)$, where the conjugate dual space $H^*$ is continuously and densely embedded into the conjugate dual space $V^*$ by a further embedding $\iota_2 : H^* \to V^*$. Then, by identifying the spaces $H$ and $H^\times$ via the RIESZ map, we combine both Hilbert pairs to obtain the triple

$$V \overset{\iota_1}{\hookrightarrow} H \overset{\iota_2}{\hookrightarrow} V^*, \tag{8.2}$$

sometimes called a *Gelfand triple*, an *evolution triple* or a *rigged Hilbert space*. The center space $H$ is called the *pivot space* of (8.2).

Triples of Hilbert spaces own a remarkable property. Let $(\cdot, \cdot)$ and $\langle \cdot, \cdot \rangle$ be the inner products in $V$ and $H$, respectively, then for fixed chosen $x \in H$ the action of the antilinear linear functional $f_x := \iota_2 x \in V^*$ to any element $v \in V$ can be represented by the inner product of $x$ and $v$ in $H$, i. e. $f_x(v) = \langle x, \iota_1 v \rangle$ for $v \in V$.

It is quite natural to examine sesquilinear forms also with respect to pairs $(V, H)$ of Hilbert spaces. Moreover, in applications it is often the case that sesquilinear forms are merely continuous but not coercive. In the setting of a Hilbert pair $(V, H)$, however, we may try to make a form $\mathfrak{a}$ coercive by adding an additional term, which is a multiplicity of the inner product in $H$.

More precisely, if $\mathfrak{a}$ is given on $V$ and the slightly modified form

$$\mathfrak{a}_\lambda(u,v) := \mathfrak{a}(u,v) + \lambda\langle \iota_1 u, \iota_1 v\rangle \quad (u,v \in V)$$

is coercive for some $\lambda \in \mathbb{R}$, that is, there is $\theta > 0$ such that

$$\operatorname{Re}\mathfrak{a}(u,u) + \lambda \|\iota_1 u\|_H^2 \geq \theta \|u\|_V^2 \quad (u \in V), \tag{8.3}$$

then $\mathfrak{a}$ is called *quasi-coercive*, *$H$-elliptic* or simply a *Gårding form*. It should be clear that if $\mathfrak{a}_{\lambda_0}$ is coercive for some $\lambda_0 \in \mathbb{R}$, then $\mathfrak{a}_\lambda$ is so for $\lambda > \lambda_0$.

Quasi-coercive sesqilinear forms can be used to derive densely defined and closed linear operators in the pivot space $H$ of the underlying Hilbert triple.

Let $\mathfrak{a}$ be a sesquilinear form defined on a Hilbert space $V$, which is the left hand side of a Hilbert triple $(V, H, V^*)$ with continuous embeddings $\iota_1 : V \to H$ and $\iota_2 : H \to V^*$. We introduce a linear operator $A_\mathfrak{a}$ with domain space $H$ and range space $H$ by restriction of the form operator $\mathcal{A}_\mathfrak{a} : V \to V^*$ of $\mathfrak{a}$ to those elements $u \in V$, for which $\mathcal{A}_\mathfrak{a} u \in H$. Indeed, the domain $\mathfrak{D}(A_\mathfrak{a})$ is defined to be the subspace of those elements $u \in V$, for which $\mathcal{A}_\mathfrak{a} u \in \iota_2(H) \subset V^*$ holds, briefly denoted

$$\mathfrak{D}(A_\mathfrak{a}) := \{\, u \in V : \mathcal{A}_\mathfrak{a} u \in H \,\}.$$

Clearly, this set can be regarded as a subset of $H$ by means of the embedding $\iota_1$.

Then for $u \in \mathfrak{D}(A)$ there exists $y \in H$ such that $\iota_2 y = \mathcal{A}_\mathfrak{a} u$. We set $x := \iota_1 u$ and $A_\mathfrak{a} x := y$ for $x \in \mathfrak{D}(A_\mathfrak{a})$ and denote $A_\mathfrak{a}$ the *operator in $H$ associated with* $(V, \mathfrak{a})$, or shortly, the *variational operator* derived from $\mathfrak{a}$. Another phrase is to say that the operator $A_\mathfrak{a}$ is the *part of $\mathcal{A}_\mathfrak{a}$ in $H$*.

The following theorem collects a couple of nice properties of the operator $A_\mathfrak{a}$ associated with a quasi-coercive form $\mathfrak{a}$, which besides the Lax-Milgram lemma is our second important theorem in this section:

Theorem 8.2. Let $(V, H, V^*)$ be a Hilbert triple and the sesquilinear form $\mathfrak{a}$ be quasi-coercive on $V$ with parameter $\lambda_0 \in \mathbb{R}$. Then the following assertions are true:

1. $A_\mathfrak{a}$ is closed.
2. $A_\mathfrak{a}$ is densely defined in $H$.
3. If the embedding of $V$ into $H$ is *compact*, then the operator $A_\mathfrak{a} + \lambda$

has a compact inverse $(A_{\mathfrak{a}} + \lambda)^{-1}$ for each $\lambda > \lambda_0$.

## §9. Linear Operator-Differential Equations

A *vector-valued* function $u : [a, b] \to X$ with values in a Banach space $X$ is said to be *strongly or classical differentiable* in $t_0 \in (a, b)$, if there is an element $u'(t_0) \in X$ such that

$$\lim_{t \to t_0} \left\| \frac{u(t) - u(t_0)}{t - t_0} - u'(t_0) \right\| = 0.$$

The function $u$ is called *differentiable in* $(a, b)$, if $u$ is so in each $t \in (a, b)$. The space $C^1(a, b; X)$ consists of functions having continuous classical derivatives in $(a, b)$. Moreover, the spaces $C^k(a, b; X)$ $(k \in \mathbb{N})$ are defined recursively as usual, whereas the space $C^\infty(a, b; X)$ is the intersection of all the spaces $C^k(a, b; X)$ for $k \in \mathbb{N}_0$.

By a linear *evolution equation* in a Banach space $X$, also called a *linear operator-differential equation*, we mean an ordinary differential equation of the form

$$u'(t) = A\,u(t) + f(t) \quad (0 < t < \tau), \tag{9.1}$$

where $A$ is a densely defined linear operator $A : \mathfrak{D}(A) \subset X \to X$ and (9.1) must be valid for each $t \in (0, \tau)$. We emphasize that the differential equation only makes sense if $u'(t) \in \mathfrak{D}(A)$ for every $t \in (0, \tau]$. We shall call 9.1 *homogeneous*, if $f = 0$.

The *homogeneous Cauchy problem* is to find a solution for every $t \in [0, \tau]$ of the initial-value problem

$$\begin{cases} u'(t) = A\,u(t) \\ u(0) = u_0 \in X. \end{cases} \tag{9.2}$$

A function $u \in C([0, \tau], \mathfrak{D}(A)) \cap C^1((0, \tau], X)$ is called a *classical solution* of (9.2), if it satisfies the initial condition and the differential equation in each point $t \in (0, \tau]$. Moreover, a continuous function $u : [0, \tau) \to X$ is called a *mild solution* of (9.2) if

$$(1)\ \tilde{u}(t) := \int_0^t u(s)\,ds \in \mathfrak{D}(A) \quad \text{and} \quad (2)\ A\,\tilde{u}(t) = u(t) - u_0$$

holds for all $0 < t \leq \tau$. One is led to the notion of mild solution in a natural way by integrating both sides of the equation $u' = A\,u$, which then becomes into

$$\int_0^t u'ds \;=\; u(t) - u(0) \;=\; \int_0^t A\,u(s)\,ds.$$

If $A$ is assumed to be closed, then the action of $A$ and performing the integration commutes, that is,

$$\int_0^t A\,u(s)\,ds \;=\; A\int_0^t u(s)\,ds,$$

and to the end we achieve the equation

$$u(t) \;=\; u(0) \;+\; A\int_0^t u(s)\,ds \qquad (t\in(0,\tau]).$$

Notice that each classical solution is a mild solution, and reversely a mild solution $u$ is classical if and only if $u$ is continuously differentiable in $(0,\tau]$, thus classical and mild solutions of homogeneous Cauchy problems only differ with respect to this regularity condition.

The homogeneous Cauchy problem (9.2) is called *correctly posed*, if there is a unique classical solution for any initial value $x_0\in\mathfrak{D}(A)$. In addition, it is denoted *well-posed*, if the solution depends continuously on the initial values, or equivalently, if for each $\tau>0$ there is $c_\tau>0$ independent of $u_0$ such that

$$\|u(t)\| \le c_\tau\,\|u_0\| \qquad \text{for all } t\in(0,\tau]$$

and all $u_0\in X$, for which a classical solution exists (the set of such elements may be the domain of definition $\mathfrak{D}(A)$ of $A$ or any other dense subspace in $X$).

The abstract Cauchy problem of *second order* is to solve the initial-value problem

$$\begin{cases} u''(t) \;+\; A\,u(t) \;=\; 0 \\ u(0) \;=\; u_0,\; u'(0) \;=\; u_1, \end{cases} \tag{9.3}$$

where $u_0\in\mathfrak{D}(A)$, but $u_1$ may be an element of an enlarged subspace within $X$.

Similar to the first order problem, a function $u\in C([0,\tau],\mathfrak{D}(A))\cap C^2((0,\tau],X)$ is called a *classical solution* of (9.3), if it satisfies the initial conditions and the differential equation in each point $t\in(0,\tau]$.

The notion of *semigroup* for bounded linear operators has already been introduced by J. A. Séguier in 1904. Later some applications to partial differential equations were detected, but the first systematic and comprehensive treatement of semigroup theory was done in

the pioneering monograph [6] by E. HILLE and R. PHILLIPS. Nowadays semigroup theory provides a powerful method in solving linear operator-differential equations with a multiplicity of applications to linear partial differential equations.

Given a Banach space $X$, a $C_0$-*semigroup* or a *strongly continuous semigroup of bounded linear operators* is an operator-valued mapping $S : [0, \infty) \to \mathscr{L}(X)$, shortly denoted $\{S(t)\}_{t \geq 0}$ and distinguished by the properties

(G-1) $S(t+s) = S(t) \circ S(s)$ for all $t, s \geq 0$ (the *semigroup property*);
(G-2) $\lim_{t \downarrow 0} S(t)\, x = x$ for all $x \in X$.

It follows from (G-2), which is nothing but strong continuity from the right-hand side in $t = 0$, that $S(0)$ is the identity mapping in $X$. Notice also that the conditions (G-1) and (G-2) together imply continuity from the right of the semigroup even in every $t > 0$. In case of $X = \mathbb{R}$, semigroups share both properties with the family $\{\mathcal{E}(t)\}_{t \geq 0}$ of exponential functions $\mathcal{E}(t) : \lambda \mapsto e^{\lambda t}$, $(\lambda \in \mathbb{R})$, so this family serves as an elementary example of a strongly continuous semigroup.

We next perform a classification on the totality of strongly continuous semigroups. For each such semigroup $\{S(t)\}_{t \geq 0}$ there are constants $M \geq 0$ and $\omega \in \mathbb{R}$ such that

$$\|S(t)\| \leq M e^{\omega t} \quad \text{for } t > 0. \tag{9.4}$$

In fact, let $M \geq 1$ such that $\|S(t)\| \leq M$ for all $s \in [0, 1]$. For $t > 0$ there is $s \in [0, 1)$ and $n \in N$ such that $t = n + s$. Then $\|S(n)\| \leq M \|S(n-1)\|$ for $n > 1$ and

$$\|S(t)\| = \|S(n+s)\| = \|S(n) \circ S(s)\| \leq$$
$$\|S(n)\| \cdot \|S(s)\| \leq M^{n+1} = M e^{n \cdot \ln M}.$$

By the *orbit map* or *trajectory* $\eta_x$ of a semigroup associated with an element $x \in X$ we mean the vector-valued mapping $\eta_x : [0, \infty) \to X$ given by $\eta_x(t) := S(t)\, x$. Due to condition (G-1), strong continuity of $\{S(t)\}_{t \geq 0}$ implies continuity of the functions $\eta_x$. But $\eta_x$ is even differentiable under a very mild assumption: if $\{S(t)\}_{t \geq 0}$ is a $C_0$-semigroup, then for each $x \in X$ the mapping $\eta_x$ is differentiable in $(0, \infty)$ if and only if $\eta_x$ is right differentiable in $t = 0$.

If $\{S(t)\}_{t \geq 0}$ is a strongly continuous semigroup, we define its *infinitesimal generator* (briefly denoted the *generator* of the semigroup) as a mapping $A$ in $X$ by means of the graph

$$\Gamma_A := \{ (x,y) \in X \times X : \ y = \lim_{h \downarrow 0} \frac{1}{h}(S(h)x - x) \text{ exists } \}.$$

Hence $\mathfrak{D}(A)$ of $A$ consists of those elements $x \in X$, for which the orbit map $\eta_x$ is differentiable from the right in $t = 0$, that is, $\mathfrak{D}(A) = \{ x \in X : \eta_x \text{ is differentiable} \}$. It is easy to show that $A$ is linear, so it is a linear operator acting in $X$.

The meaning of strongly continuous semigroups is confirmed by the solution theory of homogeneous Cauchy problems governed by linear operators, which at the same time are infinitesimal generators of $C_0$-semigroups. Indeed, the *main theorem of semigroup theory* reads as follows:

THEOREM 9.1. Let $A$ be a closed and densely defined linear operator acting in a Banach space $X$. Then the following assertions are equivalent to each other:

1. The homogeneous Cauchy problem (9.2) is well-posed.
2. The operator $-A$ generates a strongly continuous semigroup of type $(\omega, M)$.
3. The resolvent set of $A$ fulfills $(0, \infty) \subset \rho(A)$ and the resolvent function $R(A, \xi)$ satisfies

$$\|R(A,\xi)^n\| = \|(A - \xi)^{-n}\| \leq M(\xi - \omega)^{-n} \quad (n \in \mathbb{N}, \xi > 0). \tag{9.5}$$

The equivalence of the both last assertions is the content of the famous FELLER-MIYADERA-HILLE-YOSIDA *theorem.* Since the number of conditions on the operator $A$ is infinite, this theorem has limited worth with respect to applications. However, the original theorem for generators of *contraction semigroups* due to HILLE and YOSIDA in 1948 quotes that in this case only the first inequality ($n = 1$, with $\omega = 0$ and $M = 1$) of (9.5) has to be verified:

THEOREM 9.2. The linear operator $-A$ acting in a Banach space $X$ is the generator of a $C_0$-semigroup of *contractions* if and only if

1. $A$ is densely defined and closed,
2. the interval $(-\infty, 0)$ belongs to the resolvent set $\rho(A)$,
3. $\|\xi\, R(A,\xi)\| \leq 1$ for all $\xi < 0$.

In fact, if $-A$ owns the mentioned properties, then the semigroup $\{S(t)\}_{t \geq 0}$ may be constructed by

$$S(t)\,x := \lim_{n \to \infty} (1 + \frac{t}{n} A)^{-n}\, x \text{ for each } t > 0 \text{ and } x \in X.$$

Our next aim is to describe the spectral properties in theorem 9.2 by equivalent conditions, which leads to the concept of *accretive* operator.

Let $H$ be a complex Hilbert space, then a linear operator $A$ with domain of definition $\mathfrak{D}(A)$ acting in $H$ is called *accretive*, if

$$\operatorname{Re}\langle A\,x, x\rangle \geq 0 \quad \text{for all } x \in \mathfrak{D}(A),$$

or equivalently, if the numerical range $\Theta(A)$ is part of the right complex half-plane. Moreover, a linear operator $A$ is said to be *dissipative*, if $-A$ is accretive.

We first notice that if $A$ is accretive (dissipative), then $\lambda A$ is so for each $\lambda > 0$. Furthermore, since

$$\|A\,x + \lambda x\|^2 - \|\lambda x\|^2 \geq 2\lambda \operatorname{Re}\langle A\,x, x\rangle \quad \text{for all } x \in \mathfrak{D}(A),$$

the operator $A$ is accretive if and only if

$$\|(A+\lambda)x\| \geq \|\lambda x\| \quad \text{for all } \lambda > 0 \text{ and } x \in \mathfrak{D}(A). \tag{9.6}$$

This property can be used to define coercivity and dissipativity for operators also in normed linear spaces, in which no inner product is defined.

The inequality (9.6) says that for $\lambda > 0$ the operator $A + \lambda$ is bounded below and thus injective. In addition, if $A$ is closed, then theorem 5.1 implies that $A$ has closed range, and by the closed graph theorem the inverse operator $(A+\lambda)^{-1} : \mathfrak{R}(A+\lambda) \to \mathfrak{D}(A)$ is bounded with norm $\|(A+\lambda)^{-1}\| \leq 1/\lambda$, so it even forms a contraction for $\lambda \geq 1$.

In the following we assume that for $A$ being accretive there is $\lambda_0 > 0$ such that $A + \lambda_0$ is surjective. A linear operator having this property is said to fulfill the *range condition*. Then there is an open interval $I$ containing $\lambda_0$ such that $A + \lambda$ is surjective for all $\lambda \in I$. It can be proved that in this case $A + \lambda$ is even surjective for all $\lambda > 0$.

A linear operator $A$ is called *maximal accretive* or shortly *m-accretive* if it is accretive and the range condition is fulfilled. The following list contains some properties of such an operator $A$:

1. $A$ is densely defined in $H$;
2. $A$ is closed;
3. $(-\infty, 0) \subset \rho(A)$;
4. $A + \lambda$ is surjective for all $\lambda > 0$.

Our next theorem characterizes the class of m-accretive operators:

Theorem 9.3. For $A$ being accretive, the following assertions are equivalent to each other:

1. $A$ is m-accretive;
2. $A$ is accretive and there is no proper accretive extension of $A$;
3. The resolvent set $\rho(A)$ contains an element of the open left half-plane;
4. the adjoint operator $A^*$ is accretive, too.

In fact, if $A$ owns the mentioned properties, the semigroup $\{S(t)\}_{t\geq 0}$ may be constructed by

$$S(t)\,x := \lim_{n\to\infty}(1+\frac{t}{n}A)^{-n}\,x \text{ for each } t>0 \text{ and } x\in X. \qquad (9.7)$$

Recall that a m-accretive operator $A$ acting in a Hilbert space distinguishes by the fact that $\mathrm{Re}\langle A\,x,x\rangle \geq 0$ for $x\in\mathfrak{D}(A)$ and $\xi\in\rho(A)$ for $\xi<0$. Using the concept of maximal accretivity, the *Lumer-Phillips theorem* classifies the operators being the infinitesimal generators of a contraction semigroup without using any spectral theoretical concepts:

Theorem 9.4. Let $A$ be a closed linear operator in Hilbert space. Then $-A$ generates a contractive $C_0$-semigroup if and only if $A$ is m-accretive.

Notice that both conditions causes the operator $A$ to be densely defined and closed. This is also true if the operator $A$ is acting in a reflexive Banach space. But in the setting of an arbitrary Banach space, theorem 9.4 only holds for those operators, which in addition are densely defined and closed.

## Part II
# Abstract Theory of Self-Adjoint Linear Operators

In this second part we introduce and discuss some classes of linear operators in Hilbert space, which rely on the presence of the inner product and cannot be defined in arbitrary Banach spaces.

The concept of adjoint operator gives rise to a further class of linear operators in Hilbert space, which are called *normal*. More precisely, a densely defined linear operator $A$ with domain of definition $\mathfrak{D}(A)$ and acting in $H$ is denoted *normal*, if

$$A \circ A^* = A^* \circ A.$$

This definition of course implies the identity of the sets $\mathfrak{D}(A)$ and $\mathfrak{D}(A^*)$. Notice that a normal operator is closed (since $A^*$ is so), $\mathfrak{D}(A) = \mathfrak{D}(A^*)$, the normed space $(\mathfrak{D}(A^*), \|\cdot\|_{A^*}$ is complete and the norm $\|\cdot\|_A$ is equivalent to the norm $\|\cdot\|_{A^*}$.

The subsequent statements are equivalent to each other:

1. $A$ is normal;
2. $A^*$ is normal;
3. $\mathfrak{D}(A) = \mathfrak{D}(A^*)$ and $\|Ax\| = \|A^*x\|$ for each element $x \in \mathfrak{D}(A)$.

The class of normal and bounded operators in Hilbert space includes a couple of important subclasses. In summary, an *unitary* linear operator distinguishes by the property $A \circ A^* = 1$, a *symmetric operator* owns the property $A \prec A^*$, and a *self-adjoint operator* is defined by $A = A^*$. Especially the concept of self-adjoint operator have led to a well-elaborated theory with many applications in quantum mechanics and other topics of mathematical physics.

## §1. Symmetric and Half-Bounded Operators

Let $H$ be a real or complex Hilbert space. A linear operator acting in $H$ and with domain of definition $\mathfrak{D}(A) \subset H$ is called *symmetric*, if

$$\langle Ax, y\rangle = \langle x, Ay\rangle \quad \text{for all } x, y \in \mathfrak{D}(A). \tag{1.1}$$

We first consider the symmetry property of operators being continuous and everywhere defined. Let $\mathscr{L}(H)$ be the Banach space of *continuous* linear operators $A : H \to H$ furnished with the *operator norm* as defined in (1.1). Then, if $A$ is symmetric, we have

$$-\|A\| \leq \langle Ax, x\rangle \leq \|A\|,$$

for $x \in H$ with $\|x\| = 1$, or equivalently,

$$\Theta(A) := \{\langle Ax, x\rangle : x \in \mathfrak{D}(A), \|x\| = 1\} \subset [-\|A\|, \|A\|],$$

where $\Theta(A)$ is the *numerical range* of $A$. Hence one may introduce in the class of bounded symmetric operators acting in $H$ the partial order

$$A_1 \geq A_2 \text{ if and only if } \langle (A_1 - A_2)x, x\rangle \geq 0 \qquad (x \in H),$$

which implies that each symmetric linear operator $A \in \mathscr{L}(H)$ can be enclosed by the simple multiplication operators $A_- := -\|A\| 1$ and $A_+ := \|A\| 1$.

On the other hand, the HELLINGER-TOEPLITZ *theorem* quotes that if $A$ is symmetric and $\mathfrak{D}(A)$ is the whole of $H$, then $A$ must be bounded, which to the end is a consequence of the closed graph theorem. In fact, let $A$ be symmetric with $\mathfrak{D}(A) = H$, then for each convergent sequence $(x_n)$ tending to 0 and with $(Ax_n)$ tending to $x \in H$ we have

$$\langle x, x\rangle = \lim_{n\to\infty} \langle Ax_n, x\rangle = \lim_{n\to\infty} \langle x_n, Ax\rangle = \langle 0, Ax\rangle = 0,$$

which implies $x = 0$, so $A$ is closed and even continuous by theorem 3.2. Hence we may always assume that the domain of definition $\mathfrak{D}(A)$ of a non-continuous symmetric linear operator $A$ is a *proper* subspace of $H$.

The symmetry of a linear operator with domain of definition dense in $H$ can be characterized by other properties than (1.1):

THEOREM 1.1. Let $A$ be a densely defined linear operator with domain of definition $\mathfrak{D}(A)$ in Hilbert space $H$. Then the following statements are equivalent:

1. $A$ is symmetric;
2. $\langle Ax, x\rangle \in \mathbb{R}$ for all $x \in \mathfrak{D}(A)$;
3. $A \prec A^*$.

The second property is of course also meaningful for those symmetric operators, which are not densely defined in the entire space $H$.

Since the adjoint $A^*$ of a symmetric operator $A$ forms a closed extension of $A$, the operator $A$ itself is closable.

Moreover, the closure $\overline{A}$ turns out to be symmetric. In fact, to show that $\overline{A}$ is symmetric, let $(x_n)$ and $(y_n)$ be sequences in $\mathfrak{D}(A)$ with $x_n \to x \in \mathfrak{D}(\overline{A})$, $Ax_n \to \overline{A}x$ and $y_n \to y \in \mathfrak{D}(\overline{A})$, $Ay_n \to \overline{A}y$. Since $\langle Ax_n, y_n\rangle = \langle x_n, Ay_n\rangle$ for all $n \in \mathbb{N}$, taking the limit for $n \to \infty$ shows that the closure of $A$ must be symmetric, too.

Recall that a number $\lambda \in \mathbb{C}$ is called an *eigenvalue* of a linear operator $A$, if there is $x \in \mathfrak{D}(A)$, called the *eigenvector* of $A$ with respect to $\lambda$, such that $Ax = \lambda x$. The following properties of eigenvalues and eigenvectors of a *symmetric* linear operator $A$ are easy to prove:

1. each eigenvalue $\lambda$ of $A$ is real;
2. eigenvectors $x_1, x_2$ belonging to different eigenvalues $\lambda_1 \neq \lambda_2$ are *orthogonal* to each other, that is, $\langle x_1, x_2\rangle = 0$.

We finally consider an important subclass of symmetric operators. Any symmetric operator $A$ with domain of definition $\mathfrak{D}(A) \subset H$ is called *half-bounded* if there is $\theta \in \mathbb{R}$ such that

$$\langle Ax, x\rangle \geq \theta \|x\|^2 \quad \text{for all } x \in \mathfrak{D}(A). \tag{1.2}$$

The operator $A$ is called *positive* (*strictly positive*), if $\theta = 0$ ($\theta > 0$) can be chosen. In case of $\theta > 0$, however, (1.2) together with the Cauchy-Schwarz inequality (2.2) implies that $A$ is also bounded below, that is,

$$\|Ax\| \geq \theta \|x\| \quad \text{for all } x \in \mathfrak{D}(A), \tag{1.3}$$

so $A$ is injective by theorem 5.1. Furthermore, this theorem also claims that if $A$ is *closed*, then $\mathfrak{R}(A)$ is closed in $H$. Thus, to prove bijectivity of $A$ it suffices to show that $\mathfrak{N}(A^*) = \{0\}$, since $H = \mathfrak{N}(A^*)^\perp = \overline{\mathfrak{R}(A)} = \mathfrak{R}(A)$ in this case.

## §2. The Concept of Self-Adjointness

*Self-adjointness* of a linear operator acting in a Hilbert space is an extremely desirable property, since it makes the operator accessible for spectral analysis, which is the study of the spectrum and the resolvent function, respectively.

To motivate the definition of this concept, let $A$ be a symmetric linear operator in Hilbert space $H$. Then every other symmetric linear operator $S$ with $A \prec S$ is called a *symmetric extension* of $A$. Notice that $A \prec S$ implies $S^* \prec A^*$, which is even true if $A$

and $S$ are non-symmetric. Furthermore, let $S_1, \ldots, S_k$ be a finite sequence of symmetric extensions of $A$, where $S_{i+1}$ extends $S_i$ for $i = 1, \ldots, k-1$. Since $S_k \prec S_k^*$ and $S_{i+1}^* \prec S_i^*$ for $i = 1, \ldots, k-1$, we finally obtain by concatenating all these extensions the chain

$$A \prec S_1 \prec S_2 \prec \ldots \prec S_k \prec S_k^* \prec \ldots \prec S_2^* \prec S_1^* \prec A^*.$$

Thus one may hope to find a *maximal* symmetric extension $S_{sa}$ of $A$ such that

$$A \prec S_{sa} = S_{sa}^* \prec A^*,$$

which means that $S_{sa}$ cannot be extended to a larger symmetric operator. This expectation lead us to introduce the following notion: $A$ is called *self-adjoint*, if $A^* \prec A$, or equivalently, if $A = A^*$. For *bounded* and *everywhere defined* linear operators, however, there is of course no difference between symmetry and self-adjointness.

It should be clear that a self-adjoint operator $A$ is closed, since $A^*$ is so. On the other hand, the closure of a symmetric linear operator is not necessarily self-adjoint. Hence we denote a closable and symmetric operator $A$ *essentially self-adjoint*, if $\overline{A}$ is self-adjoint. It should be clear that an essentially self-adjoint operator $A$ has exactly one self-adjoint extension, which is nothing but the closure of $A$.

The *range condition* serves as the standard criterion to verify (essentially) self-adjointess of a symmetric linear operator $A$:

THEOREM 2.1. A symmetric linear operator $A$ with dense domain of definition $\mathfrak{D}(A)$ in $H$ is

1. self-adjoint if and only if $\mathfrak{R}(A+i) = \mathfrak{R}(A-i) = H$,
2. essentially self-adjoint if and only if $\overline{\mathfrak{R}(A+i)} = \overline{\mathfrak{R}(A-i)} = H$.

Proving the first assertion, let $A$ be self-adjoint. We make use of the fundamental relationship between the range of $A - i$ and the kernel of $(A-i)^* = A^* + i$, namely

$$\overline{\mathfrak{R}(A-i)} = \mathfrak{N}(A^*+i)^\perp.$$

Let $y \in \mathfrak{N}(A^*+i)$, that is, $A^*y = -iy$. Then

$$\langle y, y\rangle = \operatorname{Im}\langle -iy, y\rangle = \operatorname{Im}\langle A^*y, y\rangle = \operatorname{Im}\langle y, Ay\rangle = 0,$$

since $\langle y, Ay\rangle$ is real-valued, which implies $y = 0$. This shows that $\mathfrak{N}(A^*+i) = \{0\}$ and $\mathfrak{R}(A-i)$ is dense in $H$. But since $A - i$ is

bounded below and thus has closed range, we have $\mathfrak{R}(A-i)=H$. A similar computation with $i$ replaced by $-i$ shows that also $\mathfrak{R}(A+i)=H$.

Reversely, let the range condition be valid. We show self-adjointness of $A$ as follows: Since $A \prec A^*$, we have only to show $A^* \prec A$, or equivalently, $y \in \mathfrak{D}(A^*)$ implies $y \in \mathfrak{D}(A)$.
Now, since $(A-i)^*x = (A^*+i)x$ for $x \in \mathfrak{D}(A)$, we have

$$\langle (A-i)x, y\rangle = \langle Ax, y\rangle - i\langle x, y\rangle = \langle x, A^*y\rangle - \langle x, -iy\rangle = \langle x, (A^*+i)y\rangle.$$

But since $\mathfrak{R}(A+i)=H$, there exists $z \in \mathfrak{D}(A)$ such that $(A+i)z = (A^*+i)y$. Inserting this in the right-hand side of the last equation, we obtain

$$\langle (A-i)x, y\rangle = \langle x, (A+i)z\rangle = \langle (A-i)x, z\rangle.$$

But since in addition $\mathfrak{R}(A-i)=H$, we obtain $\langle u, y-z\rangle = 0$ for every vector $u := (A-i)x \in H$, which implies $y=z$. □

Essentially self-adjointness of an operator can also be characterized by means of its adjoint: a linear operator $A$ is essentially self-adjoint if and only if $A$ is symmetric and $A^*$ is symmetric, and $\overline{A} = A^*$ holds in this case.

## §3. Self-Adjoint Extensions of Symmetric Operators

We review some aspects of JOHN V. NEUMANN's *extension theory* for symmetric operators. First notice that the closure of a symmetric linear operator is not necessarily self-adjoint. To ensure self-adjointess of a symmetric operator $A$, the kernels of both operators $A^* \pm i$ are of special interest. In fact, if these are trivial, then $A \pm i$ are surjective and $\pm i$ are both regular values of $A$.

F. RIESZ and J. V. NEUMANN have developed an extension theory, which answers the question in what way a symmetric and closed operator can be extended to a self-adjoint operator, that is, under which conditions a symmetric and closed but *not* self-adjoint operator can be extended in order the extension to be self-adjoint. Moreover, the uniqueness of such an extension if of fundamental interest.

Remember first that for a densely defined and closed symmetric operator $A$ the relation $\mathfrak{D}(A) \subset \mathfrak{D}(A^*)$ holds. But even more is valid, since due to VON NEUMANN the domain of definition $\mathfrak{D}(A^*)$ of the

adjoint operator can be represented as sum of three complementary subspaces [3] :

$$\mathfrak{D}(A^*) = \mathfrak{D}(A) \oplus \mathfrak{N}(A^* + i) \oplus \mathfrak{N}(A^* - i). \tag{3.1}$$

Thus the linear operators $A^* + i$ and $A^* - i$ are of special interest. The numbers

$$n_+ := \dim \mathfrak{N}(A^* + i), \qquad n_- := \dim \mathfrak{N}(A^* - i)$$

are called the *defect indices* of $A$, where the value $n_\pm = \infty$ is admissible, too. One easily derives from (3.1) that a symmetric and closed linear operator $A$ with defect indices $(n_+, n_-)$ can be extended to a self-adjoint operator iff $n_+ = n_-$, and that this operator is self-adjoint iff in addition $n_+ = n_- = 0$ holds.

To extend in case of $n_+ = n_-$ the symmetric linear operator $A$ to a self-adjoint one, it suffices to have a isometric and linear mapping

$$U : \mathfrak{N}(A^* + i) \to \mathfrak{N}(A^* - i).$$

Then, if $U$ is given in such a way, we define $A_s x := Ax$ for $x \in \mathfrak{D}(A)$ and

$$A_s z := iz + -iUz \text{ for } z \in \mathfrak{N}(A^* + i),$$

which implies that $A_s$ is symmetric and hence forms a symmetric extension of $A$. One can show that $A_s = A^*$, so $A_s$ is even self-adjoint. The set of all self-adjoint extensions is given by the set of isometric linear operators $U$ as defined above.

It is often not easy to find a self-adjoint extension of a symmetric linear operator, But it is helpful to start with the symmetric sesquilinear form associated with a symmetric linear operator $A$ to derive a self-adjoint operator $A_{sa}$ extending $A$.

For $\mathfrak{a}$ being a symmetric sesquilinear form on a Hilbert space $H$, we define its *adjoint form* $\mathfrak{a}^*$ by virtue of

$$\mathfrak{a}^*(u, v) := \overline{\mathfrak{a}(v, u)} \qquad (u, v \in H).$$

Clearly, if $\mathfrak{a}$ is continuous, then its adjoint $\mathfrak{a}^*$ is so. Moreover, a sesquilinear form $\mathfrak{a}$ is called *symmetric*, if $\mathfrak{a} = \mathfrak{a}^*$. A well-known example of a symmetric form is the inner product $\langle \cdot, \cdot \rangle_H$ in a Hilbert space $H$, which differs from arbitrary symmetric sesquilinear

[3] The subspaces $V_1$, $V_2$ of $V$ are called a complementary decomposition of $V$ (notation: $V = V_1 \oplus V_2$), if $\langle x, y \rangle = 0$ for all $x \in V_1$, $y \in V_2$ and $V$ is generated by $V_1 \cup V_2$.

forms by positive definiteness.

THEOREM 3.1. If $\mathfrak{a}$ is symmetric, then $A_{\mathfrak{a}}$ is self-adjoint and even half-bounded with lower bound $\lambda_0$.

In the following we want to elaborate that each half-bounded linear operator can be extended to a self-adjoint one by means of its associated sesquilinear form.

We first consider a symmetric sesquilinear form $\mathfrak{a} : \mathfrak{D}(\mathfrak{a}) \times \mathfrak{D}(\mathfrak{a}) \to \mathbb{C}$, whose domain of definition $\mathfrak{D}(\mathfrak{a})$ is a dense subspace in $H$. Recall that $\mathfrak{a}$ is said to be symmetric if

$$\overline{\mathfrak{a}(y,x)} = \mathfrak{a}(x,y) \quad \text{for all } x,y \in \mathfrak{D}(\mathfrak{a}),$$

thus $\mathfrak{a}(x,x)$ is real-valued for $x \in \mathfrak{D}(\mathfrak{a})$. Just as for operators, such a form is said to be *half-bounded* or *bounded below* if

$$\mathfrak{a}[x] := \mathfrak{a}(x,x) \geq \theta \|x\|^2 \text{ for all } x \in \mathfrak{D}(\mathfrak{a})$$

and some $\theta \in \mathbb{R}$. Clearly, a half-bounded linear operator $A$ with domain of definition $\mathfrak{D}(A) \subset H$ gives rise to a half-bounded sesquilinear form $\mathfrak{a}_A$ by letting

$$\begin{cases} \mathfrak{D}(\mathfrak{a}_A) := \mathfrak{D}(A), \\ \mathfrak{a}_A(x,y) := \langle Ax, y\rangle \quad \text{for } x,y \in \mathfrak{D}(\mathfrak{a}_A). \end{cases}$$

But $\mathfrak{a}_A$ only becomes into an inner product on the subspace $\mathfrak{D}(\mathfrak{a}_A)$, if $\theta > 0$ or a further term is added to make $\mathfrak{a}_A$ into a positive definite form. In fact, let $A$ be half-bounded with lower bound $\theta$, then for each $\lambda > -\theta$ the subspace $\mathfrak{D}(\mathfrak{a}_A)$ becomes into a pre-Hilbert space with inner product $[\cdot,\cdot]_{\mathfrak{a},\lambda}$ given by

$$[x,y]_{\mathfrak{a}_A,\lambda} := \langle Ax, y\rangle + \lambda \langle x, y\rangle, \quad x,y \in \mathfrak{D}(\mathfrak{a}_A).$$

and derived norm

$$\|x\|_{\mathfrak{a}_A,\lambda} := \sqrt{[x,y]_{\mathfrak{a}_A,\lambda}} = \sqrt{\langle Ax,x\rangle + \lambda \cdot \|x\|^2} = \sqrt{\langle Ax + \lambda x, x\rangle}, \tag{3.2}$$

called the *energetic norm* induced by the operator $A$. It should be noted that if $\mu > -\theta$ is another real number, then $\|x\|_{\mathfrak{a}_A,\lambda}$ and $\|x\|_{\mathfrak{a}_A,\mu}$ are equivalent norms, that is, they own the same Cauchy sequences in their common domain of definition.

However, the domain $\mathfrak{D}(\mathfrak{a}_A)$ of the form $\mathfrak{a}_{A,\lambda}$ is not necessarily a complete normed space with respect to (3.2). Hence we define the

space $H_{\mathfrak{a}_A,\lambda}$ to be the completion of the domain $\mathfrak{D}(\mathfrak{a}_A)$ with respect to the topology induced by the norm $\|\cdot\|_{\mathfrak{a}_A,\lambda}$:

$$H_{\mathfrak{a}_A,\lambda} := \overline{\mathfrak{D}(\mathfrak{a}_A)}_{\|\cdot\|_{\mathfrak{a}_A,\lambda}}.$$

The sesquilinear form $[\cdot,\cdot]_{\mathfrak{a}_A,\lambda}$, defined on the subspace $\mathfrak{D}(\mathfrak{a}_A)$, may be extended to the space $H_{\mathfrak{a}_A,\lambda}$ by

$$\overline{[x,y]_{\mathfrak{a}_A,\lambda}}(x,y) := \lim_{n\to\infty} [x_n,y_n]_{\mathfrak{a}_A,\lambda}\,, \tag{3.3}$$

where $(x_n)$ and $(y_n)$ are approximating sequences of $(x,y)\in H_{\mathfrak{a}_A,\lambda}\times H_{\mathfrak{a}_A,\lambda}$ with respect to the energetic norm and with $x_n, y_n \in \mathfrak{D}(\mathfrak{a}_A)$ for all $n\in\mathbb{N}$. Moreover, it can be shown that first the definition of the expression $\overline{[x,y]}_{\mathfrak{a}A,\lambda}$ is independent from the choice of such sequences and second $H_{\mathfrak{a}_A,\lambda} = H_{\mathfrak{a}_A,\mu}$ holds for all $\lambda,\mu > -\theta$, which gives rise to introduce $H_A := H_{\mathfrak{a}_A,\lambda}$ for some $\lambda > -\theta$.

The set $H_A$ endowed with the inner product (3.3) forms a Hilbert space in its own right and is called the *energetic space* of the half-bounded operator $A$. Its elements can be identified with elements in Hilbert space $H$: let $x_0 \in H_{\mathfrak{a}_A}$, then there is $(x_n)$ with elements $x_n \in \mathfrak{D}(\mathfrak{a}_A)$ such that $(x_n)$ is Cauchy with respect to the energetic norm. But then $(x_n)$ is also Cauchy with respect to the norm $\|\cdot\|$ of the space $H$, thus there is $x_1 \in H$ such that $\|x_1 - x_n\| \to 0$. The mapping $x_0 \in H_A \mapsto x_1 \in H$ proves to be injective and hence forms a continuous embedding of $H_A$ into $H$, so we can regard $(H_A, H)$ as a pair of Hilbert spaces in the sense of section 12.

The operator $A_F$ associated with the form $\mathfrak{a}_{A,\lambda}$ surely forms an extension of $A$ and proves to be self-adjoint by theorem 8.2. It is called the FRIEDRICHS *extension* of $A$. Moreover, the domain of definition of the operator $A_F$ is given by the linear subspace $\mathfrak{D}(A_F) = H_A \cap \mathfrak{D}(A^*)$.

The operator $A_F$ owns a remarkable property: if one adds to the half-bounded operator $A$ the operator $\lambda 1$ ($\lambda\in\mathbb{R}$), then the FRIEDRICHS extensions of $A$ and $A+\lambda 1$ are related to each other by the equality $(A+\lambda 1)_F = A_F + \lambda 1$.

The FRIEDRICHS extension method is an important tool to construct self-adjoint exensions of half-bounded linear operators, which can be applied to various linear partial differential operators appearing in applications, see part III for some examples of such constructions.

## §4. Perturbation Theory

*Perturbation theory* deals with operators of the form $A+B$, where $A$ is assumed to be self-adjoint and $B$ to be symmetric with domain of definition $\mathfrak{D}(B)$ being a superset of $\mathfrak{D}(A)$, so $A+B$ is well defined with domain of definition $\mathfrak{D}(A)$. It is the aim to decide under which conditions on the operator $B$ the sum $A+B$ proves to be self-adjoint. This question is especially of interest for applications in quantum mechanics.

Let $H$ be a complex Hilbert space and the operator $A:\mathfrak{D}(A)\to H$ be self-adjoint. Then the following concept is useful: a symmetric operator $B:\mathfrak{D}(S)\to H$ with $\mathfrak{D}(A)\subset\mathfrak{D}(B)$ is called *$A$-bounded*, if there are constants $\theta, c\geq 0$ such that

$$\|Bx\| \leq \theta\|Ax\| + c\|x\| \quad \text{for all } x\in\mathfrak{D}(A). \tag{4.1}$$

In order to ensure $A$-boundedness of $B$, it is sufficient to ensure (4.1) only for elements $x$ in a core $A_0$ of $A$ (a *core* of a closed operator $A$ is any restriction $A_0$ to a smaller domain, whose closure coincides with $A$).

The number $\theta$ is called the *$A$-bound* of $B$. Notice that if $B$ is a bounded operator, then it is $A$-bounded with $a<1$ for any $A$. In addition, the validness of the estimate

$$\|Bx\|^2 \leq \theta^2\|Ax\|^2 + c^2\|x\|^2 \quad \text{for all } x\in\mathfrak{D}(A).$$

is a sufficient condition for (4.1) to be true.

Let $A$ be self-adjoint in Hilbert space $H$ and $B$ a further symmetric operator having domain $\mathfrak{D}(B)\supset\mathfrak{D}(A)$. The following KATO-RELLICH-*theorem* provides a criterion on the self-adjointness of the operator $A+B$, which has been independently found by T. KATO and F. RELLICH:

THEOREM 4.1. Let the linear operator $A$ be self-adjoint and the linear operator $B$ with $\mathfrak{D}(B)\supset\mathfrak{D}(A)$ be symmetric and $A$-bounded with $\theta<1$. Then $A+B$ with domain of definition $\mathfrak{D}(A)$ is self-adjoint.

PROOF: Clearly, $A+S$ is symmetric. For $\mu>0$ we have

$$\left\|(A-i\mu)^{-1}x\right\| \leq \frac{1}{\mu}x, \quad (x\in H).$$

Moreover, an elementary computation yields $\|Ax - i\mu x\|^2 = \|Ax\|^2 + |\mu|^2 \|x\|^2$. Let $y := (A - i\mu)x$, then $x = (A - i\mu)^{-1}y$, and we obtain

$$\|y\|^2 = \left\|A(A - i\mu)^{-1}y\right\|^2 + |\mu|^2 \|x\|^2 \geq \left\|A(A - i\mu)^{-1}y\right\|^2$$

or equivalently $\|y\| \geq \left\|A(A - i\mu)^{-1}y\right\|$. Now consider for any $y \in H$ the operator equation

$$z + S(A - i\mu)^{-1}z = y. \tag{4.2}$$

Since $S$ is $A$-bounded,

$$\left\|S(A - i\mu)^{-1}z\right\| \leq \theta \left\|A(A - i\mu)^{-1}z\right\| + c \left\|(A - i\mu)^{-1}z\right\|.$$

Inserting the estimations above into this equation, we obtain

$$\left\|S(A - i\mu)^{-1}z\right\| \leq \theta \|z\| + \frac{c}{|\mu|} \|z\|,$$

which implies that the operator $S(A - i\mu)^{-1}$ is bounded. Especially for $\mu$ small enough it follows that $S(A - i\mu)^{-1}$ is a contraction, so by Banach*'s fixpoint theorem* the equation (4.2) has a unique solution $z$. Finally, letting $v := (A - i\mu)^{-1}z$ leads to

$$(A + S - i\mu)v = Av - i\mu x + Sx = z + S(A - i\mu)^{-1}z = y,$$

which implies that $\mathfrak{R}(A + S - i\mu) = H$. But this is the range condition for the operator $A + S$. □

A similar result is obtained for the question, whether *essentially self-adjointness* of a linear operator $A$ is preserved by a symmetric and $A$-bounded perturbation $B$:

Theorem 4.2. Let $A : \mathfrak{D}(A) \to H$ be essentially self-adjoint and assume that $B$ is symmetric and $A$-bounded, that is, $\mathfrak{D}(A) \subset \mathfrak{D}(B)$ and (4.1) holds with bound $\theta < 1$. Then the operator $A + B$ is essentially self-adjoint on $\mathfrak{D}(A)$, where $\overline{A + B} = \overline{A} + \overline{B}$.

## §5. The Spectrum of Symmetric and Self-Adjoint Operators

A great portion of spectral theory deals with the study of the resolvent set and the spectrum of certain classes of closed or continuous linear operators. Important examples are constituted by the classes of symmetric and self-adjoint operators.

In order to investigate the spectrum of a symmetric operator, we first start with the study of linear operators of the form $A - \lambda$, where $A$ is closed, densely defined and symmetric.

Remember that a number $\lambda \in \mathbb{C}$ with $\operatorname{Im} \lambda \neq 0$ does not belong to the numerical range $\Theta(A)$ of $A$, hence $\lambda$ is contained in the field $\Delta(A)$ of semi-Fredholm regularity. This means that $A - \lambda$ is bounded below, and since we assume $A$ to be closed, $A - \lambda$ is injective and normally solvable by theorem 5.2, thus upper semi-Fredholm with $\alpha(A - \lambda) = 0$. By the punctured neighbourhood theorem 5.3, the Fredholm index $\chi(A - \lambda)$ is locally constant for each fixed $\lambda \notin \mathbb{R}$. In addition, it can be shown that it is even globally constant in the upper and lower complex half-plane, respectively.

Before proceeding with the study of the range of $A - \lambda$, notice that by using the orthogonal relations (4.1) this space and the kernel of $A^* - \overline{\lambda}$ are coupled by

$$H = \mathfrak{R}(A - \lambda) \oplus \mathfrak{N}(A^* - \overline{\lambda}).$$

We now distinguish between two cases with respect to surjectivity of $A - \lambda$, or equivalently, to the injectivity of $A^* - \overline{\lambda}$:

a) either $\mathfrak{R}(A - \lambda) = H$, which implies $\mathfrak{N}(A^* - \overline{\lambda}) = \{0\}$, then $A - \lambda$ is invertible, $\lambda$ is a regular value of $A$, and $\mathbb{H}^+$ and $\mathbb{H}^-$ belongs to $\rho(A)$, respectively,
b) or $\mathfrak{R}(A - \lambda) \neq H$, then $\mathfrak{N}(A^* - \overline{\lambda}) \neq \{0\}$ and $\lambda$ belongs to the residual spectrum $\sigma_r(A)$, so $\mathbb{H}^+$ and $\mathbb{H}^-$ are subspaces of $\sigma_r(A)$, respectively.

As it has already been indicated, both cases may each happen for the upper and the lower complex half plane $\mathbb{H}^+$ and $\mathbb{H}^-$. So in combination there are four possibilities how the spectrum $\sigma(A)$ of a closed symmetric operator $A$ looks like:

1. $\sigma(A) = \mathbb{C}$ in case of $\mathbb{H}^+ \cup \mathbb{H}^- \subset \sigma_r(A)$,
2. $\sigma(A) = \overline{\mathbb{H}^+}$ in case of $\mathbb{H}^+ \subset \sigma_r(A)$ and $\mathbb{H}^- \subset \rho(A)$,
3. $\sigma(A) = \overline{\mathbb{H}^-}$ in case of $\mathbb{H}^- \subset \sigma_r(A)$ and $\mathbb{H}^+ \subset \rho(A)$,
4. $\sigma(A) \subset \mathbb{R}$ in case of $\mathbb{H}^+ \cup \mathbb{H}^- \subset \rho(A)$.

In summary, we have $\sigma(A) \in \{\mathbb{C}, \overline{\mathbb{H}^+}, \overline{\mathbb{H}^-}, S\}$, where $S$ is any subset of the real line. But if the resolvent set $\rho(A)$ contains at least one element $\xi \in \mathbb{R}$, then only the last case in the previous enumeration can occur. In fact, the property $\sigma(A) \subset \mathbb{R}$ leads us

again to the notion of *self-adjointness*, since both properties prove to be equivalent to each other.

Our next subject is the study of the spectrum of a self-adjoint operator. The first result is the claim that self-adjoint operators differ from other operators by the specific location of their spectra:

THEOREM 5.1. A closed linear operator $A$ is self-adjoint if and only if $\sigma(A) \subseteq \mathbb{R}$, and in this case we have for $\operatorname{Im}\xi \neq 0$ the resolvent estimation

$$\|R(A,\xi)\| \leq \frac{1}{|\operatorname{Im}\xi|}.$$

Indeed, $\sigma(A)$ being a subset of the real line means that $\mathbb{H}^+$ and $\mathbb{H}^-$ belong to the resolvent set of $A$, which is true if and only if the range condition is fulfilled, that is, $A \pm i$ are both surjective. The resolvent estimation is a consequence of the inequality $\|R(A,\xi)\| \leq \delta(\xi,\sigma(A))^{-1}$, where $\delta(\xi,\sigma(A)$ denotes the positive distance between $\xi \in \rho(A)$ and the spectrum of $A$.

Remember the subsets of the spectrum $\sigma(A)$ of a linear operator $A$ defined in section 7. They have been introduced by the criterion how the perturbed operator $A-\lambda$ fails to be bijective. This approach has led us to the notion of *point spectrum* $\sigma_p(A)$, *continuous spectrum* $\sigma_c(A)$ and *residual spectrum* $\sigma_r(A)$, respectively.

The next result shows that the division of the spectrum for self-adjoint operators is simpler than in the arbitrary case:

THEOREM 5.2. If $A$ is self-adjoint, then the residual spectrum $\sigma_r(A)$ is empty, thus $\sigma(A) = \sigma_p(A) \cup \sigma_c(A)$.

This can be concluded by the orthogonality relation $\mathfrak{N}(A) = \mathfrak{R}(A)^\perp$, so no spectral value may belong to the residual spectrum $\sigma_r(A)$.

When dealing with a self-adjoint linear operator $A$, it is advantageous to breakup the set $\sigma(A)$ into other subsets as for arbitrary linear operators.

First, the *pure point spectrum* $\sigma_{pp}(A)$ is

$$\sigma_{pp}(A) := \{\lambda \in \mathbb{C} : \mathfrak{R}(A-\lambda) = \overline{\mathfrak{R}(A-\lambda)} \neq H\}.$$

Second, by *points embedded in continuum* are meant the elements of the set

$$\sigma_{pec}(A) := \{\lambda \in \mathbb{C} : \mathfrak{R}(A-\lambda) \neq \overline{\mathfrak{R}(A-\lambda)} \neq H\}.$$

Both definitions do not require $A-\lambda$ to be injective, so $\sigma_{pp}(A)$ and $\sigma_{pec}(A)$ possibly contain eigenvalues of $A$. We join both subsets together and redefine the point specturm $\sigma_p(A) := \sigma_{pp}(A) \sqcup \sigma_{pec}(A)$, that is, $\sigma_p(A)$ for $A$ being self-adjoint consists of all $\lambda \in \mathbb{C}$, for which $\overline{\mathfrak{R}(A-\lambda)} \neq H$. This leads to the following

THEOREM 5.3. The point spectrum $\sigma_p(A)$ is exactly the set of eigenvalues of $A$.

Third, the *purely continuous spectrum* is the set

$$\sigma_{pc}(A) := \{\lambda \in \mathbb{C} : \mathfrak{R}(A-\lambda) \neq \overline{\mathfrak{R}(A-\lambda)} = H\}.$$

We alternatively introduce the *continuous spectrum* $\sigma_c(A) := \sigma_{pec}(A) \sqcup \sigma_{pc}(A)$, which of course is the set

$$\{\lambda \in \mathbb{C} : \mathfrak{R}(A-\lambda) \neq \overline{\mathfrak{R}(A-\lambda)}\}.$$

In opposite to the former given definition of $\sigma_c(A)$ (see the paragraph on spectral theory), this set may again contain elements $\lambda$, for which $A-\lambda$ is not injective.

For a self-adjoint operator $A$ it is also customary to denote $\sigma_{pp}(A)$ the *discrete spectrum* and its complement relatively to $\sigma(A)$ the *essential spectrum*, i. e.

$$\sigma_{ess}(A) := \sigma(A) \setminus \sigma_{pp}(A),$$

so we have $\sigma_c(A) \subset \sigma_{ess}(A)$.

The *discrete spectrum* of $A$ consists of those values $\lambda \in \mathbb{C}$, for which there is a terminating sequence $\{e_n\}$ in $H$ consisting of pairwise disjoint unit elements $e_n$. The essential spectrum of $A$ consists of those values $\lambda \in \mathbb{C}$, for which there is a non-terminating sequence $\{e_n\}$ in $H$ consisting of pairwise disjoint unit elements $e_n$, called a *Weyl sequence* for $\lambda \in \sigma_{ess}(A)$, such that

$$\lim_{n\to\infty} \|Ae_n - \lambda e_n\| = 0. \tag{5.1}$$

In summary, any number $\lambda \in \mathbb{C}$ belongs to the spectrum of a self-adjoint operator $A$, if it owns a (possibly terminating) Weyl sequence fulfilling (5.1).

## §6. Projection-Valued Measures and Spectral Integrals

The spectral theorem is definitely the most important assertion with respect to self-adjoint operators. Another way to claim this theorem reads as follows: each self-adjoint linear operator $A$ is a *spectral operator*, whose underlying *projection-valued measure* $P$ can be retrieved from $A$. In the following we want to make this assertion precise by going on step by step.

As the projection theorem shows, projection mappings acting in a Hilbert space $H$ are in one-to-one relationship to closed linear subspaces in $H$. So it is nearby to call two projections $\pi_1, \pi_2$ *orthogonal* to each other, if their related closed subspaces have this property, and in this case the identity $\pi_1 \circ \pi_2 = \pi_2 \circ \pi_1 = 0$ holds, where $0$ is the projection related with the trivial subspace $\{0\} \subset H$.

In general, a projection mapping $\pi$ has the following properties:

1. $\pi$ is idempotent: $\pi \circ \pi = \pi$.
2. $\pi$ is linear on $H$.
3. $\pi$ is bounded with $\|\pi\| \leq 1$, hence $\pi \in \mathscr{L}(H)$.
   If $\mathcal{A}_\pi := \pi(H) = \{0\}$, then $\|\pi\| = 0$, else $\|\pi\| = 1$. The only possible eigenvalues of $\pi$ are the real numbers 0 and 1.
4. Let $\mathcal{A}_1, \mathcal{A}_2$ be closed subspaces of $H$ with $\mathcal{A}_1 \subseteq \mathcal{A}_2$, then for the associated projections the equation $\pi_{\mathcal{A}_1} \circ \pi_{\mathcal{A}_2} = \pi_{\mathcal{A}_2} \circ \pi_{\mathcal{A}_1} = \pi_{\mathcal{A}_1}$ holds. In this case, the operator $\pi_{\mathcal{A}_2} - \pi_{\mathcal{A}_1}$ is also a projection, and related subspace is given by the set $\mathcal{A}_2 \ominus \mathcal{A}_1 := \mathcal{A}_2 \cap \mathcal{A}_1^\perp$.
5. $\pi = \pi^*$, that is, $\pi$ is symmetric. Conversely, each symmetric and idempotent linear operator is a projection.
6. If $\mathcal{A}$ is a *finite*-dimensional subspace of $H$, then $\mathcal{A}$ is closed, and the projection $\pi_{\mathcal{A}}$ can be represented by means of a set of base vectors $e_1, \ldots, e_n$ in $\mathcal{A}$, namely

$$\pi_{\mathcal{A}} x \;=\; \sum_{i=1}^{n} \langle x, e_i \rangle \, e_i .$$

Finally, let $\mathscr{P}(H)$ be the set of projections in $H$, then there is a partial order relation $\geq$ on $\mathscr{P}(H)$, defined by means of inclusion between the related closed subspaces:

$$\pi_{\mathcal{A}_1} \geq \pi_{\mathcal{A}_2} \quad :\Longleftrightarrow \quad \mathcal{A}_2 \subseteq \mathcal{A}_1 .$$

By a *projection-valued measure* we mean a mapping $P : \mathcal{B}_{\mathbb{R}} \to \mathscr{L}(H)$, where $\mathcal{B}_{\mathbb{R}}$ is the ordinary Borel $\sigma$-algebra in $\mathbb{R}$ and $P$ fulfills the

following properties:

(P-1) $P(\Omega)$ is a projection operator for each $\Omega \in \mathcal{B}_{\mathbb{R}}$;

(P-2) $P(\mathbb{R}) = 1_H$, the identity operator in $H$;

(P-3) $P$ is *$\sigma$-additive*, that is, if $\Omega_1, \Omega_2, \ldots$ are measurable sets in the real line and $\Omega := \bigsqcup_{n \in \mathbb{N}} \Omega_n$, then

$$P(\Omega) = \lim_{n \to \infty} \sum_{k=1}^{n} P(\Omega_k),$$

where this kind of convergence has to be understood in the sense of the strong operator topology in $\mathscr{L}(H)$.

The projection-valued measure $P$ is also called a *spectral measure* or a *resolution of the identity*. Notice that intervals of the form $(-\infty, \lambda]$ are Borel-measurable, so each projection $P((-\infty, \lambda])$ is well-defined. In fact, the one-parameter family of these projections constitutes a projection-valued measure $P$.

In the sequel it is our aim to introduce a *functional calculus* for measurable functions $f : \mathbb{R} \to \mathbb{C}$ with respect to a projection-valued measure $P$.

Let $B(\mathbb{R})$ be the linear space of *bounded* measurable functions $f : \mathbb{R} \to \mathbb{C}$ endowed with the norm

$$\|f\|_\infty := \sup_{x \in \mathbb{R}} |f(x)| < \infty,$$

which in fact is a Banach space. Then a function $e \in B(\mathbb{R})$ is called *simple*, if it adopts only finitely many values $e_1, \ldots, e_N$. Moreover, the set

$$S(\mathbb{R}) := \{\, f : \mathbb{R} \to \mathbb{C} : f \text{ is simple} \,\}$$

forms dense subspace of $B(\mathbb{R})$, and the supremum norm of $e \in S(\mathbb{R})$ is given by $\|e\|_\infty = \max\{|e_1|, \ldots, |e_N|\}$.

After these preliminaries, the setup of the spectral integral will be carried out by the following five steps:

1. First of all, each spectral measure $P$ gives rise to a complex and real-valued Borel measure on the real line. For arbitrary $x, y \in H$ let the complex measure $\mu_{x,y}$ be defined by

$$\begin{cases} \mu_{x,y} : \mathcal{B}_{\mathbb{R}} \to \mathbb{C}, \\ \Omega \mapsto \mu_{x,y}(\Omega) := \langle x, P(\Omega) y \rangle. \end{cases}$$

Due to the symmetry of $P(\Omega)$, the derived measure $\mu_x(\Omega) :=$

$\mu_{x,x}(\Omega)$ for $\Omega \in \mathcal{B}_{\mathbb{R}}$ is real-valued.

2. We next make sense of the expression $\int_{\mathbb{R}} f\,dP$ for measurable functions $f : \mathbb{R} \to \mathbb{C}$. Let us start with the case, where $f$ is *simple*, that is, $f$ takes values in the finite set $\{f_1, \ldots, f_N\}$. Define

$$\int_{\mathbb{R}} f\,dP := \sum_{n=1}^{N} f_n P(\chi^{-1}(f_n)),$$

where $\chi^{-1}(f_n)$ is the preimage of $f_n$ in $\mathcal{B}_{\mathbb{R}}$. We further mention that

$$I_{0,P} : f \to \int_{\mathbb{R}} f\,dP$$

is a continuous linear mapping from the normed space $S(\mathbb{R})$ consisting of simple functions to the Banach space $\mathscr{L}(H)$ with operator norm $\|I_{0,P}\| = 1$.

3. We shall continue with the integration of *bounded measurable functions* with respect to $P$. Due to the B.L.T. theorem, there exists a unique extension $I_P$ from $I_{0,P}$ to the closure $B(\mathbb{R}) = \overline{S(\mathbb{R})}$, where $\|I_P\| = \|I_{0,P}\|$. The mapping $I_P$ is also linear and fulfills

$$\begin{cases} I_P(1_{\mathbb{R}}) = 1_H \\ I_P(fg) = I_P(f) \circ I_P(g) \qquad\qquad f, g \in B(\mathbb{R}), \\ I_P(\overline{f}) = I_P(f)^* \end{cases}$$

where $1_{\mathbb{R}}$ is the identity in $\mathbb{R}$, thus $I_P$ even forms a C*-algebra homomorphism between the function algebra $B(\mathbb{R})$ and the operator algebra $\mathscr{L}(H)$.

4. We will also define $I_P$ for an *arbitrary measurable function* $f : \mathbb{R} \to \mathbb{C}$. But this is only possible for those elements $x \in H$ fulfilling

$$\int_{\mathbb{R}} |f|^2\,d\mu_x < \infty. \tag{6.1}$$

The set $D_f$ of elements $x \in H$ satisfying (6.1) forms a linear subspace lying dense in $H$. For $f$ we introduce an approximating sequence of functions $f_n \in B(\mathbb{R})$ given by

$$f_n := \chi_{\{t \in \mathbb{R} : |f(t)| \leq n\}} \cdot f \quad (n \in \mathbb{N}).$$

One can show that $f_n$ is Cauchy in $L^2(\mathbb{R})$, hence $I_P(f_n)$ is Cauchy in $H$, where $I_P$ is the previously defined operator acting on the

space $B(\mathbb{R})$.

5. To the end, for a fixed measurable function $f$ we may define an operator $S(P,f)$, called the *spectral operator* with respect to $P$, by virtue of

$$\int_{\mathbb{R}} f\,dP : D_f \mapsto H\,, \qquad x \in D_f \mapsto (\int_{\mathbb{R}} f\,dP\,)\,x \;:=\; y\,, \tag{6.2}$$

which enjoys the property

$$(\int_{\mathbb{R}} f\,dP)^* \;=\; \int_{\mathbb{R}} \overline{f}\,dP.$$

Especially for $f = 1_{\mathbb{R}}$ the self-adjoint operator $S_P := S(P,1)$ is defined by

$$\mathfrak{D}(S_P) := D_{1_{\mathbb{R}}}, \quad S_P x \;:=\; (\int_{\mathbb{R}} 1_{\mathbb{R}}\,dP)\,x.$$

## §7. The Spectral Theorem

In linear algebra the *main axis theorem* quotes that each hermitean matrix is diagonalizable and has purely real-valued eigenvalues. The generalization of this assertion, substituting hermitean matrices by densely defined self-adjoint operators acting in abstract Hilbert space, leads to the so-called *spectral theorem.*

Remember from integration theory that for a given measure space $(X,\mathcal{B},\mu)$ the linear space $L^2(X)$ of measurable and square-integrable functions $f : X \to \mathbb{C}$ forms a Hilbert space with inner product

$$\langle f,g\rangle \;:=\; \int_X f\,\overline{g}\,d\mu.$$

If $m : X \to \mathbb{C}$ is a measurable function, then the linear mapping $M_m$ defined by

$$\begin{cases} \mathfrak{D}(M_m) & := \{\, f \in L^2(X) : m\cdot f \in L^2(X)\,\}, \\ (M_m f)(x) & := m(x)\cdot f(x) \;\text{ for } x \in X \text{ and } f \in \mathfrak{D}(M_m) \end{cases}$$

is called a *multiplication operator* with *generator* $m$. It turns out that the mapping $M_m$ is symmetric iff $m$ is real-valued, and since the domains of definition of the operators $M_m$ and $M_m^*$ coincide, each multiplication operator with real-valued generator is self-adjoint.

The *spectral theorem for self-adjoint operators* claims that the reverse assertion is also true:

Theorem 7.1. To each self-adjoint operator $A : \mathfrak{D}(A) \subset H \to H$ there is assigned

(1) a measure space $(X, \mathcal{B}, \mu)$,
(2) a multiplication operator $M_m$ in $L^2(X)$ with generator function $m : X \to \mathbb{R}$,
(3) a bijective and isometric linear operator $U : H \to L^2(X)$

such that $A = U^{-1} \circ M_m \circ U$ holds.

The existence of the operator $U$ gives rise to speak of *unitary equivalence* between the operators $A$ and $M_m$. If some operators $A_1$ and $A_2$ are unitary equivalent, then their spectra $\sigma(A_1)$ and $\sigma(A_2)$ are identical.

In summary, theorem 7.1 implies that an operator $A$ is self-adjoint in Hilbert space $H$ iff it is unitary equivalent to a real-valued multiplication operator $M_m$ on $L^2(X)$, where $X$ is the base set of a suitable chosen measure space $(X, \mathcal{B}, \mu)$.

Using the notion of projection-valued measure, we are now able to quote a further variant of the *spectral theorem for self-adjoint operators*:

Theorem 7.2. Given a self-adjoint linear operator $A$, there is a uniquely defined projection-valued measure $P$ such that $A$ is the spectral operator related to $P$:

$$A = \int_{\mathbb{R}} 1 \, dP.$$

This version of the spectral theorem is extremely useful to define operator-valued mappings $f : \sigma(A) \to f(A)$ by

$$f(A) := \int_{\mathbb{R}} f \, dP,$$

where $f(A)$ will become into a densely defined operator acting in $H$. Indeed, passing from a self-adjoint linear operator to its family of projection-valued measures and then to the spectral operators defined by (6.2) is nothing but a *functional calculus* for the class of self-adjoint operators. For example, if $A$ is self-adjoint and *positive* at once, that is, $\langle Ax, x \rangle \geq 0$ for all $x \in \mathfrak{D}(A)$, then we may define operators like $\sqrt{A}$, $A^n$, $\exp(A)$, $\sin(A)$, $\cos(A)$ and so on.

It is also possible to formulate the spectral theorem by means of so-called *spectral resolutions* instead of projection-valued measures. Preliminary, to each projection-valued measure $P$ in Hilbert space $H$ there is related the concept of *spectral resolution*, which is nothing but a one-parameter family $\{P_\lambda\}_{\lambda\in\mathbb{R}}$ of linear operators $P_\lambda : H \to H$ such that

(R-1) $P_\lambda$ is a projection for each $\lambda\in\mathbb{R}$.
(R-2) For each element $x\in H$ the mapping $\lambda \mapsto P_\lambda x$ is continuous from the right.
(R-3) $\lambda < \mu$ implies $P_\lambda \le P_\mu$, that is, $\langle P_\lambda x, x\rangle \le \langle P_\mu x, x\rangle$ for all $x\in H$.
(R-4) $\lim_{\lambda\to-\infty} P_\lambda\, x = 0$ and $\lim_{\lambda\to\infty} P_\lambda\, x = x$ for all $x\in H$.

Each projection $P_\lambda$ of a given spectral resolution is related to a closed linear subspace $\mathcal{A}_\lambda \subset H$ and the difference $P_\mu - P_\lambda$ for $\mu \ge \lambda$ also forms a projection related to the closed subspace $\mathcal{A}_{\mu+0} \ominus \mathcal{A}_\lambda$.

Now, using spectral resolutions instead of projection-valued measures, the spectral theorem claims that each self-adjoint operator $A$ owns a spectral resolution $\{P_\lambda\}_{\lambda\in\mathbb{R}}$ such that

$$\mathfrak{D}(A) \;=\; \{\, x\in H : \int_{\mathbb{R}} \lambda^2\, d\langle P_\lambda\, x, x\rangle < \infty \,\},$$

$$\langle A\, x, y\rangle \;=\; \int_{\mathbb{R}} \lambda\, d\langle P_\lambda\, x, y\rangle \quad \text{for } x\in\mathfrak{D}(A) \text{ and } y\in H.$$

We next want to investigate the relationship between the shape of the spectrum and the spectral resolution of a self-adjoint operator.

In fact, the following theorem shows that the resolvent set and the different kinds of spectra of a self-adjoint linear operator $A$ are determined by its spectral resolution:

THEOREM 7.3. Let $A$ be self-adjoint in $H$ and $\mathcal{A}_\mu$ be the closed subspace in $H$ associated with the projection $P_\mu$ for some $\mu\in\mathbb{R}$. Then

1. $\mu\in\mathbb{R}$ belongs to the *point spectrum* $\sigma_p(A)$ iff $0 < \dim(\mathcal{A}_{\mu+0} \ominus \mathcal{A}_\mu) < \infty$.
2. $\mu\in\mathbb{R}$ belongs to the *continuous spectrum* $\sigma_c(A)$ iff for each $\epsilon > 0$ the subspace $\mathcal{A}_{\mu+\epsilon} \ominus \mathcal{A}_{\mu-\epsilon}$ is infinite-dimensional.
3. $\mu\in\mathbb{R}$ is a *regular value* of $A$ iff there is an open interval $(\alpha,\beta)\subset\mathbb{R}$ such that $\mu\in(\alpha,\beta)$ and $P_\mu$ is constant in $(\alpha,\beta)$.

Especially the resolvents $R(A,\xi)$ for $\xi \in \rho(A)$ of a self-adjoint operator $A$ can be represented by means of the spectral resolution:

$$R(A,\xi)\,x \;=\; \int_{-\infty}^{\infty} \frac{1}{\lambda-\xi}\, dP_\lambda x, \quad (x \in H,\ \operatorname{Im}\xi \neq 0).$$

On the other hand, Stone*'s formula* provides a constructive method to describe the spectral resolution of $A$ in an interval $[a,b]$ by means of the resolvents $R(A,\xi)$. Let $x \in H$ be arbitrary, then

$$\frac{1}{2}\{P[a,b]+P(a,b)\}\,x \;=\; \lim_{\epsilon \downarrow 0}\, \frac{1}{2\pi i} \int_a^b \{R(A,t+i\epsilon)\,x - R(A,t-i\epsilon)\,x\}\, dt.$$

## §8. Self-Adjoint Operators with Pure Point Spectrum

We now turn to linear operators $k$ acting in a *separable* Hilbert space $H$, which are compact and symmetric at the same time. Since compactness implies continuity, it may be assumed that $k$ is everywhere defined in $H$.

Of course, if an operator is compact-symmetric, the properties of compactness and symmetry complement, so we may quote that the spectrum $\sigma(k)$ of such an operator consists of a sequence $\{\lambda_n\}$ of *real* eigenvalues of $k$, which either terminates or converges to 0. Due to the compactness of $k$, the related eigenspaces $E_{\lambda_n}$ are finite-dimensional. By the Gram-Schmidt orthogonalization method, one can define a subset $\mathfrak{O}$ of the totality of eigenvectors of $k$ such that $\mathfrak{O}$ spans a dense subspace in $H$ and consequently constitutes an *orthonormal base* in $H$.

It is the assertion of the Hilbert-Schmidt *spectral theorem*, also often called the *discrete spectral theorem*, that there is a representation of a symmetric-compact operator $k$ in a separable Hilbert space $H$ as a *countable* sum of operators $k_{\lambda_n}$, where each of them is non-trivial on a finite-dimensional subspace of $H$ and acts there like a simple multiplication operator $k_{\lambda_n} := \lambda_n 1$. In other words, the theorem provides an orthogonal decomposition of $H$ into a sequence of $k$-invariant subspaces:

Theorem 8.1. Let the Hilbert space $H$ be separable. Given a symmetric-compact linear operator $k : H \to H$, there is a null sequence $\{\lambda_n\}$ consisting of real eigenvalues and an orthonormal base in $H$ consisting of eigenvectors $(x_n)$ of $k$ such that for all $x \in H$ the following representation holds:

$$k\,x \;=\; \sum_{n=1}^{\infty} \lambda_n \,\langle x, x_n\rangle\, x_n. \tag{8.1}$$

We finally consider linear operators $A$ having the property $\sigma(A) = \sigma_p(A)$, that is, the continuous and residual spectrum of $A$ are both empty and $\sigma(A)$ only consists of eigenvalues. Such operators are of outstanding meaning in applied analysis, since they appear as linear differential operators defined on *bounded* domains $G \subset \mathbb{R}^d$.

A closed operator $A$ with domain of definition $\mathfrak{D}(A)$ is called *having compact resolvents*, if at least one resolvent $A-\xi$ with $\xi \in \rho(A)$ is compact. The resolvent equation (6.2) shows that in this case every other resolvent of $A$ is compact.

There is a useful criterion to check whether $A$ has this distinguished property:

THEOREM 8.2. Let the domain of definition $\mathfrak{D}(A) \subset X$ of a densely defined operator $A$ in $X$ be equipped with the graph norm defined by (3.1). Then $A$ has compact resolvents if and only if the natural embedding $\iota : (\mathfrak{D}(A), \|\cdot\|_A) \hookrightarrow (X, \|\cdot\|)$ is compact.

One also says that an operator having compact resolvents is an *operator with pure point spectrum*, since each $\lambda \in \sigma(A)$ is an eigenvalue of finite multiplicity. This follows from the fact that $x \in X$ is a solution of the equation $Ax = \lambda x$ if and only if $x$ solves the equation *in resolvent form*

$$(\lambda - \xi)^{-1} x \;=\; R(A,\xi)x\,, \quad \xi \in \rho(A),$$

so both equations are equivalent to other. Then the left hand side of this equation must be of the form $(\lambda - \xi)^{-1}x = \zeta_n(\xi)x$ for some $n \in \mathbb{N}$, where $\{\zeta_n(\xi)\}_{n\in\mathbb{N}}$ is the sequence of eigenvalues of $R(A,\xi)$. This implies that $\sigma(A)$ is a subset of the set $\{\,\xi + \zeta_n(\xi)^{-1} : n \in \mathbb{N}\,\}$.

It is quite natural to order the set of eigenvalues of $A$ by their moduli. Hence one obtains a sequence $\{\lambda_n\}$ of eigenvalues $\lambda_n$ such that

$$|\lambda_1| \;\le\; |\lambda_1| \;\le\; |\lambda_3| \,\ldots\, \to \infty.$$

With respect to the solution theory of the equation $Ax = y$, an operator $A$ is called *completely solvable*, if $A$ has compact resolvents and $\xi = 0$ is a regular value of $A$, so $A^{-1}$ is bounded and has range $\mathfrak{R}(A^{-1}) = \mathfrak{D}(A)$.

Let $A$ be a self-adjoint linear operator with pure point spectrum. Then each resolvent of $A$ is symmetric and compact, and by the Hilbert-Schmidt spectral theorem 8.1 the set of eigenvectors forms an orthonormal base in $H$. But these eigenvectors are precisely those, that also belong to the eigenvalues of $A$.

We summarize the preceeding considerations to the following *spectral theorem for self-adjoint operators with pure point spectrum*:

Theorem 8.3. Let $A$ with domain of definition $\mathfrak{D}(A) \subset H$ be self-adjoint and with pure point spectrum $\sigma(A) = \{\lambda_n : n \in \mathbb{N}\} \subset \mathbb{R}$. Then $|\lambda_n| \to \infty$ as $n \to \infty$, and

$$Ax = \sum_{n=1}^{\infty} \lambda_n \langle x, x_n \rangle x_n \quad \text{for} \quad x \in \mathfrak{D}(A) := \{x \in H : \sum_{n=1}^{\infty} \lambda_n^2 |\langle x, x_n \rangle|^2 < \infty\}.$$

This theorem is of fundamental meaning in the investigation of eigenvalue problems on bounded domains with respect to formally self-adjoint linear differential operators.

## §9. Evolution Equations Governed by Self-Adjoint Operators

The functional calculus for self-adjoint operators permits us to construct solutions for the homogeneous Cauchy problem of first and second order. For this we first have to generalize the concept of classical derivative for functions with values in a Banach space $X$.

The analysis of linear operator-differential equations In the Hilbert space setting can directly be attacked for the case, where the differential equations is governed by a positive self-adjoint operator. The following existence and uniqueness theorems all rely on the spectral theorem and the functional calculus for self-adjoint unbounded operators.

In the sequel we deal with first order problems and shall show how one can solve the homogeneous Cauchy problem *of first order*

$$\begin{cases} u'(t) = A\,u(t) \\ u(0) = u_0 \in \mathfrak{D}(A), \end{cases} \tag{9.1}$$

where $A$ is self-adjoint and positive.

Remember that a linear operator $A$ acting in a Hilbert space is called *positive*, if $A$ is symmetric and $\langle Ax, x \rangle \geq 0$ for all $x \in \mathfrak{D}(A)$.

If $A$ is positive, then any solution of the problem (9.1) satisfies the estimate

$$\|u(t)\| \leq \|u(0)\| \quad \text{for all } t > 0\,, \tag{9.2}$$

which is a consequence of the inequality $\|u(t)\|^2 = -2\langle Au, u\rangle \leq 0$ and causes the solution of the problem to be unique.

To verify the existence of a solution for (9.1), let $(P_\lambda)_{\lambda\in\mathbb{R}}$ be the spectral resolution of the self-adjoint operator $A$. Notice that $P_\lambda = 0$ for $\lambda < 0$, since $A$ is positive. Then the functional calculus for the continuous function $t \mapsto e^{-t}$ gives rise to the spectral operator

$$L_t(x) := e^{-At}x = \int_0^\infty e^{-\lambda t}\, dP_\lambda x \qquad (t \geq 0,\ x \in H). \tag{9.3}$$

Obviously, we have $L_0 = 1$, and (9.2) implies $\|L_t\| \leq 1$ for all $t > 0$.

In conclusion, the existence of the solution of (9.1) is ensured by the following

THEOREM 9.1. The function $u : t \in [0,\infty) \mapsto L_t(u_0)$ is continuous. Moreover, it is continuously differentiable in $(0,\infty)$. and forms a solution of (9.1).

We also apply this method to the homogeneous Cauchy problem *of second order*

$$\begin{cases} u''(t) = A\,u(t) \\ u(0) = u_0 \in \mathfrak{D}(A) \\ u'(0) = u_1 \in H, \end{cases} \tag{9.4}$$

where again the operator $A$ is assumed to be self-adjoint and positive. Notice that under this condition the square root operator $\sqrt{A}$ is well-defined.

Each solution of this Cauchy problem satisfies the energy conservation law

$$\|u'(t)\|^2 + \left\|\sqrt{A}\,u(t)\right\|^2 = \|u'(0)\|^2 + \left\|\sqrt{A}\,u(0)\right\|^2 \quad (t > 0),$$

which implies that the solution of the problem must be unique.

Rather similar to the first order Cauchy problem, the existence of a solution for (9.4) can be solved by the functional calculus method for self-adjoint operators.

Let $(P_\lambda)$ be the spectral resolution of a self-adjoint operator $A$. Then we define the function $u : [0,\infty) \to H$ by applying the functional

calculus to the functions $x \mapsto \cos(\sqrt{x})$ and $x \mapsto \sin(\sqrt{x})/\sqrt{x}$. In fact, the function $u$ is built up as the sum of two spectral operators, namely

$$u(t) := L_t(u_0, u_1) := \int_0^\infty \cos\sqrt{\lambda}t \, dP_\lambda \, u_0 + \int_0^\infty \frac{\sin\sqrt{\lambda}t}{\sqrt{\lambda}} \, dP_\lambda \, u_1 \qquad (t > 0). \tag{9.5}$$

When defining the family of linear operators $\{L_t(u_0, u_1)\}$ this way, we obtain the following existence and uniqueness theorem for second order problems:

Theorem 9.2. The mapping $t \mapsto L_t(u_0, u_1)$ is continuous in $[0, \infty)$ and twice continuously differentiable in $(0, \infty)$. Furthermore, it solves problem (9.4) in a unique manner.

The functional calculus for self-adjoint operators can also be applied to further classes of initial-value problems. By the *abstract Schrödinger equation* we mean the linear operator-differential equation

$$u'(t) + iA\,u(t) = 0\,, \tag{9.6}$$

where $A$ is a self-adjoint operator in Hilbert space $H$ (the operator $iA$ is called *skew-adjoint* in this case).

Each solution $u$ of (9.6) satisfies the so-called *energy equation*

$$\|u(t)\| = \|u(0)\| \text{ for } t > 0\,, \tag{9.7}$$

which immediately follows from

$$\frac{d}{dt} \|u(t)\|^2 = \langle -iAu(t), u(t)\rangle + \langle u(t), -iAu(t)\rangle = 0.$$

If an initial value $u(0) = u_0$ is given, then (9.7) implies that the solution of the initial-value problem for (9.6) is unique.

To construct a solution for problem (9.6), we define the spectral operator

$$L_t(x) := e^{-iAt}x = \int_{-\infty}^\infty e^{-i\lambda t} dP_\lambda x \qquad (t > 0), \tag{9.8}$$

where $P_\lambda$ again is the spectral resolution of $A$. Then the family of operators $L_t$ consitutes the solution of the abstract Schrödinger problem:

Theorem 9.3. The mapping $t \mapsto L_t(u_0)$ forms a solution of the

Cauchy problem for (9.6), where each operator $L_t$ for $t > 0$ is unitary.

## §10. Positive Self-Adjoint Operators Generate Semigroups

There are strong relationships between the functional calculus for half-bounded self-adjoint operators and the generation of strongly continuous semigroups by maximal accretive operators- It is the aim of this section to elaborate the similarities between these approaches.

Given a Banach space $X$, a *$C_0$-semigroup* or a *strongly continuous semigroup of bounded linear operators* is an operator-valued mapping $S : [0, \infty) \to \mathscr{L}(X)$, shortly denoted $\{S(t)\}_{t \geq 0}$ and distinguished by the properties

(G-1) $S(t + s) = S(t) \circ S(s)$ for all $t, s \geq 0$ (the *semigroup property*);

(G-2) $\lim_{t \downarrow 0} S(t)\, x = x$ for all $x \in X$.

It follows from property (G-2), which is nothing but strong continuity from the right-hand side in $t = 0$, that $S(0)$ is the identity mapping in $X$. Notice also that the conditions (G-1) and (G-2) together imply continuity from the right of the semigroup even in every $t > 0$. In case of $X = \mathbb{R}$, semigroups share both properties with the family $\{\mathcal{E}(t)\}_{t \geq 0}$ of exponential functions

$$\mathcal{E}(t) : \lambda \mapsto e^{\lambda t} \qquad (\lambda \in \mathbb{R}),$$

thus $\{\mathcal{E}(t)\}_{t \geq 0}$ serves as an elementary example of a strongly continuous semigroup.

By the *orbit map* or *trajectory* $\eta_x$ of a semigroup associated with a start element $x \in X$ we mean the vector-valued mapping $\eta_x : [0, \infty) \to X$ given by $\eta_x(t) := S(t)\, x$. Due to condition (G-1), strong continuity of $\{S(t)\}_{t \geq 0}$ implies continuity of the functions $\eta_x$. But $\eta_x$ is even differentiable under a very mild assumption:

THEOREM 10.1. Let $\{S(t)\}_{t \geq 0}$ be a $C_0$-semigroup. Then for each $x \in X$ the mapping $\eta_x$ is differentiable in $(0, \infty)$ if and only if $\eta_x$ is right differentiable in $t = 0$.

Given a $C_0$-semigroup $\{S(t)\}_{t \geq 0}$, we define the *infinitesimal generator* (briefly denoted the *generator* of the semigroup) as a mapping $A : \mathfrak{D}(A) \subset X \to X$ by means of its graph

$$\Gamma_A := \{\,(x,y) \in X \times X : \; y = \lim_{h \downarrow 0} \frac{1}{h}(S(h)x - x) \text{ exists } \}.$$

Hence the domain of definition $\mathfrak{D}(A)$ of $A$ consists of those elements $x \in X$, for which the orbit map $\eta_x$ is differentiable from the right in $t = 0$. By theorem 10.1, this is equivalent to $\mathfrak{D}(A) = \{\, x \in X :$ $\eta_x$ is differentiable $\}$. It is easy to show that $A$ is linear, hence it is a linear operator acting in $X$.

In the following we indicate the relationship between self-adjoint operators and strongly continuous semigroups.

Let $\{S(t)\}_{t \geq 0}$ be a strongly continuous semigroup. If one defines $U(t) := S(t)$ for $t \geq 0$ and $U(t) := S(-t)$ for $t < 0$, then a *group* $\{U(t)\}_{t \in \mathbb{R}}$ of *unitary* operators is defined, which fulfills the property $U(s+t) = U(s) \circ U(t)$ for all $s, t \in \mathbb{R}$. Reversely, Stone*'s theorem* claims that the infinitesimal generator of a unitary group is skew-adjoint:

Theorem 10.2. Each unitary group $\{U_t\}_{t \in \mathbb{R}}$ in a Hilbert space is infinitesimally generated by a linear operator of type $iA$, where $A$ is self-adjoint.

We finally want to show that a positive self-adjoint operator is the generator of a contraction group.

It should be clear that if $A$ is positive and self-adjoint, then it is m-accretive at the same time. Hence $A$ being positive and self-adjoint is the infinitesimal generator of a semigroup of contractions. Of course, the family $\{L_t\}_{t \geq 0}$ of spectral operators given by (9.3) coincides with the semigroup $\{S(t)\}_{t \geq 0}$ generated by $A$. Thus the solution of the homogeneous Cauchy problem can be obtained either by the functional calculus for self-adjoint operators or by the generator theorem 9.4.

## Part III
# Self-Adjoint Realizations of Differential Expressions

This final section contains applications of the theory of self-adjoint operators to linear partial differential operators and the boundary and initial-value problems associated with them.

We first introduce the fundamental spaces of functions, in which the analysis of linear partial differential operators is performed. In fact, the most important class of normed linear spaces used in functional analysis, harmonic analysis and PDE theory is constituted by so-called *Lebesgue spaces.*

For $p \in [1, \infty)$, let $(X, \mathcal{B}, \mu)$ be a measure space and $\mathcal{L}^p(X)$ be the $\mathbb{C}$-linear space of measurable functions $f : X \to \mathbb{C}$, for which $|f|^p$ is Lebesgue-integrable. For $f, g \in \mathcal{L}^p(X)$ we introduce the equivalence relation $f \sim g$ according to $f = g$ $\mu$-almost everywhere. Then the elements of $L^p(X)$ are the equivalence classes in $\mathcal{L}^p(X)$ with respect to $\sim$, that is, $L^p(X) := \mathcal{L}^p(X)/\sim$. Now, it is possible to define a norm in $L^p(X)$ according to

$$\|f\|_p := \left( \int_X |f|^p \, d\mu \right)^{\frac{1}{p}},$$

and the FISCHER-RIESZ *theorem* ensures that $(L^p(X), \|\cdot\|_p)$ is a complete normed space.

The case $p = 2$ gives rise to a special attention: here it is possible to derive the norm $\|\cdot\|_2$ of $L^2(X)$ from the inner product

$$\langle f, g \rangle := \int_X f \, \overline{g} \, d\mu$$

by $\|f\|_2 := \langle f, f \rangle^{1/2}$, which of course makes $L^2(X)$ into a Hilbert space. Spaces of type $L^2(X)$, where $X$ is the underlying set of a measure space $(X, \mathcal{B}_X, \mu)$, form an important class of Hilbert spaces. Especially in the case, where $X = G$ is an open subset of one of the Euclidean spaces $\mathbb{R}^d$, we obtain the Hilbert spaces $L^2(G)$, which proves to be separable.

We finally enumerate the usual orthonormal bases of some Hilbert spaces $L^2(I)$, where $I \subset \mathbb{R}$ denotes a finite or infinite open interval:
1. If $I = (-\pi, \pi)$, the *trigonometrical functions* given by the functions

$$\frac{1}{\sqrt{2\pi}}, \quad \frac{1}{\sqrt{\pi}}\sin(x), \quad \frac{1}{\sqrt{\pi}}\cos(x), \quad \frac{1}{\sqrt{\pi}}\sin(2x), \quad \frac{1}{\sqrt{\pi}}\cos(2x), \quad \dots$$

form an orthonormal base in the Hilbert space $L^2((-\pi,\pi))$.

2. In case of $I=(-1,1)$, an orthonormal base, founded on polynomials, is given by the so-called *Legendre polynomials*

$$P_n(x) := \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n, \quad n\in\mathbb{N},$$

together with the constant function $P_0(x) := 1$. These functions form the only orthogonal system in $L^2(-1,1)$ containing polynomials of any order.

3. Let $I$ be the half-line $(0,\infty)$. Then the sequence of functions

$$L_n^\alpha(x) := x^{\alpha/2}e^{-x/2}\cdot\mathcal{L}_n^\alpha(x), \quad n\in\mathbb{N}$$

forms an orthonormal base in the space $L^2(0,\infty)$, where $\mathcal{L}_n^\alpha$ are called the *Laguerre polynomials*, defined by the equations

$$\mathcal{L}_n^\alpha(x) := \frac{x^{-\alpha}e^x}{n!}\frac{d^n}{dx^n}(e^{-x}x^{n+\alpha}), \quad n\in\mathbb{N}.$$

4. In case of $I$ being the entire set $\mathbb{R}$, an orthonormal base of $L^2(\mathbb{R})$ is given by the sequence of *Hermite functions*

$$\mathcal{H}_n(x) := e^{-\frac{x^2}{2}}H_n(x), \quad n\in\mathbb{N},$$

where $H_n$ are the *Hermite polynomials*

$$H_n(x) := (-1)^n \sum_{k_1+2k_2=n} \frac{n!}{k_1!\,k_2!}(-1)^{k_1+k_2}(2x)^{k_1}, \quad n\in\mathbb{N}.$$

## §1. Fourier-Plancherel Transformation

The mapping known as *Fourier transformation* is an important tool in advanced analysis and has many applications, especially to the investigation of linear differential operators with constant coefficients considered in one of the Euclidean spaces $\mathbb{R}^d$.

Rather Fourier transformation can be defined in various linear spaces, it is customary to introduce this mapping first in Lebesgue space $L^1(\mathbb{R}^d)$. Preliminary, for elements $f,g\in L^1(\mathbb{R}^d)$ a multiplication operation $f*g$ is defined according to

$$(f*g)(x) := \int_{\mathbb{R}^d} f(t)\cdot g(x-y)\,dy,$$

called *convolution* in $L^1(\mathbb{R}^d)$. This mapping is commutative, associative and satisfies the estimate

$$\|f * g\|_1 \leq \|f\| \cdot \|g\| \quad \text{for all } f, g \in L^1(\mathbb{R}^d).$$

Since $L^1(\mathbb{R}^d, +)$ is a Banach space, the so-called *convolution algebra* $(L^1(\mathbb{R}^d), +, *)$ constitutes a commutative but non-unital Banach algebra.

Now Fourier transformation is a mapping $\mathscr{F} : f \to \hat{f}$ in $L^1(\mathbb{R}^d)$, which sends each $f \in L^1(\mathbb{R}^d)$ to $\hat{f} \in L^\infty(\mathbb{R}^d)$, called the *Fourier transform* of $f$ and defined by

$$\hat{f}(\xi) := \int_{\mathbb{R}^d} e^{-2\pi i \langle \xi, x \rangle} f(x)\, dx \qquad (\xi \in \mathbb{R}^d), \tag{1.1}$$

where $\langle \xi, x \rangle$ is the usual inner product of $\xi, x \in \mathbb{R}^d$.

We collect some remarkable properties of the mapping $\mathscr{F}$ to the following list:

1. For each $f \in L^1(\mathbb{R}^d)$ the function $\hat{f}$ is continuous on $\mathbb{R}^d$, so $\hat{f} \in C(\mathbb{R}^d)$.
2. $\hat{f}$ is vanishing at infinity, that is, Fourier transformation constitutes a mapping $\mathscr{F} : L^1(\mathbb{R}^d) \to C_0(\mathbb{R}^d)$ (*Riemann-Lebesgue lemma*).
3. $\mathscr{F}$ is injective, i. e. $\mathscr{F}(f) \neq \mathscr{F}(g)$ for $f \neq g$.
4. $\|\mathscr{F}(f)\|_\infty \leq \|f\|_1$ for $f \in L^1(\mathbb{R}^d)$.
5. $\mathscr{F}(\alpha f + \beta g) = \alpha\, \mathscr{F}(f) + \beta\, \mathscr{F}(g)$ for $\alpha, \beta \in \mathbb{C}$, thus $\mathscr{F}$ is linear.
6. $\mathscr{F}(f * g) = \mathscr{F}(f) \cdot \mathscr{F}(g)$, so $\mathscr{F}$ is an algebra homomorphism between the Banach algebras $L^1(\mathbb{R}^d, +, *)$ and $C_0(\mathbb{R}^d, +, \cdot)$.
7. If $f \in L^1(\mathbb{R}^d)$ and $g_f : x \in \mathbb{R}^d \mapsto |x| f(x) \in L^1(\mathbb{R}^d)$, then $\hat{f} \in C^1(\mathbb{R}^d)$ and
$$\partial_j f(x) = \widehat{-i\xi_j f(\xi)} \qquad (j = 1, \ldots, d).$$

It is possible to extend $L^1$-Fourier transformation to the Hilbert space $L^2(\mathbb{R}^d)$. If the functions $u, v : \mathbb{R}^d \to \mathbb{C}$ are Lebesgue-integrable then by definition they belong to the space $L^1(\mathbb{R}^d)$, and their *Fourier transforms* are well-defined by formula (1.1).

However, if is not only a member of $L^1(\mathbb{R}^d)$, but also $u \in L^2(\mathbb{R}^d)$, we define Fourier transformation a little bit different:

$$\hat{u}(\xi) := \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{-2\pi i\langle \xi, x\rangle}\, u(x)\, dx. \tag{1.2}$$

Then for functions $u, v \in L^2(\mathbb{R}^d)$ the inner product between $u$ and $v$ and the inner product of their Fourier transforms $\hat{u}$ and $\hat{v}$ are related by PLANCHEREL*'s formula*

$$\langle u, v\rangle \;=\; \langle \hat{u}, \hat{v}\rangle.$$

Hence *Fourier transformation* $\mathscr{F} : u \mapsto \hat{u}$ constitutes a partial isometry from the space $L^{1,2}(\mathbb{R}^d) := L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$ to the space $L^2(\mathbb{R}^d)$.

It is a crucial fact now that $L^{1,2}(\mathbb{R}^d)$ proves to be dense in $L^2(\mathbb{R}^d)$. Since the mapping $\mathscr{F}$ is linear and continuous on $L^{1,2}(\mathbb{R}^d)$ , it is even *uniformly continuous* on that space and can be extended to an *unitary*, that is, a bijective and isometric linear operator $\mathscr{F}_P : L^2(\mathbb{R}^d) \to L^2(\mathbb{R}^d)$, called the FOURIER-PLANCHEREL *operator* or the $L^2$*-Fourier transformation* in $L^2(\mathbb{R}^d)$.

But in general a function $u \in L^2(\mathbb{R}^d)$ is not an element of $L^1(\mathbb{R}^d)$. Thus it is not possible to define the $L^2$-Fourier transform $\hat{u}$ of every $u \in L^2(\mathbb{R}^d)$ by means of formula (1.2). However, for $u \in L^2(\mathbb{R}^d)$ and $R > 0$ the function

$$\hat{u}_R(\xi) := \frac{1}{(2\pi)^{d/2}} \int_{|x|\le R} e^{-2\pi i\langle x, \xi\rangle}\, u(x)\, dx$$

is well-defined, and one may obtain the $L^2$-Fourier transform $\hat{u}$ as the limit of the functions $\hat{u}_R$ for $R \to \infty$ with respect to the 2-norm, that is, $\lim_{R\to\infty} \|\hat{u} - \hat{u}_R\|_2 = 0$.

## §2. Hilbert-Sobolev Spaces

We continue with linear partial differential expressions and the notation of *weak differentiability* for a function $f : G \to \mathbb{C}$, where $f$ is assumed to be *locally integrable* on a domain $G \subset \mathbb{R}^d$, that is, $f$ is Lebesgue-integrable on each compact subset of $G$. Let $L^1_{loc}(G)$ be the linear space of all such functions.

Preliminary, by a $d$-dimensional *multi-index* is meant a $d$-tuple $\alpha = (\alpha_1, \ldots, \alpha_d)$ with values $\alpha_i \in \mathbb{N}_0$, whose *modulus* is the sum $|\alpha| := \alpha_1 + \ldots + \alpha_d$. Multi-indices are used to shorten multiple partial derivatives $\partial^\alpha f := \partial_1^{\alpha_1}\partial_2^{\alpha_2} \ldots \partial_d^{\alpha_d} f$ of a sufficiently often differentiable function $f : G \to \mathbb{C}$, but also to denote the multiple powers $\xi^\alpha := \xi_1^{\alpha_1} \cdot \xi_2^{\alpha_2} \cdot \ldots \cdot \xi_d^{\alpha_d}$ of a complex vector $\xi = (\xi_1, \xi_2, \ldots, \xi_d) \in \mathbb{C}^d$.

Then $f \in L^1_{loc}(G)$ is called *weakly differentiable* with respect to the multi-index $\alpha$, if there is a further function $g \in L^1_{loc}(G)$ such that

$$\int_G g(x)\,\phi(x)\,dx \;=\; (-1)^{|\alpha|} \int_G f(x)\,\partial^\alpha \phi(x)\,dx \qquad \text{for all } \phi \in C_c^\infty(G).$$

We denote $\partial^\alpha f := g$ in this case, and weak differentiability arises as a linear mapping $\partial^\alpha : D_\alpha \subset L^1_{loc}(G) \to L^1_{loc}(G)$, where $D_\alpha$ is a suitable linear subspace of $L^1_{loc}(G)$.

The concept of weak derivative is compliant with the classical derivative in the following sense: if $f$ has (classical) derivative $\partial^\alpha f$, then $f$ is also weakly differentiable and both notions of derivatives coincide. Clearly, the reverse statement does not hold as the simple example $f : x \mapsto |x|$ for $x \in (-1,1)$ shows.

Sobolev spaces play an important role in PDE theory, since the solutions of partial differential equations are frequently searched in these spaces.

We preliminary define for any integer $k \in \mathbb{N}$ the *Hilbert-Sobolev space* $H^k(G)$ as the set of those functions $f \in L^2(G)$, whose weak derivatives $\partial^\alpha f$ exist and belong to $L^2(G)$ for all $\alpha$ with $|\alpha| \le k$, briefly written

$$H^k(G) \;:=\; \{\, f \in L^2(G) : \partial^\alpha f \in L^2(G) \text{ for } |\alpha| \le k \,\}.$$

The spaces $H^k(G)$ are of course linear spaces, and specifically for $k = 0$ the space $H^0(G)$ is the Lebesgue space $L^2(G)$ itself.

For functions $f \in H^k(G)$ let the mapping $f \mapsto |f|_k$ be defined by

$$|f|_k \;:=\; \Big( \sum_{|\alpha| \le k} \int_G |\partial^\alpha f|^2\,dx \Big)^{1/2}. \tag{2.1}$$

It fulfills all the properties of a norm, thus the space $H^k(G)$ is actually a normed space, and the norm defined by (2.1) is usually called a *Sobolev norm.* With respect to the question of completeness, it can be shown that the spaces $H^k(G)$ are complete. But they also can be endowed with an inner product according to

$$\langle f, g \rangle_k \;:=\; \sum_{|\alpha| \le k} \int_G \partial^\alpha f\; \partial^\alpha g\,dx\,, \qquad f, g \in H^k(G),$$

so $H^k(G)$ actually forms a Hilbert space.

There is still another way to define Sobolev spaces. Consider the subspace

$$C_k^\infty(G) := \{ f \in C^\infty(G) : |f|_k < \infty \},$$

and because this space is *not* complete with respect to the norm $|\cdot|_k$, we define

$$\mathcal{H}^k(G) := \overline{C_k^\infty(G)}.$$

But notice that the space $\mathcal{H}^k(G)$ is defined by a purely topological operation, i. e. taking the closure of a smaller space, so for elements in this space there is no instrinsic notion of derivative. Hence one introduces the notion of *strong derivative* for $u \in \mathcal{H}^k(G)$: given a multi-index $\alpha$ ($|\alpha| \leq k$), a further function $\partial^\alpha u \in \mathcal{H}^k(G)$ is defined to be the strong derivative of $u$ with respect to $\alpha$, if there is a sequence $\{\phi_n\} \in C_k^\infty(G)$ such that $|\partial^\alpha u - \partial^\alpha \phi_n)|_k \to 0$ for $n \to \infty$.

The MEYERS-SERRIN *theorem* claims $\mathcal{H}^k(G) \cong H^k(G)$ in the sense of Banach space isomorphisms, so the notions of strong and weak derivative are equivalent to each other.

In order to model boundary conditions for partial differential equations, a further class of Sobolev spaces is of interest. Let $C_c^k(G)$ be the space of $k$-times continuously differentiable functions $f : G \to \mathbb{C}$ with compact support in $G$. Since this space is not dense in $H^k(G)$ with respect to the norm $|\cdot|_k$, it is nearby to define the closure

$$H_0^k(G) := \overline{C_c^k(G)}_{\|\cdot\|_k},$$

which is also called a *Hilbert-Sobolev space*, since an inner product in $H_0^k(G)$ can be defined by

$$\langle f, g \rangle_k := \sum_{|\alpha \leq k, |\beta| \leq k} \int_G \partial^\alpha f \; \partial^\alpha g \, dx \,, \qquad f, g \in H^k(G)$$

such that $|f|_1 = \langle f, f \rangle_k^{1/2}$. The space $H_0^k(G)$ forms a closed subspace of $H^k(G)$, so we may regard $H_0^k(G)$ as a Hilbert space in its own right.

The reason, why compact operators are of outstanding meaning in the analysis of linear problems, is founded by RELLICH's *embedding theorem* for Sobolev spaces:

THEOREM 2.1. If $G \subset \mathbb{R}^d$ is *bounded*, then for each $k \in \mathbb{N}$ there is a compact embedding

$$k_m : H_0^k(G) \hookrightarrow H_0^{k-1}(G).$$

This theorem implies that all Hilbert-Sobolev spaces $H_0^k(G)$ are compactly embedded into the space $L^2(G)$. Hence each bounded sequence within $H_0^k(G)$ owns a convergent subsequence with respect to the norm $\|\cdot\|_2$ of $L^2(G)$. Notice that in theorem 2.1 there are no restrictions regarding the smoothness or other properties of the boundary $\partial G$.

## §3. Realizations of Linear Partial Differential Operators

In the sequel we deal with the general *linear partial differential expression*

$$\mathcal{A}(x,u) := \sum_{|\alpha|\le k} c_\alpha(x)\,\partial^\alpha u(x) \tag{3.1}$$

of order $k\in\mathbb{N}$ (also denoted a *linear partial differential operator*), whose coefficients $c_\alpha$ are real or complex-valued functions $c_\alpha : \overline{G}\to\mathbb{C}$ satisfying certain regularity conditions like measurability, boundedness, continuity or differentiability.

We assign to a linear differential expression $\mathcal{A}$ two further possible features. First, $\mathcal{A}$ is said to be *essentially real* if the top-order coefficients $c_\alpha$ with $|\alpha|=k$ are real-valued. Second, if each function $c_\alpha$ is constant, then $\mathcal{A}$ is called a *differential expression with constant coefficients.*

We denote a densely defined linear operator $A$ with domain of definition $\mathfrak{D}(A)$ and acting in a Banach space $X$ of functions $u: G\to\mathbb{C}$ a *realization* of the expression $\mathcal{A}$, if $Au=\mathcal{A}u$ for all $u\in\mathcal{D}$, where

1. the set $\mathcal{D}$ is at least a dense subspace of $\mathfrak{D}(A)$,
2. the expression $\mathcal{A}u$ has a meaning either in the classical or in the weak sense.

Both conditions are especially true for the choice $\mathcal{D}=C_c^\infty(G)$, so the space of test functions forms a *core* of any realization of $\mathcal{A}$.

Since we are interested in operators acting in Hilbert spaces, we define the linear operator $\mathcal{A}_0$ acting in Hilbert space $L^2(G)$ to be the realization of $\mathcal{A}$ with dense domain of definition $C_c^\infty(G)\subset L^2(G)$, i. e.

$$\begin{cases} \mathfrak{D}(\mathcal{A}_0) := C_c^\infty(G)\subset L^2(G) \\ \mathcal{A}_0\,\phi := \mathcal{A}\,\phi \ \text{ for all } \phi\in\mathfrak{D}(\mathcal{A}_0). \end{cases}$$

Supposed the operator $\mathcal{A}_0$ is closable, then the closure $\mathcal{A}_{min} := \overline{\mathcal{A}_0}$ is called the *minimal operator* of $\mathcal{A}$ in $L^2(G)$.

The following theorem illustrates that the theory of closable (and closed) linear operators is applicable to a wide range of linear partial differential expressions:

Theorem 3.1. If a linear partial differential operator $\mathcal{A}$ of order $n$ has smooth coefficients, then the realization $A$ acting in $L^2(G)$ with domain of definition $\mathfrak{D}(A) = C^n(G) \cap L^2(G)$ is closable.

Proof: Let $(u_j)$ be a sequence with $u_j \in \mathfrak{D}(A)$ for $j \in \mathbb{N}$ and $u_j \to 0$ for $j \to \infty$. Also, let $(Au_j)$ be the sequence in $L^2(G)$ with $Au_j \to f$ for $j \to \infty$. By the standard criterion for closability, we have to show that $f = 0$. Since the coefficients of $\mathcal{A}$ are sufficiently smooth, there is the formal adjoint $\mathcal{A}^\dagger$ associated with $\mathcal{A}$ such that

$$\langle \mathcal{A}\, u_j, \phi \rangle \;=\; \int_G \mathcal{A}\, u_j\, \phi \, dx \;=\; \int_G u_j\, \mathcal{A}^\dagger\, \phi \, dx \;=\; \langle u_j, \mathcal{A}^\dagger\, \phi \rangle$$

for all $\phi \in C_c^n(G)$. By the continuity of the inner product in $L^2(G)$, performing the limit for $j \to \infty$ leads to

$$\langle f, \phi \rangle \;=\; \lim_{j \to \infty} \langle \mathcal{A}\, u_j, \phi \rangle \;=\; \lim_{j \to \infty} \langle u_j, \mathcal{A}^\dagger\, \phi \rangle \;=\; \langle 0, \mathcal{A}^\dagger\, \phi \rangle \;=\; 0.$$

Since $\phi \in C^n(G)$ is arbitrary and $C^n(G)$ is dense in $L^2(G)$, we have $f = 0$. □

This theorem is especially true if $\mathcal{A} = \mathcal{A}_c$ has constant coefficients. In this case it is possible to determine the domain of definition of the minimal operator $(\mathcal{A}_c)_{min}$. Indeed, the definition of Sobolev spaces $H^k(G)$ shows that if $\mathcal{A}_c$ is of order $k$, then $(\mathcal{A}_c)_{min}$ has domain of definition $H_0^k(G)$.

## §4. Formally Self-Adjoint Differential Expressions

It is often useful to combine the study of a linear differential operator with that of the so-called *formally adjoint* operator. Let $C^m(G)$ be the linear space of at least $m$-times continuously differentiable complex-valued functions and $C_c^\infty(G)$ be the subspace of *test functions* consisting of indefinitely differentiable functions $\phi : G \to \mathbb{C}$ *with compact support*, that is, the set

$$\operatorname{supp}(\phi) \;:=\; \overline{\{\, x \in G : \phi(x) \neq 0 \,\}}$$

is bounded in $\mathbb{R}^d$, so the distance between $\operatorname{supp}(\phi)$ and $\partial G$ is strictly positive.

Supposed the linear partial differential operator $\mathcal{A}$ of order $2m$

$$\mathcal{A}(x,u) := \sum_{|\alpha|\le 2m} c_\alpha(x)\, \partial^\alpha u(x) \tag{4.1}$$

has coefficients $c_\alpha \in C^{|\alpha|}(G)$, then for $u \in C^m(G)$ and $\psi \in C_c^\infty(G)$ we find by repeated partial integration

$$\int_G \sum_{|\alpha|\le 2m} (c_\alpha\, \partial^\alpha u)\, \overline{\psi}\, dx \;=\; \int_G u \sum_{|\alpha|\le 2m} \overline{\partial^\alpha(\overline{c_\alpha}\,\psi)}\, dx.$$

Hence there is a *formally adjoint* differential expression

$$\mathcal{A}^\dagger(x,u) := \sum_{|\alpha|\le 2m} (-1)^{|\alpha|} \partial^\alpha(\overline{c_\alpha(x)}\, u(x)) \tag{4.2}$$

such that the identity

$$\int_G \mathcal{A}\phi\, \overline{\psi}\, dx \;=\; \int_G \phi\, \overline{\mathcal{A}^\dagger \psi}\, dx \tag{4.3}$$

holds for $\phi, \psi \in C_c^\infty(G)$. Using LEIBNIZ's rule to compute all the derivatives $\partial^\alpha(\overline{c_\alpha}\, u)$ in (4.2), it can be shown that there are functions $c_\alpha^* : G \to \mathbb{C}$ such that

$$\mathcal{A}^\dagger(x,u) \;=\; \sum_{|\alpha|\le 2m} c_\alpha^*(x)\, \partial^\alpha u(x).$$

Thus $\mathcal{A}^\dagger$ is uniquely defined by $\mathcal{A}$ and also forms a linear differential expression of type (4.1). It is easy to verify that $\mathcal{A}^\dagger$ is elliptic if and only if $\mathcal{A}$ is so. Assigning to each expression $\mathcal{A}$ its formally adjoint $\mathcal{A}^\dagger$ is a self-dual operation, so $\mathcal{A}^{\dagger\dagger} = \mathcal{A}$. Especially for all *elementary* differential operators $\partial_j := d/dx_j$ of order one, a simple computation yields ${\partial_j}^\dagger = -\partial_j$ for $j = 1, \ldots, d$.

Notice that formula (4.3) holds since at least one of the functions $\phi, \psi$ belongs to the space $C_c^\infty(G)$. If we replace the test functions $\phi, \psi \in C_c^\infty(G)$ by more general functions $u, v \in C^m(\overline{G})$, the space of at least $m$-times continuously differentiable functions $u : \overline{G} \to \mathbb{C}$, the identity (4.3) becomes into

$$\mathcal{A}u\, \overline{v} \;-\; u\, \overline{\mathcal{A}^\dagger v} \;=\; \sum_{j=1}^{d} \partial_j C_j(u, \overline{v}), \tag{4.4}$$

where the functions $C_j$ are bilinear expressions in $u, \overline{v}$ and their derivatives up to order $2m-1$. Then performing integration over $G$, (4.4) leads to

$$\int_G (\mathcal{A}u\,\overline{v} - u\,\overline{\mathcal{A}^\dagger v})\,dx = \int_{\partial G} \sum_{j=1}^{d} \nu_j C_j(u,\overline{v})\,d\sigma, \tag{4.5}$$

where $d\sigma$ is the surface measure on the boundary $\partial G$. Equation (4.5) is called Green's *formula* associated with the linear differential operator $\mathcal{A}$.

We denote a linear differential expression $\mathcal{A}$ *formally self-adjoint*, if $\mathcal{A}^\dagger = \mathcal{A}$ holds. If $\mathcal{A}$ has constant coefficients, then it is formally self-adjoint if and only if $c_\alpha \in \mathbb{R}$ for $|\alpha|$ even and $c_\alpha \in i\,\mathbb{R}$ for $|\alpha|$ odd.

## §5. The Momentum Operator on Various Intervals

For an arbitrary interval $I \subset \mathbb{R}$ we consider the first order differential operator

$$A_1 := i\frac{d}{dx},$$

known from theoretical physics as the *momentum operator*. Its classical domain of definition is the space $C^1(I)$ consisting of continuously differentiable complex-valued functions. The momentum operator is the simplest differential operator one can imagine, so it is natural to study this operator first. In advance, the momentum operator defined on domains with homogeneous boundary conditions proves to be symmetric but no half-bounded, so we cannot apply the Friedrichs extension theory to obtain self-adjoint extensions of $A_1$.

Basically, for each $\lambda \in \mathbb{C}$ the function $f_\lambda(x) := e^{-i\lambda x}$ is a solution of the equation $A_1 f = \lambda f$, hence $f_\lambda$ is an eigenvector of $A_1$ with respect to the eigenvalue $\lambda$. Thus the momentum operator has eigenvectors for any complex number, but they do not necessarily belong to the Hilbert space $L^2(I)$.

We want to determine the spectrum and the possible self-adjointness of the momentum operator on various intervals in the real line and will start with the case $I = (-\infty, \infty)$. If the domain of definition $\mathfrak{D}(A_1)$ is the space $C_c^\infty(\mathbb{R})$ of test functions, then $A_1$ is a densely defined operator in Hilbert space $L^2(\mathbb{R})$. For $f, g \in \mathfrak{D}(A_1)$ we compute

$$\langle A_1 f, g\rangle \;=\; i\int_{\mathbb{R}} f(t)\overline{g(t)}\,dt \;=\; if\bar{g}\,|_{-\infty}^{+\infty} - i\int_{\mathbb{R}} f'(t)\overline{g(t)}\,dt \;=\; \langle f, A_1 g\rangle,$$

so $A_1$ is symmetric on $\mathfrak{D}(A_1)$. Recall that the Sobolev spave $H^1(\mathbb{R})$ is the closure of $C_c^\infty(\mathbb{R})$ with respect to the Sobolev norm $|\cdot|_1$, which is also the graph norm of the momentum operator. Hence the domain of definition of the operator $\overline{A_1}$ is the space $H^1(\mathbb{R})$.

Now, for $f \in H^1(\mathbb{R})$ the Fourier transform of the function $A_1 f$ is given by

$$\widehat{A_1 f}(\xi) \;=\; \int_{\mathbb{R}} e^{-it\xi}\,(i\frac{d}{dt}f(t))\,dt \;=\; \xi\,\hat{f}(\xi),$$

so the momentum operator is unitarily equivalent to the multiplication operator $M_1$ generated by the identity mapping on $\mathbb{R}$. As we have seen, the operator $M_1$ is self-adjoint on $L^2(\mathbb{R})$ and has pure continuous spectrum $\sigma(M_1) = \mathbb{R}$. Due to the unitary equivalence of $M_1$ and $A_1$, both assertions must also be true for the momentum operator $A_1$ with domain of definition $H^1(\mathbb{R})$. Clearly, each operator defined on the space $C_c^\infty(\mathbb{R})$ is a core of its closure, thus the momentum operator with domain of definition $C_c^\infty(\mathbb{R})$ is essentially self-adjoint on the Hilbert space $L^2(\mathbb{R})$.

Unfortunately, the results for a bounded or half-bounded interval are not as satisfying as in the case described above. In the following, we shall compute the deficiency spaces and deficiency indices $n_\pm$ of the closure of $A_1$ for the intervals $(0,\infty)$ and $(0,1)$. Remember that a symmetric linear operator can only have self-adjoint extensions iff $n_+ = n_-$.

1. The interval $I := (0,\infty)$. The operator $A_1$ is symmetric on the dense domain of definition $C_c^\infty(0,\infty)$. The domain of definition of its closure $\overline{A_1}$ is the Sobolev space $H_0^1(0,\infty)$, and the domain of definition of the operator $\overline{A_1}^*$ is the Sobolev space $H^1(0,\infty)$. We have $u \in D_+ := \mathfrak{N}(\overline{A_0}^* - i)$, iff $u$ is the solution of the differential equation $iu' = iu$, whose solution $u(t) = e^t$ is not a $L^2$-function, so we have $D_+ = \{0\}$ and $n_+ = 0$, respectively.

   Similarly, $v \in D_- := \mathfrak{N}(\overline{A_0}^* + i)$ iff $v$ is the solution of the differential equation $iv' = -iv$. But since its solution $v(t) = e^{-t}$ is $L^2$-integrable, if follows that $D_-$ is a non-trivial subspace of $L^2(0,\infty)$ and $n_- = 1$. In summary, the deficiency indices of $\overline{A_1}$ are $(0,1)$, thus there is no self-adjoint extension of this operator.
2. The interval $I := (0,1)$. The operator $A_1$ is symmetric on the

dense domain of definition $C_c^\infty(0,1)$ and closed if $\mathfrak{D}(A_1)$ is the Sobolev space $H_0^1(0,1)$. Moreover, the domain of definition of the adjoint operator $\overline{A_1}^*$ is the Sobolev space $H^1(0,1)$.

Let us compute the deficiency spaces $D_\pm$ of $\overline{A_1}^*$. We have $u \in D_\pm$ if and only if $u$ satisfies the differential equation

$$i\,\frac{du}{dt} \;=\; \pm\, i\,u,$$

which has the solutions $u_\pm = e^{\pm t}$. Hence the deficiency indices are $n_+ = n_- = 1$, so $\overline{A_1}$ owns self-adjoint extensions.

Indeed, there is a one-parameter family $S_\alpha$ $(\alpha \in [0, 2\pi))$ of self-adjoint extensions for this operator, and the domain of definition of each extension is described by the set

$$\mathfrak{D}(S_\alpha) \;:=\; \{\, f \in AC[0,1] \;:\; f(1) = e^{i\alpha} f(0) \,\}.$$

Remember that the set $AC[0,1]$ consists of almost everywhere differentiable functions, so we have $S_\alpha f := if'$ for $f \in \mathfrak{D}(S_\alpha)$.

Clearly, since there is no unique self-adjoint extension, the operator $A_1$ with domain of definition $\mathfrak{D}(A_1) = C_c^\infty(0,1)$ cannot be essentially self-adjoint in Hilbert space $L^2(0,1)$.

## §6. Sturm-Liouville Operators

A further example of a linear differential operator (in divergence form) is given by the STURM-LIOUVILLE *operator*

$$\mathcal{A}_{SL}(u,x) \;:=\; -(p(x)\,u'(x))' \;+\; q(x)\,u(x), \qquad x \in (a,b)$$

with $p(x) > 0$ and $q(x) \geq 0$ for $x \in [a,b]$, where $-\infty < a < b < \infty$. This operator models a vibrating string being fixed at the endpoints of a finite interval $[a,b]$.

The corresponding Dirichlet form reads as

$$B_{\mathcal{A}_{SL}}(u,v) \;=\; \int_a^b (p\,u'\,\overline{v}' \;+\; q\,u\,\overline{v})\,dx,$$

which shows that $\mathcal{A}_{SL}$ is symmetric and half-bounded on the domain of definition $C_c^\infty(G)$. The energetic space $H_{B_{SL}}$ is the Hilbert-Sobolev space $H_0^1(a,b)$, containing all nearly everywhere differentiable functions $u \in C[a,b]$ with $u(a) = u(b) = 0$.

The self-adjoint extension $\mathcal{A}_{SL,F}$ of the Sturm-Liouville operator in Hilbert space $L^2(a,b)$ has domain

$$\mathfrak{D}(\mathcal{A}_{SL,F}) = H^2(a,b) \cap H_0^1(a,b).$$

It is a linear operator with pure point spectrum, since $(a,b)$ is a bounded subset of the real line. Especially for $p(x) = 1$ and $q(x) = 0$ for $x \in (a,b)$ it is possible to compute explicitely the eigenvalues and eigenfunctions of the operator $\mathcal{A}_{SL,F} = -d^2/dx^2$. In fact, in this case the eigenvalue distribution is represented by

$$\lambda_n = \frac{n^2\pi^2}{(b-a)^2} \qquad (n \in \mathbb{N}),$$

where each $\lambda_n$ owns a one-dimensional eigenspace, and the related eigenfunctions $u_n$ are the mappings

$$u_n(x) := \sin\left(\frac{n\pi x}{b-a}\right) \text{ for } x \in (a,b)$$

forming an orthonormal base in Hilbert space $L^2(a,b)$. Thus, the smallest eigenvalue is the number $\pi^2/(b-a)^2$, and the asymptotical behaviour of the eigenvalues is $\lambda_n \sim n^2$.

## §7. The Laplacian in Bounded and Unbounded Domains

This subsection is devoted to the *Laplace operator*, which is the most important second order differential operator in mathematical physics. In general, let $G \subset \mathbb{R}^d$ be an open domain, then in the classical sense the Laplace operator (briefly denoted *Laplacian*) is the linear mapping $-\Delta : C^2(G) \to C(G)$ defined by

$$\Delta u := \frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_1^2} + \ldots + \frac{\partial^2 u}{\partial x_n^2}.$$

This operator can be considered as a prototype of a second order *elliptic* differential operator.

In order to apply the theory of self-adjoint operators to the investigation of the Laplacian, we firstly have to find a suitable dense subspace of the Hilbert space $L^2(G)$, on which $\Delta$ can be defined and fulfills the symmetry property. This will be the space $C_c^\infty(G)$ of test functions. Notice that the Laplacian is an unclosed operator on this domain.

Using Green's second formula, and due to the boundary condition $\phi(x) = 0$ for $x \in \partial G$ and $\phi \in C_c^\infty(G)$, we compute for $\phi, \psi \in C_c^\infty(G)$

$$\int_G \Delta\phi\,\overline{\psi}\,dx \;=\; -\sum_{i=1}^{n}\int_G \frac{\partial^2\phi}{\partial x_i^2}\,\overline{\psi}\,dx \;=\; -\sum_{i=1}^{n}\int_G u\,\overline{\frac{\partial^2\phi}{\partial x_i^2}}\,dx \;=\; \int_G \phi\,\overline{\Delta\psi}\,dx,$$

so $\langle\Delta\phi,\psi\rangle = \langle\phi,\Delta\psi\rangle$, which implies symmetry of the operator $\Delta$ on the space $C_c^\infty(G)$.

Moreover, we get for the operator $-\Delta$ the inequality

$$\langle -\Delta\phi,\phi\rangle \;=\; \sum_{i=1}^{n}\int_G \frac{\partial^2\phi}{\partial x_i^2}\,\overline{\phi}\,dx \;=\; \sum_{i=1}^{n}\int_G |\frac{\partial\phi}{\partial x_i}|^2 dx \;\geq 0,$$

so $-\Delta$ is positive. Since the Laplacian is a half-bounded operator on the domain $C_c^\infty(G)$ with lower bound $c_A = 0$, it fulfills all the prerequisites for Friedrichs' extension method. Thus we can state the following

Theorem 7.1. The operator $A := -\Delta$ with domain $\mathfrak{D}(A) := C_c^\infty(G)$ has at least one self-adjoint extension.

It should be mentioned that when working with test functions $\phi \in C_0^\infty(G)$ there was no need to make a distinction between bounded and unbounded domains in the computations above.

In the following we consider some realizations of the Laplacian in various domains:

1. *Laplacian in* $\mathbb{R}^d$. Let $G$ be the entire Euclidean space $\mathbb{R}^d$. The analysis of the Laplacian will be as easy as for the momentum operator in this case. Since the operator $-\Delta$ is unitarily equivalent to the multiplication operator $M_{-\Delta}$ generated by the function

$$m_2 : (x_1,\ldots,x_n) \mapsto (x_1^2 + \ldots + x_n^2), \quad x \in \mathbb{R}^d$$

and the operator $M_{m_2}$ is real-valued and self-adjoint, it follows that the operator $-\Delta$ with domain of definition $H^2(\mathbb{R}^d)$ must be self-adjoint, too. The spectrum of $-\Delta$ is that of the operator $M_{m_2}$, that is, it is purely continuous with $\sigma_c(-\Delta) = [0,\infty)$. Moreover, since $-\Delta$ with domain of definition $H^2(\mathbb{R}^d)$ is the closure of $-\Delta$ with domain of definition $C_c^\infty(\mathbb{R}^d)$, the operator $-\Delta$ with domain of definition $C_c^\infty(\mathbb{R}^d)$ is essentially self-adjoint on the Hilbert space $L^2(\mathbb{R}^d)$.

The energetic norm of the Laplacian for a function $\phi \in \mathfrak{D}(-\Delta_F)$ is given by the formula

$$\|\phi\|_{-\Delta} := \int_{\mathbb{R}^d} |\phi|^2 dx + \sum_{i=1}^{n} \int_{\mathbb{R}^d} \frac{\partial \phi}{\partial x_i}\, dx.$$

Since it is the well-known Sobolev norm $|\phi|_1$, the domain of definition $\mathfrak{D}(-\Delta_F)$ is the Sobolev space $H_0^1(\mathbb{R}^d)$, which in case of $G = \mathbb{R}^d$ is identical to the space $H^1(\mathbb{R}^d)$.

The domain of definition of the adjoint operator $-\Delta^*$ is the subspace

$$\mathfrak{D}(-\Delta^*) = \{ f \in L^2(\mathbb{R}^d) : -\Delta f \in L^2(\mathbb{R}^d) \},$$

which is equal to the space $H^2(\mathbb{R}^d)$ itself. Recall that a half-bounded operator $A$ is self-adjoint on the intersection set $\mathfrak{D}(A^*) \cap \mathfrak{D}(T_F)$, so we conclude that the Laplacian must be self-adjoint on the domain

$$\mathfrak{D}(-\Delta_F) = H^2(\mathbb{R}^d) \cap H_0^1(\mathbb{R}^d) = H^2(\mathbb{R}^d)$$

in compliance with the statement above.

2. *Dirichlet-Laplacian.* Let $G \subset \mathbb{R}^d$ be a *bounded* domain. Consider the Laplacian $-\Delta$ with domain $\mathfrak{D}(-\Delta) := C_c^\infty(G)$. The related sesquilinear form $Q$ to the Laplacian is given by

$$Q(\phi, \psi) := \int_G \nabla\phi \, \nabla\psi \, dx, \qquad \phi, \psi \in \mathfrak{D}(-\Delta),$$

which can continuously be extended to a sesquilinear form $\overline{Q}$ on the domain $\mathfrak{D}(\overline{Q}) := H_0^1(G) \times H_0^1(G)$.

Due to the FRIEDRICHS extension procedure, there is a self-adjoint operator $A_D$ on the Hilbert space $L^2(G)$ related to $\mathbb{Q}$ and defined by

$$\begin{cases} \mathfrak{D}(A_D) & := \{ f \in H_0^1((G) : \Delta f \in L^2(G) \} \\ A_D f & := -\Delta f, \quad f \in \mathfrak{D}(A_D). \end{cases}$$

The operator $A_D$ is called the *Laplace operator with Dirichlet boundary conditions*, often it is shortly called the *Dirichlet-Laplacian.* Obviously, we have

$$H_0^1((G) \cap H^2(G) \subset \mathfrak{D}(A_D).$$

One can prove that if $G$ is bounded and has a $C^2$-boundary or $G = \mathbb{R}^n_+$, then the reversed inclusion is also true, and in this case $\mathfrak{D}(A_D) = H_0^1(G) \cap H^2(G)$. This equality still holds in the case in which $G$ is convex and bounded.

The main result concerning the Dirichlet-Laplacian is given by the following

THEOREM 7.2. The operator $A_D$ as defined above is self-adjoint in Hilbert space $L^2(G)$. The domain of definition $\mathfrak{D}(A_D)$ equipped with the graph norm of $A_D$ is continuously embedded in $H_0^1(G)$. If $G$ has $C^1$-boundary, then $A_D$ is compactly embedded in $L^2(G)$.

From the last assertion it follows immediately that for a bounded set with $C^1$-boundary the operator $A_D$ has pure point spectrum. It is an important property of the Dirichlet-Laplacian in this case that 0 does not belong to the spectrum, so $A_D$ is invertible, and the inverse operator $A_D^{-1}$ is bounded.

3. *The Neumann-Laplacian.* Let $G \subset \mathbb{R}^d$ be a domain with $C^1$-boundary $\partial G$, let $H$ be the Hilbert space $L^2(G)$ and let the operator $A_N$ be defined by

$$\begin{cases} \mathfrak{D}(A_N) & := \{f \in H^2(G) : f_v(x) = 0, x \in \partial G\} \\ A_N f & := -\Delta f, \quad f \in \mathfrak{D}(A_N). \end{cases}$$

The operator $A_N$ is called the *Laplace operator with Neumann boundary condition* or briefly the *Neumann-Laplacian.* In distinction to the Dirichlet-Laplacian, the Neumann-Laplacian has 0 as an eigenvalue, and all constant functions in $\mathfrak{D}(A_N)$ are corresponding eigenfunctions.

## §8. Schrödinger Operators

By the Schrödinger operator is meant the linear partial differential operator given by

$$\mathcal{A}_q(u, x) := (-\Delta u)(x) + q(x),$$

defined in the entire Euclidean space $\mathbb{R}^d$. In the special case, where $q = 0$, self-adjointness of the operator $-\Delta$ acting in $L^2(\mathbb{R}^d)$ and with domain of definition $H^2(\mathbb{R}^d)$ is well-known from the last section.

As an application of theorem 4.2 coming from perturbation theory, consider the case, where $d = 3$ and the function $q$ belongs to $L^2(\mathbb{R}^3) + L^\infty(\mathbb{R}^3)$, that is,

$$q = q_1 + q_2$$

for some functions $q_1, q_2$ with $q_1 \in L^2(\mathbb{R}^3)$ and $q_2 \in L^\infty(\mathbb{R}^3)$. Then the function $q$ gives rise to a multiplication operator with domain of definition $C_c^\infty(\mathbb{R}^3)$. Since the negative Laplacian $-\Delta$ is essentially self-adjoint on that domain and $q$ is $-\Delta$-bounded with bound $\theta < 1$, the operator $\mathcal{A}_q = -\Delta + q$ with domain of definition $C_c^\infty(\mathbb{R}^3)$ is also essentially self-adjoint. The self-adjoint extension $\mathcal{H}_q$ of $\mathcal{A}_q$ is the generator of the unitary group $\{U(t)\}_{t\in\mathbb{R}}$ with

$$U(t) := e^{it\mathcal{H}_q} \qquad (t \in \mathbb{R}),$$

which uniquely solves the *time-dependent* Schrödinger problem

$$\begin{cases} u_t = i \cdot \mathcal{H}_q u, \\ u(0) = u_0. \end{cases}$$

The investigation of *time-independent solutions* of the Schrödinger equation leads to the eigenvalue problem

$$\mathcal{H}_q u := -\Delta u + qu = \lambda u, \tag{8.1}$$

which means to find those real numbers $\lambda$, for which there are non-trivial solutions $u$ of problem (8.1). Thus the discrete spectrum $\sigma_d(\mathcal{H}_q)$ of the Schrödinger operator is of interest. In fact, the non-trivial eigenfunctions belonging to eigenvalues $\lambda \in \sigma_d(\mathcal{H}_q)$ describe the stationary states of a quantum mechanical system.

The presence of a non-trivial potential function $q$ can fairly change the spectrum of a Schrödinger operator like the example of the *harmonic oscillator* shows: the spectrum of the Laplacian equals the connected set $[0, \infty)$, but the spectrum of the operator $-\Delta + q$ with $q : x \mapsto \frac{1}{2}x^2$ consists of discrete points only.

It is a further and interesting problem in quantum mechanics to determine lower and upper bounds for the discrete spectrum of the operator $\mathcal{H}_q$. In the following we present three assertions, which contain information on the bounds of the set $\sigma_d(\mathcal{H}_q)$, namely the BIRMAN-SCHWINGER *bound*, the LIEB-CWIKEL-ROZENBLUM *bound*, and KATO*'s theorem.*

1. BIRMAN-SCHWINGER *bound.* The following assertion presents bounds for the set $\sigma_d(\mathcal{H}_q)$, which are valid for certain potentials $q$ in case of $d = 3$. Preliminary, let $q \in L_\infty(\mathbb{R}^3)$ and

$$B_q := \int_{\mathbb{R}^3}\int_{\mathbb{R}^3} \frac{q(x)q(y)}{|x-y|^2}\, dx\, dy < +\infty.$$

Then, if $N(\mathcal{H}_q)$ is the total number of eigenvalues of the operator $\mathcal{H}_q$, then

$$N(\mathcal{H}_q) \leq \frac{1}{16\pi^2} B_q.$$

2. The Lieb-Cwikel-Rozenblum *theorem* provides an estimate on the number $N_(\mathcal{H}_q)$ of negative eigenvalues of the Schrödinger operator: let $d > 3$ and $q$ be an element of the space $L^\infty_{loc}(\mathbb{R}^d)$, then

$$N(\mathcal{H}_q) \leq c_n \int_{\mathbb{R}^d} \min\{q(x), 0\}^{n/2}\, dx,$$

where $c_n$ is a constant depending only from the space dimension $d$.

3. The subsequent result is often denoted as Kato*'s theorem.* Indeed, it does not describe bounds of the discrete spectrum, but quotes the absence of positive eigenvalues under relative mild assumptions on the potential function $q$: if $q \in L^\infty_{loc}(\mathbb{R}^d)$ and

$$\lim_{|x| \to \infty} |x|\, q(x) = 0,$$

then the operator $-\Delta + q$ does not own any positive eigenvalue.

## §9. Strongly Elliptic Differential Operators of Order $2m$

A *linear elliptic partial differential operator* $\mathcal{A}$ of even order $2m$ on a domain $G \subseteq \mathbb{R}^d$ is a mapping of the form $\mathcal{A}u = \sum_{|\alpha| \leq 2m} a_\alpha D^\alpha u$, where the functions $a_\alpha$ are for examples members of the space $C^{2m}(G)$ and whose *principal symbol*

$$a(x, \xi) = \sum_{|\alpha| = 2m} a_\alpha(x)\, \xi^\alpha, \quad \xi \in \mathbb{R}^d$$

satisfies $a(x, \xi) \neq 0$ for all $x \in G$ and $\xi \in \mathbb{R}^d \setminus \{0\}$. We will briefly call $\mathcal{A}$ an *elliptic operator.* Furthermore, $\mathcal{A}$ is called *uniformly strongly elliptic*, if there is a constant $c > 0$ such that

$$\sum_{|\alpha| = |\beta| = m} a_{\alpha\beta}(x)\, \xi^\alpha \xi^\beta \geq c \sum_{|\alpha| = m} \xi^{2\alpha}.$$

Elliptic operators are usually considered under the following assumptions:

(E-1) The functions $a_\alpha$ are real-valued for $|\alpha| = 2m$, and either $a(x, \xi) < 0$ or $a(x, \xi) > 0$ holds for all $x \in G$ and

all $\xi \in \mathbb{R}^d \setminus \{0\}$. This implies that for any compact set $K \subset G$ there is a constant $C_K > 0$ such that

$$a(x,\xi) \geq C_K |\xi|^{2m} \qquad (x \in G,\ \xi \in \mathbb{R}^d).$$

(E-2) The expression $\mathcal{A}$ can be written in *divergence form*

$$\mathcal{A}(x,\cdot) = \sum_{|\alpha|\leq m, |\beta|\leq m} (-1)^\alpha D^\alpha (a_{\alpha\beta}(x)\, D^\beta),$$

where $a_{\alpha\beta} = \overline{a_{\beta\alpha}}$ for all $\alpha$ and $\beta$. Notice that every linear differential operator with sufficiently smooth coefficient functions can be written in divergence form.

As for the differential operators mentioned in the previous sections, a formally self-adjoint and uniformly strongly elliptic differential operator $\mathcal{A}$ causes a symmetric and quasi-coercive sesquilinear form defined on the space $H_0^m(G) \times H_0^m(G)$, so the FRIEDRICHS extension theory gives rise to a self-adjoint realization $\mathcal{A}_F$ in $L^2(G)$ with domain of definition

$$\mathfrak{D}(\mathcal{A}_F) = \{u \in H_0^m(G) : \mathcal{A}_F\, u \in L^2(G)\}.$$

If the boundary of $G$ is sufficiently smooth, however, then

$$\mathfrak{D}(\mathcal{A}_F) = H^{2m}(G) \cap H_0^m(G)$$

holds. It is also true that $\mathcal{A}_F$ has pure point spectrum, if $G$ is bounded in $\mathbb{R}^d$.

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